

AP EAMCET-2019-Engineering-20 Apr-Shift-2 Mathematics

1. If $f : \mathbb{R} - \left\{ \frac{3}{7} \right\} \rightarrow \mathbb{R} - \left\{ \frac{3}{7} \right\}$ is given by $f(x) = \frac{3x+5}{7x-3}$, then the

statement which is not true is

(1) $f^{-1}(x) = f(x)$

(2) $(f \circ f)(x) = x$

(3) $(f \circ f \circ f)(x) = x$

(4) $(f \circ f \circ f \circ f)(x) = x$

2. If $f : A \rightarrow B, g : B \rightarrow C$ are functions such that $g \circ f : A \rightarrow C$ is onto, then a necessary condition is

(1) f is onto

(2) g is onto

(3) both f, g are onto

(4) neither f nor g is onto.

3. The sum of the first n terms of the series $\frac{3}{5} + \frac{21}{25} + \frac{117}{125} + \dots$ is

$$(1) \ n + \frac{2^{n+1}}{3 \times 5^n} - \frac{2}{3}$$

$$(2) \ n - \frac{2^{n+1}}{3 \times 5^n} - \frac{2}{3}$$

$$(3) \ n + \frac{2^{n+1}}{3 \times 5^n} + \frac{2}{3}$$

$$(4) \ n - \frac{2^{n+1}}{3 \times 5^n} + \frac{2}{3}$$

4. The sum of the values of x so that the matrix

$$\begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix} - x \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ is singular is,}$$

$$(1) \ 3$$

$$(2) \ 5$$

$$(3) \ 7$$

$$(4) \ 9$$

$$5. \text{ If } A = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}, \text{ then } (\text{Adj } A)^{-1} =$$

- (1) $A + I$
- (2) $A - I$
- (3) A
- (4) $\text{Adj}(A^{-1})$

6. If $abc \neq 0$ and the system of equation $x + 7ay + 2az = 0$, $a + 6by + 2bz = 0$, $x + 5cy + 2cz = 0$ has a non-trivial solution, then a, b, c are in

- (1) Harmonic Progression
- (2) Geometric Progression
- (3) Arithmetic Progression
- (4) Arithmetic-geometric Progression

7. z_1, z_2 are complex numbers with $|z_1 - z_2| < k$. If the complex number z satisfies the condition $|z - z_1| + |z - z_2| = k$, then z lies on

- (1) a parabola
- (2) an ellipse
- (3) a circle
- (4) a hyperbola

8. In a complex plane $\mathbb{C}$, the set $\left\{ z \in \mathbb{C} : \arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4} \right\}$

represents

- (1) a straight line
- (2) a circle
- (3) a parabola
- (4) an ellipse

9. For $a, b, c, d \in \mathbb{R}$, if $z_1 = a + ib$, $z_2 = c + id$ are such that $|z_1| = |z_2| = 1$ and $\operatorname{Re}(z_1 \bar{z}_2) = 0$, then the pair of complex numbers $w_1 = a + ic$ and $w_2 = b + id$ satisfy

- (1) $\operatorname{Re}(w_1 \bar{w}_2) = 0$
- (2) $\operatorname{Re}(w_1 \bar{w}_2) = 1$
- (3) $|w_1| \neq |w_2|$
- (4) $|w_1| = |w_2| = 0$

10. If $(1+x)^n = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ and

$$a_0 - a_2 + a_4 - a_6 + \dots = k \cos \frac{n\pi}{4}, \text{ then } k =$$

- (1) 2^n
- (2) 2^{2n}
- (3) $\frac{2^n}{2}$
- (4) $2^{n/2}$

11. For $x > 2$, the equation $\sqrt{x+2} - \sqrt{x-2} = \sqrt{4x-2}$ has

- (1) One Solution
- (2) Two solutions
- (3) More than two solutions
- (4) No solution

12. If $x \in \mathbb{R}$, then the range of $\frac{x}{x^2 - 5x + 9}$ is

- (1) $\left(-\frac{1}{11}, 1\right)$
- (2) $\left(-\infty, \frac{-1}{11}\right) \cup (1, \infty)$

$$(3) \left[-\frac{1}{11}, 1 \right]$$

$$(4) \left[-1, \frac{1}{11} \right]$$

13. If α, β, γ are the roots of the equation $x^3 - ax^2 + bx - c = 0$ then

$$\sum \alpha^2 (\beta + \gamma) =$$

$$(1) ab - 3c$$

$$(2) \frac{ab - 3c}{c}$$

$$(3) \frac{b^2 - 2ac}{c^2}$$

$$(4) \frac{a^2 - 2b}{c^2}$$

14. The sum of all the rational roots of the equation

$$6x^6 - 25x^5 + 31x^4 - 31x^2 + 25x - 6 = 0 \text{ is}$$

$$(1) 3$$

$$(2) 3.5$$

$$(3) \frac{25}{6}$$

(4) 2.5

15. The number of five digit numbers that are divisible by 6 which can be formed by choosing digits from $\{0,1,2,3,4,5\}$, when repetition is allowed, is

(1) 648

(2) 540

(3) 1296

(4) 1080

16. A person invites 8 guests to a dinner and places 5 of them at one table and the remaining 3 at another, both the tables being round. The number of ways in which the guest can be arranged is

(1) 40320

(2) 2688

(3) 8064

(4) 13440

17. If 15^k divides $47!$ but 15^{k+1} does not divide it, then $k =$

- (1) 15
- (2) 12
- (3) 10
- (4) 5

18. If $(1 + x + x^2)^n = c_0 + c_1x + c_2x^2 + \dots$, then the value of $c_0c_1 - c_1c_2 + c_2c_3 - \dots$ is

- (1) $(-1)^n$
- (2) 0
- (3) 2^n
- (4) 3^n

19. If x is so small that x^2 and higher power of x can be neglected,

then the approximate value of $\left(1 + \frac{3}{4}x\right)^{\frac{1}{2}} \left(1 - \frac{2x}{3}\right)^{-2}$ is

- (1) $\frac{41 + 24x}{41}$
- (2) $\frac{41 - 24x}{41}$

(3) $\frac{24 + 41x}{24}$

(4) $\frac{24 - 41x}{24}$

20. If $\frac{x^4}{(x-a)(x-b)(x-c)} = P(x) + \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$, then

$P(0) + A(a-b)(a-c) =$

(1) $a^4 + b^4 + c^4 + a$

(2) $a + b + c$

(3) $a^4 - a - b - c$

(4) $a + b + c + a^4$

21. If $\sin \alpha = p$, then the quadratic equation whose roots are $\tan \frac{\alpha}{2}$,

$\cot \frac{\alpha}{2}$ is

(1) $px^2 - 2x + p = 0$

(2) $px^2 + 2x + p = 0$

(3) $px^2 + x + p = 0$

(4) $px^2 - x + p = 0$

22. $\cos 66^\circ + \sin 84^\circ =$

(1) $\frac{1}{4}(\sqrt{3} + \sqrt{5})$

(2) $\frac{1}{4}\sqrt{5}(\sqrt{3} + 1)$

(3) $\frac{1}{4}(\sqrt{3} + 1)(\sqrt{5} + 1)$

(4) $\frac{1}{4}\sqrt{3}(\sqrt{5} + 1)$

23. If $\tan \theta + \tan 2\theta + \sqrt{3} \tan \theta \tan 2\theta = \sqrt{3}$, then the general value of θ are

(1) $(3n + 1)\frac{\pi}{3}, n \in \mathbb{Z}$

(2) $(3n + 1)\frac{\pi}{9}, n \in \mathbb{Z}$

(3) $(3n + 1)\frac{\pi}{6}, n \in \mathbb{Z}$

(4) $(2n + 1)\frac{\pi}{9}, n \in \mathbb{Z}$

24. The sum of the values of x satisfying the equation

$$\sin^{-1}\left(\frac{3x}{5}\right) + \sin^{-1}\left(\frac{4x}{5}\right) = \sin^{-1}(x), \text{ is}$$

(1) 0

(2) $\frac{1}{2}$

(3) $\frac{1}{3}$

(4) $\frac{1}{4}$

25. If $\theta = \frac{\pi}{6}$ and $x = \log\left[\cot\left(\frac{\pi}{4} + \theta\right)\right]$ then $\sinh(x) =$

(1) $\sqrt{3}$

(2) $\frac{1}{\sqrt{3}}$

(3) $-\sqrt{3}$

(4) $-\frac{1}{\sqrt{3}}$

26. In $\triangle ABC$, if the median AD drawn through A is perpendicular to the side AC , then $3ca \cos A \cos C + 2a^2 =$

- (1) c^2
- (2) $2c^2$
- (3) $3c^2$
- (4) $4c^2$

27. Let Δ denote the area of a $\triangle ABC$. If α, β, γ are the lengths of the altitudes of the $\triangle ABC$, then $\alpha^{-2} + \beta^{-2} + \gamma^{-2} =$

- (1) $\frac{4}{\Delta}(\tan A + \tan B + \tan C)$
- (2) $\frac{1}{\Delta}(\cot A + \cot B + \cot C)$
- (3) $\frac{\Delta^2}{2}(\tan A + \tan B + \tan C)$
- (4) $\frac{\Delta^2}{4}(\cot A + \cot B + \cot C)$

28. In $\triangle ABC$, $(r_2 + r_3) \cot\left(\frac{B+C}{2}\right) =$

- (1) $a + b + c$

(2) a

(3) b

(4) c

29. If the vectors $\overrightarrow{AB} = \vec{i} + 3\vec{j} + 4\vec{k}$, $\overrightarrow{AC} = 5\vec{i} + \vec{j} + 2\vec{k}$ are two sides of triangle ABC, whose centroid is G, then $|\overrightarrow{AG}| =$

(1) $\frac{2}{3}\sqrt{22}$

(2) $\sqrt{22}$

(3) $\frac{1}{3}\sqrt{22}$

(4) $\sqrt{18}$

30. The point of intersection of the lines represented by

$$\vec{r} = (\vec{i} + 2\vec{j} - \vec{k}) + \lambda(2\vec{i} + 3\vec{j} + 4\vec{k}) \quad \text{and}$$

$$\vec{r} = (-\vec{i} - 3\vec{j} + 7\vec{k}) + \mu(\vec{i} + 2\vec{j} - \vec{k}) \text{ is}$$

(1) $3\vec{i} + 5\vec{j} + 3\vec{k}$

(2) $5\vec{i} + 8\vec{j} + 7\vec{k}$

(3) $-\vec{i} - \vec{j} - 5\vec{k}$

(4) $-3\bar{i} - 4\bar{j} - 9\bar{k}$

31. If A, B, C and D are points whose position vectors are $\bar{i} + \bar{j} + \bar{k}$, $4\bar{i} - \bar{j} + 2\bar{k}$, $5\bar{i} + \bar{j}$, $7\bar{i} + 2\bar{j} + 3\bar{k}$ respectively, then the projection of $\overline{AB}$ on $\overline{CD}$ is

(1) $\frac{4}{3}$

(2) $\sqrt{\frac{2}{7}}$

(3) $\frac{3}{4}$

(4) $\sqrt{\frac{7}{2}}$

32. Let $\bar{a}$ and $\bar{b}$ be unit vectors with θ as the acute angle between

them. If $\frac{1}{2}|\bar{a} - \bar{b}| = \sin \lambda \theta$, then $4\lambda^2 =$

(1) 4

(2) 1

(3) 3

(4) 2

33. Let $\vec{a} = 2\vec{i} - 3\vec{j} + 4\vec{k}$, $\vec{b} = \vec{i} + 2\vec{j} - 2\vec{k}$ and $\vec{c} = 3\vec{i} - \vec{j} + \vec{k}$. The volume (in cubic units) of the parallelepiped having $\vec{a} + \vec{b} + \vec{c}$, $\vec{a} - \vec{b} + \vec{c}$ and $\vec{a} + \vec{b} - \vec{c}$ as coterminal edges is

- (1) 6
- (2) 7
- (3) 28
- (4) 36

34. If $|\vec{a}| = 1$, $|\vec{b}| = 1$, $|\vec{c}| = 2$ and $\vec{a} \times (\vec{a} \times \vec{c}) + \vec{b} = \vec{0}$, then $(\vec{a} \cdot \vec{c})^2 =$

- (1) 1
- (2) 2
- (3) 4
- (4) 3

35. The variance of the observations 2, 3, 5, 7, 11, 13, 17, 22 is

- (1) 43.75
- (2) 48.25
- (3) 80
- (4) 350

36 The coefficient of variation and standard deviation of an ungrouped data are 60 and 21 respectively. If 15 is added to every observation of the data, then the coefficient of variation of the new data is

- (1) 30
- (2) 42
- (3) 40
- (4) 20

37. Each of the two boxes A and B contains 10 chits numbered 1 to 10. If one chit is drawn at random from each of A and B, then the probability that the number on the chit drawn from A is smaller than the number on the chit drawn from B, is

- (1) $\frac{9}{10}$
- (2) $\frac{9}{20}$
- (3) $\frac{19}{20}$

(4) $\frac{17}{20}$

38. An envelope is known to have come from either 'LONDON' or 'CLIFTON'. On the postal mark only two successive letters ON are legible. The probability that the envelope comes from LONDON is

(1) $\frac{12}{17}$

(2) $\frac{5}{17}$

(3) $\frac{3}{17}$

(4) $\frac{2}{5}$

39. The probability function of a random variable X is given by $P(X = k) = ck^2$, where c is a constant and $k \in \{0, 1, 2, 3, 4\}$. If σ^2 is the variance of X and μ is the mean of X , then $\sigma^2 + \mu^2 =$

(1) 3.33

(2) 11.8

(3) $\frac{1}{30}$

(4) 354

40. In a book of 250 pages, there are 200 typographical errors. Assuming that the number of errors per page follow the Poisson law, then the probability that a random sample of 5 pages will contain no typographical error is

(1) e^{-4}

(2) $e^{-\frac{4}{5}}$

(3) $e^{-\frac{16}{25}}$

(4) e^{-1}

41. The locus of the point $P(x, y)$ satisfying the equation

$$\sqrt{(x-2)^2 + y^2} + \sqrt{(x+2)^2 + y^2} = 4 \text{ is}$$

(1) an ellipse

(2) a parabola

(3) a line segment

(4) a circle

42. If the origin is shifted to $(2,3)$ and the axes are rotated through an angle 45° about that point, then the transformed equation of $2x^2 + 2y^2 - 8x - 12y + 18 = 0$ is

(1) $x^2 - 7y^2 - 14xy - 2 = 0$

(2) $x^2 + y^2 = 4$

(3) $x^2 - y^2 = 4$

(4) $8x^2 - 2y^2 = 9$

43. A line passing through $P(4,2)$ cuts the coordinate axes at A and B respectively. If O is the origin, then the locus of the centre of the circum-circle of $\triangle OAB$ is

(1) $x^{-1} + y^{-1} = 2$

(2) $2x^{-1} + y^{-1} = 1$

(3) $x^{-1} + 2y^{-1} = 1$

(4) $2x^{-1} + 3y^{-1} = 1$

44. The centroid of the triangle formed by the lines $x + y - 1 = 0$, $x - y - 1 = 0$, $x - 3y + 3 = 0$ is

(1) $\left(\frac{4}{3}, 1\right)$

(2) $\left(-\frac{4}{3}, 1\right)$

(3) $\left(\frac{8}{3}, 3\right)$

(4) $\left(\frac{-8}{3}, 3\right)$

45. The quadratic equation whose roots are the coordinates of the circumcentre of the triangle formed by the points $(-2, -1)$, $(6, -1)$, $(2, 5)$ is

(1) $x^2 - 5x + 6 = 0$

(2) $2x^2 - 9x + 9 = 0$

(3) $3x^2 - 8x + 4 = 0$

(4) $6x^2 - 13x + 6 = 0$

46. The straight line $x + y + 1 = 0$ bisects an angle between a pair of lines, of which one is $2x - 3y + 4 = 0$. Then the equation of the other line in that pair is

(1) $2x + 3y + 4 = 0$

(2) $x - y + 1 = 0$

(3) $5x - 5y + 9 = 0$

(4) $3x - 2y + 5 = 0$

47. If the pairs of straight lines represented by $3x^2 + 2hxy - 3y^2 = 0$ and $3x^2 + 2hxy - 3y^2 + 2x - 4y + c = 0$ form a square then $(h, c) =$

(1) $(4, -1)$

(2) $(-1, 4)$

(3) $(-4, 1)$

(4) $(1, -4)$

48. The equation of the bisectors of the angles between the lines joining the origin to the points of intersection of the curve $x^2 + xy + y^2 + x + 3y + 1 = 0$ and the line $x + y + 2 = 0$ is

(1) $x^2 + 4xy - y^2 = 0$

(2) $2x^2 + 5xy - y^2 = 0$

(3) $x^2 + 6xy - 2y^2 = 0$

(4) $2x^2 - 4xy + 2y^2 = 0$

49. A is the centre of the circle $x^2 + y^2 - 2x - 4y - 20 = 0$. If the tangents drawn at the points B(1,7) and D(4,-2) on the circle meet at the point C, the area of the quadrilateral ABCD (in square units) is

(1) 75

(2) 64

(3) 56

(4) 45

50. The point of concurrence of all conjugate lines of the line $5x + 7y - 78 = 0$ with respect to the circle $x^2 + y^2 + 6x + 8y - 96 = 0$ is

(1) (-2,3)

(2) (3,-2)

(3) (3,2)

(4) (2,3)

51. The number of common tangents to the circles

$$x^2 + y^2 + 4x - 6y - 12 = 0 \text{ and } x^2 + y^2 - 8x + 10y + 5 = 0 \text{ is}$$

- (1) 4
- (2) 3
- (3) 2
- (4) 1

52. If two circles $(x-1)^2 + (y-3)^2 = r^2$ and

$$x^2 + y^2 - 8x + 2y + 8 = 0 \text{ intersect at two distinct points, then}$$

- (1) $2 < r < 8$
- (2) $1 < r < 9$
- (3) $r = 2$
- (4) $r = 8$

53. If the circle $x^2 + y^2 + 6x - 2y + k = 0$ bisects the circumference of the circle $x^2 + y^2 + 2x - 6y - 15 = 0$ then $k =$

- (1) 21
- (2) -21
- (3) -23
- (4) 23

54. The equation of the circle which cuts the circles

$$x^2 + y^2 + 4y - 7 = 0,$$

$$2x^2 + 2y^2 + 3x + 5y - 9 = 0,$$

$x^2 + y^2 + y = 0$ orthogonally is

(1) $x^2 + y^2 - 4x - 2y - 1 = 0$

(2) $x^2 + y^2 - 4x - 6y - 3 = 0$

(3) $x^2 + y^2 - 4x - 2y - 3 = 0$

(4) $x^2 + y^2 - 2x - 4y - 1 = 0$

55. The vertex and the focus of the parabola $2y^2 + 5x - 6y + 1 = 0$ are respectively

(1) $\left(\frac{7}{10}, \frac{3}{2}\right), \left(\frac{3}{40}, \frac{3}{2}\right)$

(2) $\left(-\frac{7}{10}, \frac{3}{2}\right), \left(\frac{53}{40}, \frac{3}{2}\right)$

(3) $\left(\frac{7}{10}, -\frac{3}{2}\right), \left(\frac{7}{10}, \frac{7}{8}\right)$

(4) $\left(\frac{-7}{10}, -\frac{3}{2}\right), \left(\frac{7}{10}, \frac{17}{8}\right)$

56. If a normal chord at point $t(\neq 0)$ on the parabola $y^2 = 9x$ subtends a right angle at its vertex then $t =$

- (1) $\sqrt{3}$
- (2) $\sqrt{5}$
- (3) $\pm\sqrt{3}$
- (4) $\pm\sqrt{2}$

57. The major and minor axes of an ellipse are along the X-axis and the Y-axis respectively. If its latus rectum is of length 4 and the distance between the foci is $4\sqrt{2}$, then the equation of ellipse is

- (1) $2x^2 + y^2 = 16$
- (2) $x^2 + 2y^2 = 16$
- (3) $\frac{x^2}{2} + \frac{y^2}{3} = 1$
- (4) $\frac{x^2}{3} + \frac{y^2}{2} = 1$

58. If $c \in \mathbb{R}$ such that the line $4x - y + c = 0$ touches the ellipse $x^2 + 4y^2 = 4$, then an equation having all such values of c among its roots is

$$(1) x^2 - (1 + \sqrt{17})x + \sqrt{17} = 0$$

$$(2) x^2 - (1 - \sqrt{17})x - \sqrt{17} = 0$$

$$(3) x^3 - x^2 - 17x + 17 = 0$$

$$(4) x^3 - x^2 + 17x + 17 = 0$$

59. If e_1 , e_2 are respectively the eccentricities of the curves

$$9x^2 - 16y^2 - 144 = 0 \text{ and } 9x^2 - 16y^2 + 144 = 0 \text{ then } \frac{e_1^2 e_2^2}{e_1^2 + e_2^2} =$$

$$(1) \sqrt{2}$$

$$(2) 1$$

$$(3) \sqrt{3}$$

$$(4) 2$$

60. If $A(2, 4, -1)$, $B(3, 6, -1)$ and $C(4, 5, 1)$ are three consecutive vertices of a parallelogram, then its fourth vertex is

$$(1) (1, 3, 3)$$

$$(2) (1, 3, -3)$$

$$(3) (3, 3, -1)$$

(4) $(3,3,1)$

61. If the line joining the points $A(2,3,1)$ and $B(3,5,-3)$ is perpendicular to the line joining the $C(1,2,3)$ and $D(3,y,7)$ then

$y =$

(1) 1

(2) 3

(3) 5

(4) 7

62. A plane is making intercepts 2,3,4 on X, Y and Z-axes respectively. Another plane is passing through point $(-1,6,2)$ and is perpendicular to the line joining the points $(1,2,3)$ and $(-2,3,4)$. Then, the angle between the two planes is

(1) 90°

(2) $\cos^{-1} \sqrt{\frac{12}{61}}$

(3) $\cos^{-1} \sqrt{\frac{11}{61}}$

(4) $\cos^{-1} \sqrt{\frac{5}{6}}$

63. If $\alpha = \lim_{x \rightarrow 0} \frac{x \cdot 2^x - x}{1 - \cos x}$ and $\beta = \lim_{x \rightarrow 0} \frac{x \cdot 2^x - x}{\sqrt{1+x^2} - \sqrt{1-x^2}}$ then

(1) $\alpha = \beta$

(2) $2\alpha = \beta$

(3) $\alpha = 2\beta$

(4) $\alpha = 3\beta$

64. If $f(x) = \begin{cases} \frac{x - |x|}{x}, & \text{when } x < 0 \\ 5x^2 + a, & \text{when } 0 \leq x \leq 1 \\ b \left(\frac{x^2 - 1}{x^2 - 3x + 2} \right), & \text{when } 1 < x < 3 \\ -14, & \text{when } x \geq 3 \end{cases}$ is a continuous

function on $\mathbb{R}$, then $(a, b) =$

(1) $\left(2, -\frac{7}{2} \right)$

(2) $(2, -14)$

(3) $\left(-\frac{7}{2}, -14\right)$

(4) $(2, 7)$

65. The number of points in the interval $(0, 2)$ at which

$f(x) = |x - 0.5| + |x - 1| + \tan x$ is not differentiable is

(1) 1

(2) 2

(3) 3

(4) 4

66. If $x = \sec \theta - \cos \theta$, $y = \sec^{10} \theta - \cos^{10} \theta$ and

$(x^2 + 4) \left(\frac{dy}{dx} \right)^2 = k(y^2 + 4)$, then $k =$

(1) $\frac{1}{100}$

(2) 1

(3) 10

(4) 100

67. If $y = e^{\sin^{-1}x}$, then $(1 - x^2)y_2 - xy_1 = 0$

- (1) 0
- (2) 1
- (3) y
- (4) $2y$

68. The semi vertical angle of a cone is 45° . If the height of cone is 20.025 cm, then the approximate value of its lateral surface area (in sq.cm) is

- (1) $401\sqrt{2}\pi$
- (2) $400\sqrt{2}\pi$
- (3) $402\sqrt{2}\pi$
- (4) $405\sqrt{2}\pi$

69. The radius of a sphere increase at the rate of 0.04 cm/sec. The rate of increase in the volume of that sphere with respect to its surface area, when its radius is 10 cm is

- (1) 16π
- (2) 25

(3) 20

(4) 5

70. If $f(x) = a \log|x| + bx^2 + x$ has extreme values at $x = -1$ and $x = 2$, then the ordered pair $(a, b) =$

(1) $(2, -1)$

(2) $\left(2, -\frac{1}{2}\right)$

(3) $(-1, 2)$

(4) $\left(-\frac{1}{2}, 2\right)$

71. If $f(x) = x^3 + px^2 + qx$ is defined on $[0, 2]$ such that

$f(0) = f(2)$ and $f'\left(1 + \frac{1}{\sqrt{3}}\right) = 0$, then $p^2 + q^2 =$

(1) 13

(2) 5

(3) $2 + \frac{1}{\sqrt{3}}$

(4) 1

72. If $x \neq \frac{-3}{\sqrt{2}}$, then $\int \frac{x^2}{2x^2 + 6\sqrt{2}x + 9} dx =$

(1) $\frac{1}{2\sqrt{2}} \left[(\sqrt{2}x + 3) - 6 \log |\sqrt{2}x + 3| - \frac{9}{\sqrt{2}x + 3} \right] + c$

(2) $\frac{1}{2\sqrt{2}} \left[(\sqrt{2}x + 3) - 6 \log |\sqrt{2}x + 3| + \frac{9}{\sqrt{2}x + 3} \right] + c$

(3) $\sqrt{2}x + 3 - 6 \log |\sqrt{2}x + 3| + c$

(4) $\log(2x^2 + 6\sqrt{2}x + 9) + c$

73. $\int \frac{\sqrt{1-x^2} \sin^{-1} x + x}{\sqrt{1-x^2}} dx =$

(1) $x \sin^{-1} x + \sqrt{1-x^2} + c$

(2) $\sin^{-1} x + \sqrt{1-x^2} + c$

(3) $x \sin^{-1} x + c$

(4) $\frac{x \sin^{-1} x}{\sqrt{1-x^2}} + c$

74. If $\int x^3 e^{2x} dx = \frac{e^{2x}}{8} f(x) + c$, then the sum of all the complex roots

of $f(x) = 1$ is

(1) $\frac{1}{2}$

(2) 3

(3) 1

(4) 2

75. $\int \sin^5 x \cos^5 x dx =$

(1) $\frac{\cos^6 x}{60} [6 \sin^4 x + 3 \sin^2 x + 1] + c$

(2) $-\frac{\sin^6 x}{60} [6 \cos^4 x - 3 \cos^2 x + 1] + c$

(3) $-\frac{\cos^6 x}{60} [6 \sin^4 x + 3 \sin^2 x + 1] + c$

(4) $\frac{\sin^6 x}{60} [6 \cos^4 x + 3 \cos^2 x + 1] + c$

76. $\lim_{x \rightarrow \infty} \left[\frac{1}{n} + \frac{n^2}{(n+1)^3} + \frac{n^2}{(n+2)^3} + \frac{n^2}{(n+3)^3} + \cdots + \frac{1}{125n} \right] =$

(1) $\frac{3}{8}$

(2) $\frac{15}{32}$

(3) $\frac{12}{25}$

(4) $\frac{35}{72}$

77. $\int_{e^{-1}}^{e^2} \left| \frac{\log x}{x} \right| dx =$

(1) $\frac{2}{5}$

(2) 2

(3) 5

(4) $\frac{5}{2}$

78. The area (in square units) bounded by $y = \tan^{-1}x$, $y = \cot^{-1}x$ and the Y-axis is

(1) $\log_e 4$

(2) $\log_e 2$

(3) $\log_e 3$

(4) $\log_e 5$

79. The order of the differential equation corresponding to the family of parabolas whose axes are along the X axis and whose foci are at the origin is

(1) 4

(2) 3

(3) 2

(4) 1

80. The general solution of the differential equation

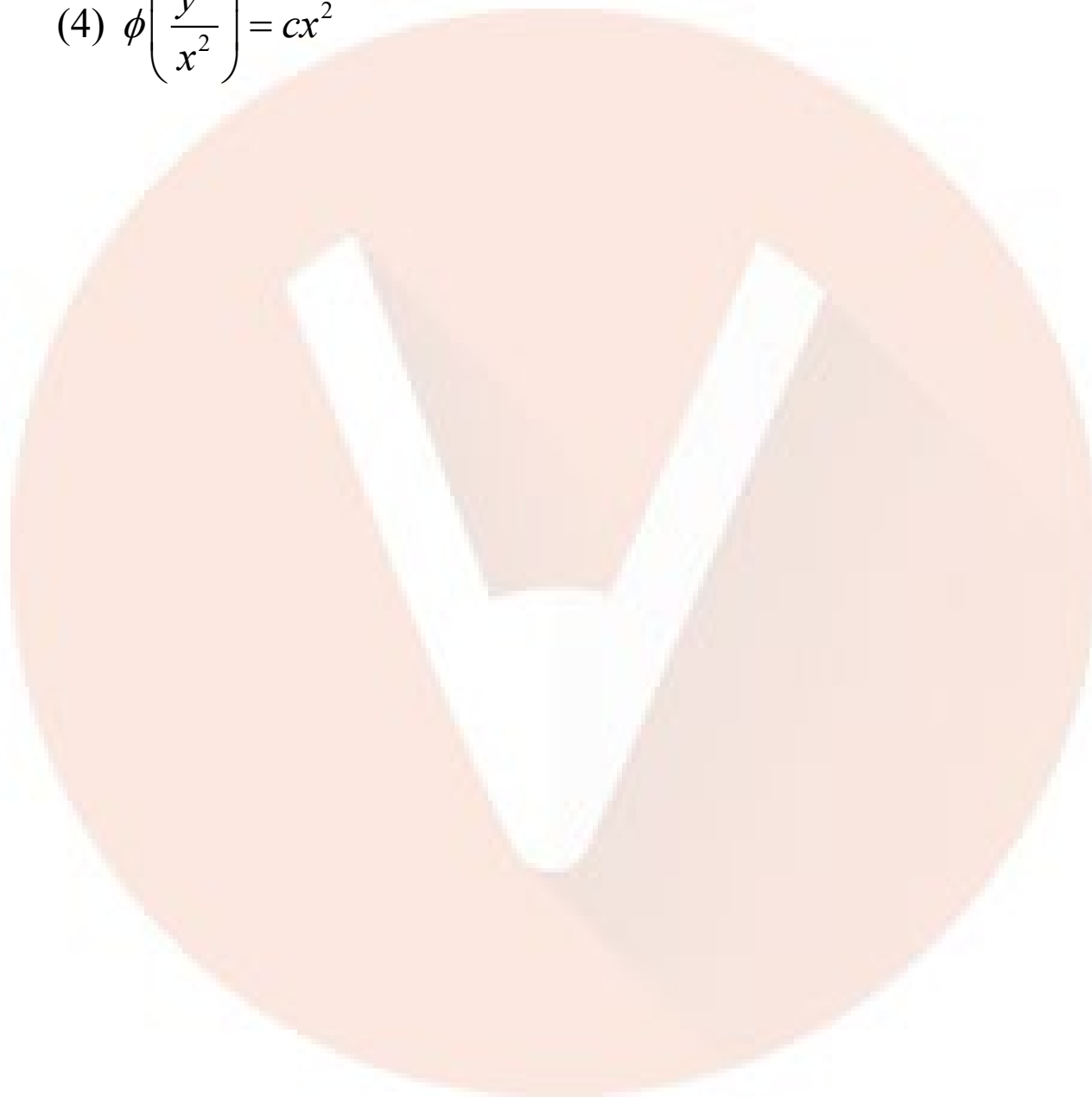
$$yy' = x \left[\frac{y^2}{x^2} + \frac{\phi\left(\frac{x^2}{y^2}\right)}{\phi'\left(\frac{x^2}{y^2}\right)} \right], \text{ where } \phi \text{ is an arbitrary function, is}$$

(1) $x\phi\left(\frac{y^2}{x^2}\right) = cy$

(2) $x^2\phi\left(\frac{y^2}{x^2}\right) = c$

$$(3) \ x^2 \phi\left(\frac{y^2}{x^2}\right) = cy^2$$

$$(4) \ \phi\left(\frac{y^2}{x^2}\right) = cx^2$$



AP EAMCET-2019-Engineering-20 Apr-Shift-2 Physics

81. A physical quantity obtained from the ratio of the coefficient of thermal conductivity to the universal gravitational constant has a dimensional formula $M^{2a}L^{4b}T^{2c}K^d$, then the value of $\frac{a+b}{c+b} - d$ is

(1) $+\frac{3}{2}$

(2) $-\frac{1}{2}$

(3) $-\frac{3}{2}$

(4) $+\frac{1}{2}$

82. A body starting from rest at $t=0$ moves along a straight line with a constant acceleration. At $t=2$ s, the body reverses its direction keeping the acceleration same. The body returns to the initial position at $t=t_0$, the t_0 is

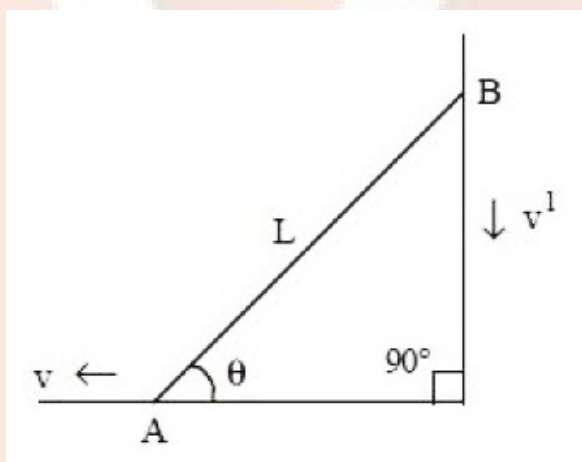
(1) 4 s

(2) $(4 + 2\sqrt{2})s$

(3) $(2 + 2\sqrt{2})s$

(4) $(4 + 4\sqrt{2})s$

83. A thin uniform rod of length L is resting against a wall and the floor as shown in the figure. Its lower end A is pulled towards left with a constant velocity ' v '. Then the downward velocity ' v' ' of the other end B when the rod makes an angle θ with the floor is



(1) v

(2) $v \cos \theta$

(3) $v \sin \theta$

(4) $v \cot \theta$

84. Two boys conducted experiments on projectile motion with stopwatch and noted some readings. As one throws a stone in air at some angle with the horizontal, the other boy observes that after 4 s the stone is moving at an 30° to the horizontal and after another 2 s it is travelling horizontally. The magnitude of the initial velocity of the stone is (Acceleration due to gravity = 10 ms^{-2})

(1) $40\sqrt{3} \text{ ms}^{-1}$

(2) $20\sqrt{3} \text{ ms}^{-1}$

(3) $10\sqrt{3} \text{ ms}^{-1}$

(4) $50\sqrt{3} \text{ ms}^{-1}$

85. A force of $(2.6\hat{i} + 1.6\hat{j}) \text{ N}$ acts on a body of mass 2 kg. If the velocity of the body at time $t = 0$ is $(3.6\hat{i} - 4.8\hat{j}) \text{ ms}^{-1}$, the time at which the body will just have a velocity along X – axis only is

(1) 1 s

(2) 2 s

(3) 3 s

(4) 6 s

86. The force required to move a body up a rough inclined plane is double the force required to prevent the body from sliding down the plane. If the angle of inclination of the plane is 60° , the coefficient of friction is

(1) $\frac{1}{3}$

(2) $\frac{1}{\sqrt{2}}$

(3) $\frac{1}{\sqrt{3}}$

(4) $\frac{1}{2}$

87. A particle moves in the X-Y plane under the action of a force

$$\vec{F} = K \left[\frac{x}{(x^2 + y^2)^{3/2}} \hat{i} + \frac{y}{(x^2 + y^2)^{3/2}} \hat{j} \right], \text{ where } K \text{ is a constant.}$$

Work done by the force when the particle moves from $(0, a)$ to $(a, 0)$ along a circular path of radius “a” about the origin is

(1) $\frac{2K\pi}{a}$

(2) $\frac{K\pi}{a}$

(3) $\frac{K\pi}{2a}$

(4) 0

88. A disc of mass 100 gram slides down from rest on an inclined plane of 30° and comes to rest after travelling a distance of 1 m along the horizontal plane. If the coefficient of friction is 0.2 for both inclined and horizontal planes, then the work done by the frictional force over the whole journey, approximately, is

(1) 0.106 J

(2) 0.05 J

(3) 0.306 J

(4) 0.2 J

89. Two identical discs are moving with the same kinetic energy. One rolls and the other slides. The ratio of their speeds is

(1) 1 : 2

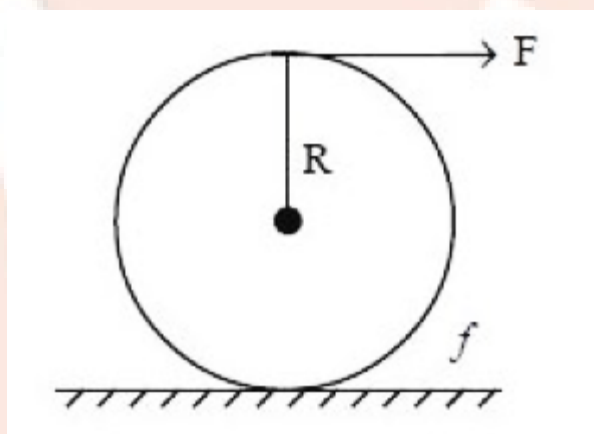
(2) 1 : 1

(3) 2 : 3

(4) $\sqrt{2} : \sqrt{3}$

90. A tangential force 'F' acts at the top of a thin spherical shell of mass 'm' and radius 'R'. The acceleration of the shell if it rolls without slipping is

(f = rolling friction)



(1) $\frac{5F}{6m}$

(2) $\frac{6F}{5m}$

(3) $\frac{3F}{2m}$

(4) $\frac{F}{6m}$

91. A simple pendulum is placed inside a lift, which is moving with a uniform acceleration. If the time periods of the pendulum while the lift is moving upwards and downwards are in the ratio 1 : 2, then the acceleration of the lift is

(Acceleration due to gravity = 10 ms^{-2})

(1) 6 ms^{-2}

(2) 0 ms^{-2}

(3) 3 ms^{-2}

(4) 2 ms^{-2}

92. Two bodies each of mass 'm' are hung from a balance whose scale pans differ in a vertical height by h. If the mean density of the earth is ρ , the error in weighing is

(1) $\frac{4\pi\rho Gmh}{3}$

(2) $\frac{3\pi\rho Gmh}{4}$

(3) $\frac{8\pi\rho Gmh}{3}$

(4) $\frac{3\pi\rho Gmh}{8}$

93. A one metre steel wire of negligible mass and area of cross-section 0.01 cm^2 is kept on a smooth horizontal table with one end fixed. A ball of mass 1 kg is attached to the other end. The ball and the wire are rotating with an angular velocity of ω . If the elongation of the wire is 2 mm , then ω is

(Young's modulus of steel = $2 \times 10^{11} \text{ Nm}^{-2}$)

- (1) 5 rad s^{-1}
- (2) 10 rad s^{-1}
- (3) 15 rad s^{-1}
- (4) 20 rad s^{-1}

94. A cylindrical tank has a hole of area 2 cm^2 at its bottom. If water is poured into the tank from a tube above it at the rate of $100 \text{ cm}^3 \text{ s}^{-1}$, then the maximum height upto which water can rise in the tank is

(Acceleration due to gravity = 10 ms^{-2})

- (1) 2.5 cm
- (2) 1.25 cm
- (3) 5.5 cm
- (4) 3.5 cm

95. The densities of wood and benzene at $0\text{ }^{\circ}\text{C}$ are 880 kg m^{-3} and 900 kg m^{-3} respectively. The coefficient of volume expansion is $1.2 \times 10^{-3}\text{ }^{\circ}\text{C}^{-1}$ for wood and $1.5 \times 10^{-3}\text{ }^{\circ}\text{C}^{-1}$ for benzene. Then the temperature at which a piece of wood just sinks in benzene is

- (1) $88\text{ }^{\circ}\text{C}$
- (2) $90\text{ }^{\circ}\text{C}$
- (3) $83.3\text{ }^{\circ}\text{C}$
- (4) $90.3\text{ }^{\circ}\text{C}$

96. A window used to thermally insulate a room from outside consists of two parallel glass sheets each of area 2.6 m^2 and thickness 1 cm separated by 5 cm thick stagnant air. In the steady state, the room-glass interface is at $18\text{ }^{\circ}\text{C}$ and the glass-outdoor interface is at $-2\text{ }^{\circ}\text{C}$. If the thermal conductivities of glass and air are respectively $0.8\text{ Wm}^{-1}\text{K}^{-1}$ and $0.08\text{ Wm}^{-1}\text{K}^{-1}$, the rate of flow of heat through the window is

- (1) 15 W
- (2) 40 W
- (3) 60 W

(4) 80 W

97. Five moles of Hydrogen gas initially at STP is compressed adiabatically so that its temperature becomes 673 K. The increase in internal energy of the gas is

($R = 8.3 \text{ Jmol}^{-1}\text{K}^{-1}$, $\gamma = 1.4$)

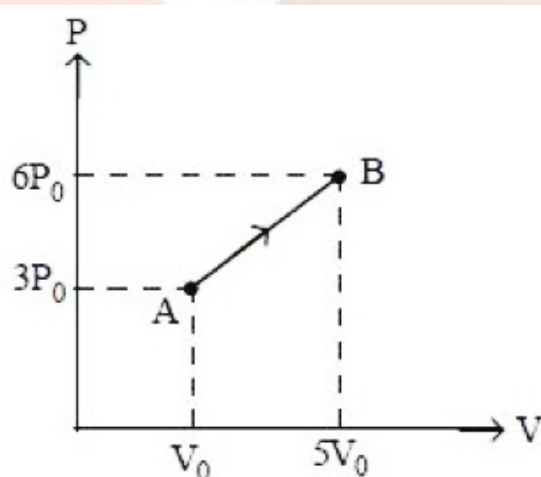
(1) 80.5 kJ

(2) 21.55 kJ

(3) 41.50 kJ

(4) 65.55 kJ

98. One mole of a monatomic ideal gas undergoes the process $A \rightarrow B$ in the given $P - V$ diagram. Specific heat capacity in the process is



(1) $\frac{13R}{3}$

(2) $\frac{13R}{6}$

(3) $\frac{7R}{3}$

(4) $\frac{2R}{3}$

99. If the speed of sound in a mixture of 2 moles of Helium and 2 moles of Hydrogen at temperature $\frac{972}{5}$ K is $n \times 100 \text{ ms}^{-1}$, then the value of n is

(Take $R = \frac{25}{3} \text{ Jmol}^{-1}\text{K}^{-1}$)

(1) 9

(2) 10

(3) 100

(4) 90

100. A siren placed at a railway platform is emitting a sound of frequency 5 kHz. A passenger sitting in a moving train 'A'

records the frequency of the siren as 5.5 kHz. During his return journey by train 'B' he records the frequency of the siren as 6 kHz. The ratio of the speed of train B to that of train A is

(1) $\frac{242}{252}$

(2) 2

(3) $\frac{5}{6}$

(4) $\frac{11}{6}$

101. The speed of a transverse wave travelling in a wire of length 50 cm, cross-sectional area 1 mm^2 and mass 5 g is 80 ms^{-1} . The Young's modulus of the material of the wire is $4 \times 10^{11} \text{ Nm}^{-2}$. The extension in the length of the wire is

(1) $8 \times 10^{-5} \text{ m}$

(2) $8 \times 10^{-4} \text{ m}$

(3) $16 \times 10^{-5} \text{ m}$

(4) $16 \times 10^{-4} \text{ m}$

102. An object is fixed at the bottom of a vessel and water is filled in the vessel upto a height of 10 cm. A plane mirror is placed at a height of 7 cm from the surface of water in such a way that its reflecting surface faces the water. The distance of the image from the mirror is

(Refractive index of water = 1.33)

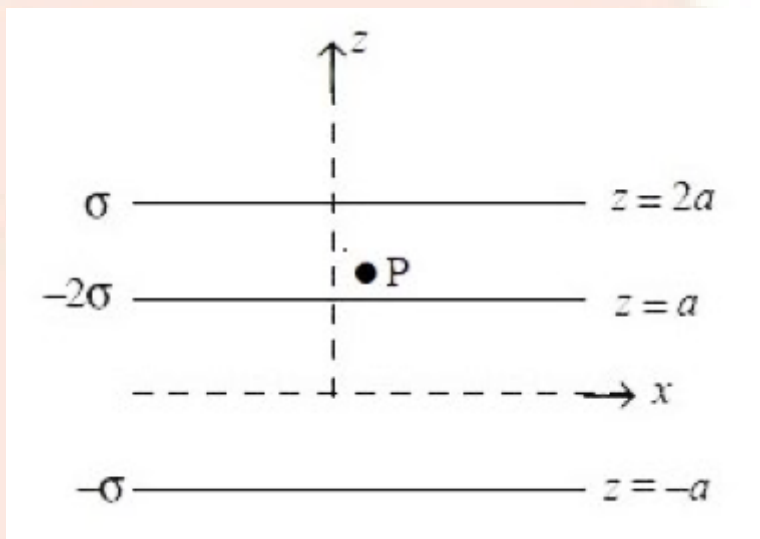
- (1) 7.5 cm
- (2) 7 cm
- (3) 14.5 cm
- (4) 21.8 cm

103. The angular deviation of 5th order dark fringe is 12° in a single slit experiment. If the width of the slit is $9 \mu\text{m}$ then the wavelength of the incident light is

- (1) $4862 \text{ \AA}$
- (2) $5892 \text{ \AA}$
- (3) $6022 \text{ \AA}$
- (4) $3768 \text{ \AA}$

104. Three infinitely long charged sheets are placed as shown in the figure. The electric force acting on a charge $-q$ placed at the point P is

(σ - surface charge density, ϵ_0 - permittivity of free space)



(1) $+\frac{2\sigma}{\epsilon_0}\hat{k}$

(2) $-\frac{2\sigma}{\epsilon_0}\hat{k}$

(3) $+\frac{4\sigma}{\epsilon_0}\hat{k}$

(4) $-\frac{4\sigma}{\epsilon_0}\hat{k}$

105. Assertion (A): Half of the charge of an electron does not exist.

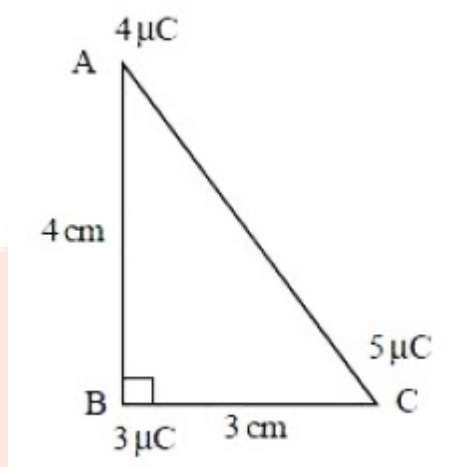
Reason (R): Electric charge is quantized.

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A).
- (2) Both (A) and (B) are correct but (R) is not the correct explanation of (A).
- (3) (A) is correct but (R) is not correct.
- (4) (A) is not correct but (R) is correct.

106. The potential difference between two points A (2, 1, 0) m and B (0, 2, 4) m in an electric field $(x\hat{i} - 2y\hat{j} + z\hat{k}) \text{ Vm}^{-1}$ is

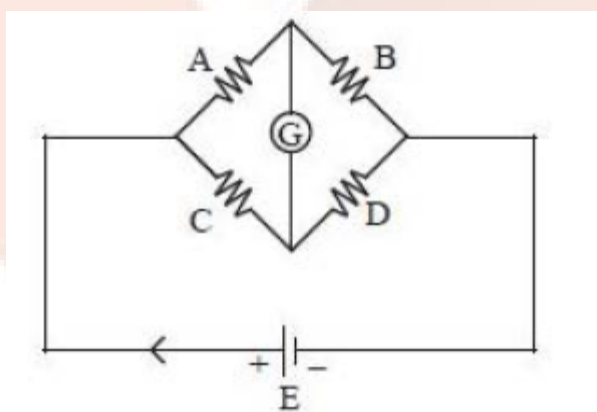
- (1) 2 V
- (2) 3 V
- (3) 1 V
- (4) 6 V

107. Three point charges of $3 \mu\text{C}$, $4 \mu\text{C}$ and $5 \mu\text{C}$ are arranged at the three corners of a right angled triangle ABC as shown in the figure. The work done in moving the charges at A and C so that the three charges are located at the three corners of an equilateral triangle of side 3 cm is



- (1) 0.3 J
- (2) 1.1 J
- (3) 2.2 J
- (4) 3.3 J

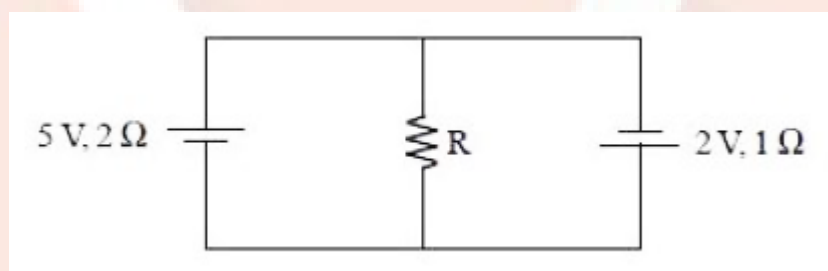
108. Four resistors A, B, C and D form a Wheatstone bridge as shown in the figure. The bridge is balanced when $C = 100\ \Omega$. The value of D is



- (1) $10\ \Omega$

- (2) $100\ \Omega$
- (3) $110\ \Omega$
- (4) $120\ \Omega$

109. In the circuit shown, if the current through the resistor R is $\frac{1}{5}\text{ A}$, the value of R is



- (1) $2\ \Omega$
- (2) $3\ \Omega$
- (3) $5\ \Omega$
- (4) $1\ \Omega$

110. An electron accelerated through a potential difference 'V', passes through a uniform transverse magnetic field and experiences a force 'F'. If the accelerating potential is increased to '2 V', the electron in the same magnetic field will experience a force

- (1) F
- (2) $\frac{F}{2}$
- (3) $\sqrt{2}F$
- (4) $2F$

111. A coil in the shape of an equilateral triangle of side 2 cm is suspended from a vertex such that it hangs in a vertical plane between the poles of a permanent magnet producing a horizontal magnetic field of 100×10^{-3} tesla. The magnetic field is parallel to the plane of the coil. For the moment of couple acting on the coil to be $2\sqrt{3} \times 10^{-5}$ Nm, the current to be passed through the coil is

- (1) 0.5 A
- (2) 1 A
- (3) 2 A
- (4) 4 A

112. A metal rod is subjected to cycles of magnetisation at the rate of 42 Hz. Density of the metal is $6 \times 10^3 \text{ kgm}^{-3}$ and its specific

heat capacity is $0.1 \times 10^{-3} \text{ cal kg}^{-1} \text{ } ^\circ\text{C}^{-1}$. If the area of its B–H loop corresponds to energy density of 10^{-2} J m^{-3} , then the rise in its temperature in one minute is

- (1) 5°C
- (2) 10°C
- (3) 15°C
- (4) 20°C

113. A coil is placed in a time varying magnetic field. The power dissipated due to current induced in the coil is P_1 . If the number of turns is doubled and radius of the wire is halved, the power dissipated is P_2 . Then $P_1 : P_2$ is

- (1) 1 : 4
- (2) 3 : 2
- (3) 2 : 1
- (4) 4 : 1

114. If the emf of an ac source is given by $6\sin \omega t + 4\sin 2\omega t$ volt then r.m.s. value of the emf is

- (1) $\sqrt{10} \text{ V}$

(2) $\sqrt{26}$ V

(3) $\sqrt{32}$ V

(4) $\sqrt{20}$ V

115. A lamp delivers a luminous flux of 100 W to an absorber of area 1 cm^2 . The force due to radiation pressure is

(1) 3.3×10^{-4} N

(2) 16.5×10^{-7} N

(3) 3.3×10^{-6} N

(4) 3.3×10^{-7} N

116. An electron of charge 'e' and mass 'm' moving with an initial velocity $v_0 \hat{i}$ is subjected to an electric field $E_0 \hat{j}$. The de-Broglie wavelength of the electron at a time 't' is

(Initial de-Broglie wavelength of the electron = λ_0)

(1) λ_0

(2) $\lambda_0 \sqrt{1 + \frac{e^2 E_0^2 t^2}{m^2 v_0^2}}$

$$(3) \frac{\lambda_0}{\sqrt{1 + \frac{e^2 E_0^2 t^2}{m^2 v_0^2}}}$$

$$(4) \frac{\lambda_0}{\left(1 + \frac{e^2 E_0^2 t^2}{m v_0^2}\right)}$$

117. Match the following List-I with List-II in connection with Bohr's atomic model

<u>List - I</u>		<u>List - II</u>
(A)	Speed of revolution of (I) electron	$\frac{1}{4\pi\epsilon_0} \frac{2\pi ze^2}{nh}$
(B)	Kinetic energy (II)	$-\left(\frac{1}{4\pi\epsilon_0}\right)^2 \frac{2\pi^2 me^4 z^2}{n^2 h^2}$
(C)	Total energy (III)	$\left(\frac{1}{4\pi\epsilon_0}\right)^2 \frac{2\pi^2 me^4 z^2}{n^2 h^2}$
(D)	Frequency (IV)	$\left(\frac{1}{4\pi\epsilon_0}\right)^2 \frac{4\pi^2 z^2 e^4 m}{n^3 h^3}$

The correct answer is

	A	B	C	D
(1)	I	III	II	IV

	A	B	C	D
(2)	II	IV	III	I

	A	B	C	D
(3)	III	I	IV	II

	A	B	C	D
(4)	III	I	II	IV

118. Half-life of a radioactive substance is 18 minutes. The time interval between its 20% decay and 80% decay in minutes is

- (1) 6
- (2) 9
- (3) 18
- (4) 36

119. In a transistor, the value of ' α ' varies between $\frac{20}{21}$ and $\frac{100}{101}$.

Then the value of ' β ' varies between

- (1) 1 and 10
- (2) 0.95 and 0.99
- (3) 20 and 100
- (4) 200 and 300

120. A TV tower has a height of 5 m in a region of population density $\frac{1000}{\pi}$ per square kilometre. Number of people that can receive the transmission is nearly (in thousands)

- (1) 128
- (2) 64
- (3) 256
- (4) 32

**AP EAMCET-2019-Engineering-20 Apr-Shift-2
Chemistry**

121. The energy associated with Bohr's orbit in the hydrogen atom

is given by the expression, $E_n = -\frac{13.6}{n^2} \text{ eV}$. The energy in eV

associated with the orbit having a radius $9r_1$ is (r_1 is the radius of the first orbit)

- (1) -13.6
- (2) -6.8

(3) -1.51

(4) -1.36

122. When a certain metal was irradiated with light of frequency $4.0 \times 10^{16} \text{ s}^{-1}$, the photoelectrons emitted had four times the kinetic energy as the kinetic energy of photoelectrons emitted when the same metal was irradiated with light of frequency $2.0 \times 10^{16} \text{ s}^{-1}$. The threshold frequency (ν_0) of the metal in s^{-1} is

(1) 2×10^{16}

(2) 4×10^{16}

(3) 2.5×10^{16}

(4) 1.33×10^{16}

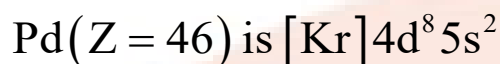
123. Which of the following statements are correct for classification of elements?

I) The properties of elements are periodic functions of their atomic numbers.

II) Non metallic elements are less in number than the metallic elements

III) The first ionization energies of elements along a period do not vary in a regular manner.

IV) The ground state electronic configuration of



- (1) I, II, III, IV
- (2) I, II, III only
- (3) II, III, IV only
- (4) I, II, IV only

124. The changes in bond length with respect to N-N and O-O, when N_2 becomes N_2^+ and O_2 becomes O_2^+ are respectively.

- (1) increases, decreases
- (2) decreases, increases
- (3) increases, increases
- (4) decreases, decreases

125. Arrange the following species in the increasing order of lone pair of electrons.

- (a) CO
- (b) NO_2^-

(c) NF_3

(d) CO_3^{-2}

(1) $(a) < (b) < (c) < (d)$

(2) $(b) < (c) < (a) < (d)$

(3) $(c) < (a) < (d) < (b)$

(4) $(a) < (b) < (d) < (c)$

126. The ratio between RMS velocities of H_2 at 50 K and O_2 at 800 K is

(1) 4 : 1

(2) 2 : 1

(3) 1 : 1

(4) 1 : 4

127. On reduction with hydrogen, 3.6 g of an oxide of metal (M) left 3.2 g of the metal. If the atomic weight of the metal is 64, the formula of the oxide is

(1) M_2O_3

(2) M_2O

(3) MO

(4) MO₂

128. At 300 K, the equilibrium constant for a reaction is 10. The standard free energy change (in kJ mol⁻¹) for the reaction is

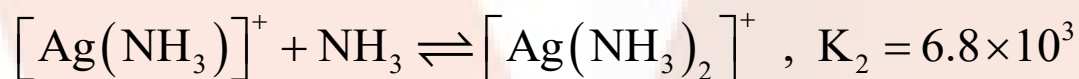
(1) -57.4

(2) -115.2

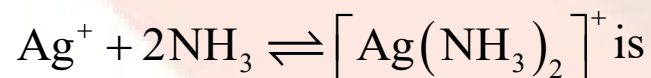
(3) +57.4

(4) -5.74

129. Observe the following equations



The equilibrium constant for the following reaction,



(1) 6.008×10^3

(2) 1.008×10^7

(3) 1.088×10^6

(4) 1.028×10^3

130. A buffer solution is prepared by mixing 10 mL of 1.0 M acetic acid and 20 mL of 0.5 M sodium acetate and then diluted to 100 mL with distilled water. The pH of the buffer solution is

(pK_a of acetic acid is 4.76)

- (1) 4.84
- (2) 5.21
- (3) 4.34
- (4) 4.76

131. Identify the correct statements from the following

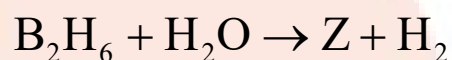
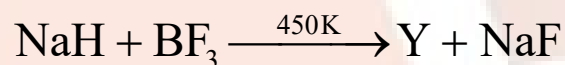
- I) B_2H_6 is an electron deficient hydride
- II) NH_3 is an electron rich hydride
- III) NaH is a covalent hydride
- IV) $YbH_{2.5}$ is an interstitial hydride

- (1) I, II, III
- (2) II, III, IV
- (3) I, II IV
- (4) I, III, IV

132. Compound A is prepared by the electrolysis of aqueous solution of B using Castner-Kellner cell with mercury as cathode and carbon as anode. A and B respectively are

- (1) NaOH, NaCl
- (2) NaCl, NaOH
- (3) NaHCO₃, NaOH
- (4) Na₂CO₃, NH₃

133. In the following reactions



X, Y and Z are respectively

- (1) B₂H₆, LiBH₂, H₃BO₃
- (2) B₃N₃H₆, B₂H₆, H₃BO₃
- (3) (BN)_n, LiBH₄, HBO₂
- (4) B₂H₆, B₂H₆, HBO₂

134. Consider the following statements

I) In diamond, each carbon atom is sp³ hybridized

II) Graphite has planar hexagonal layers of carbon atoms

III) Silicones being surrounded by non polar alkyl groups are water repelling in nature

IV) The order of catenation in group 14 elements is



(1) I, II, III

(2) II, III, IV

(3) I, II, IV

(4) I, III, IV

135. Which one of the following statements is **not** correct?

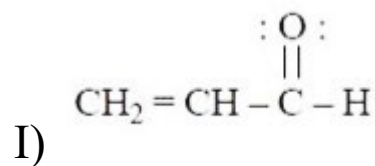
(1) NO_2 is a lung irritant

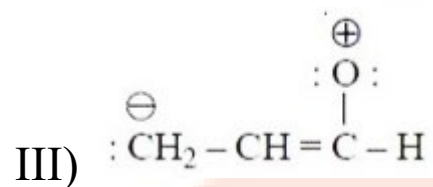
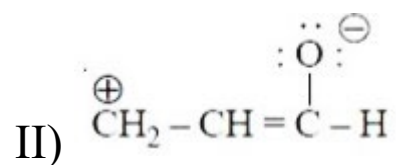
(2) The municipal sewage has BOD value of 100-4000 ppm

(3) Main source of CO is automobile exhaust fumes

(4) COD is the measure of bacteria in water.

136. The stability order of the following resonance structures is





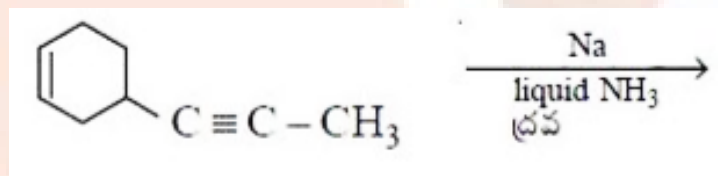
(1) $\text{III} < \text{II} < \text{I}$

(2) $\text{II} > \text{I} > \text{III}$

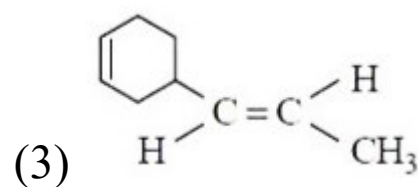
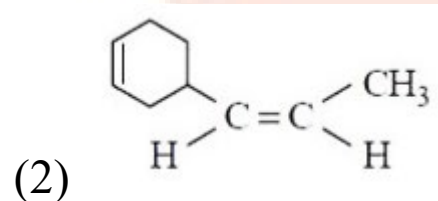
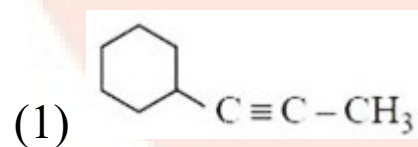
(3) $\text{II} < \text{I} < \text{III}$

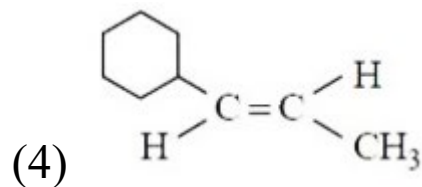
(4) $\text{II} > \text{III} > \text{I}$

137.



The product formed in the above reaction is





138. The correct order of reactivity of the following compounds, towards electrophilic substitution reactions is



- (1) (a) > (b) > (c) > (d)
- (2) (d) > (c) > (b) > (a)
- (3) (c) > (b) > (a) > (d)
- (4) (b) > (c) > (a) > (d)

139. Which one of the following statements is **not** correct?

- (1) Schottky defect in ionic solids does not change the density of the crystal
- (2) Packing efficiency is the percentage of total space filled by the particles

(3) In body centered cubic unit cell, the relationship between

atomic radius (r) and the edge length (a) is, $r = \frac{\sqrt{3}}{4}a$

(4) Photovoltaic cell is used for conversion of light energy into electrical energy.

140. 1.2 mL of acetic acid having density 1.06 gcm^{-3} is dissolved in 1 litre of water. The depression in freezing point observed for this concentration of acid was 0.041°C . The van't Hoff factor of the acid is (K_f of water = $1.86 \text{ K kg mol}^{-1}$)

(1) 0.41

(2) 1.04

(3) 0.96

(4) 1.54

141. 100 mL of 1.5% (w/v) solution of urea is found to have an osmotic pressure of 6.0 atm and 100 mL of 3.42% (w/v) solution of cane sugar is found to have an osmotic pressure of 2.4 atm. If the two solutions are mixed, the osmotic pressure of the resulting solution in atm is

(Assume that there is no reaction between urea and cane sugar)

- (1) 8.4
- (2) 16.8
- (3) 4.2
- (4) 2.1

142. The emf of the following cell

$\text{Mg}|\text{Mg}^{+2}(0.01\text{M})||\text{Sn}^{+2}(0.1\text{M})|\text{Sn}$ at 298 K in 'V' is

(Given, $E_{\text{Mg}^{+2}|\text{Mg}}^0 = -2.34 \text{ V}$, $E_{\text{Sn}^{+2}|\text{Sn}}^0 = -0.14 \text{ V}$)

- (1) 2.17
- (2) 2.23
- (3) 2.51
- (4) 2.45

143. The reaction $\text{X} \rightarrow \text{products}$ is a first order reaction. In 40 minutes, the concentration of X changes from 1.0 M to 0.25 M. What is the initial rate of reaction when $[\text{X}] = 0.1\text{M}$?

($\log 4 = 0.60$)

- (1) $1.73 \times 10^{-3} \text{ mol L}^{-1} \text{ min}^{-1}$

(2) $3.47 \times 10^{-4} \text{ mol L}^{-1} \text{ min}^{-1}$

(3) $1.73 \times 10^{-4} \text{ mol L}^{-1} \text{ min}^{-1}$

(4) $3.45 \times 10^{-3} \text{ mol L}^{-1} \text{ min}^{-1}$

144. Which of the following statements is **not** correct?

- (1) Both physical and chemical adsorptions are exothermic
- (2) Physical adsorption takes place with decrease of free energy whereas chemical adsorption occurs with increases of free energy.
- (3) Physical adsorption requires low activation energy but chemical adsorption requires high activation energy.
- (4) The magnitude of chemical adsorption increases and that of physical adsorption decreases with rise in temperature.

145. Which one of the following statements is **not** correct?

- (1) Van Arkel method is used for refining of Zirconium
- (2) Mond process is used for refining of Nickel
- (3) Zone refining is based on the principle that the impurities are more soluble in the melt than in the solid state of the metal

(4) High melting metals are refined by liquation

146. Assertion (A): P_4O_{10} cannot be used to remove moisture from ammonia gas

Reason (R): P_4O_{10} reacts with NH_3 gas

The correct answer is

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are correct and (R) is not the correct explanation of (A)
- (3) (A) is correct but (R) is not correct
- (4) (A) is not correct but (R) is correct

147. Which one of the following reactions does **not** occur?

- (1) $Cl_2 + 2Br^- \rightarrow Br_2 + 2Cl^-$
- (2) $ClF_3 + H_2O \rightarrow HCl + HOF + F_2$
- (3) Cold & Dil $2NaOH + Cl_2 \rightarrow NaCl + NaOCl + H_2O$
- (4) $Na_2SO_3 + 2HCl \rightarrow 2NaCl + SO_2 + H_2O$

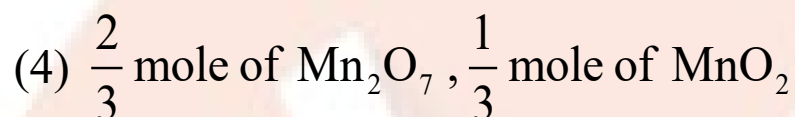
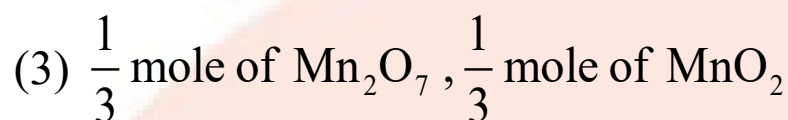
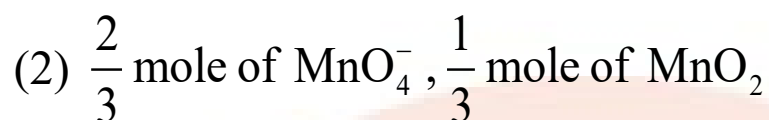
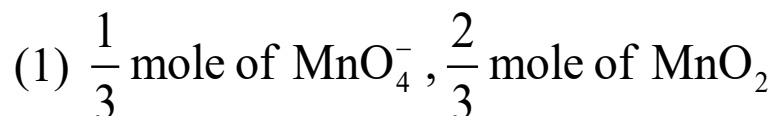
148. Which one of the following statements regarding helium is **not** correct?

- (1) It is used to fill gas balloons instead of hydrogen because it is lighter and not inflammable.
- (2) It is used in gas cooled nuclear reactors.
- (3) It is used to produce and sustain powerful super conducting magnets
- (4) It is not used as a cryogenic agent

149. Which one of the following coordination complexes exhibits the lowest value of magnetic moment (in B.M.)?

- (1) $[\text{Cr}(\text{CN})_6]^{-3}$
- (2) $[\text{Mn}(\text{CN})_6]^{-3}$
- (3) $[\text{Fe}(\text{CN})_6]^{-3}$
- (4) $[\text{Co}(\text{CN})_6]^{-3}$

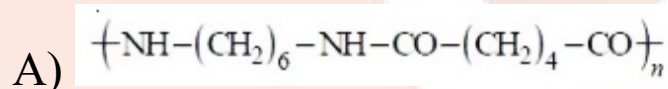
150. Disproportionation products of one mole of MnO_4^{2-} in aqueous acidic medium are



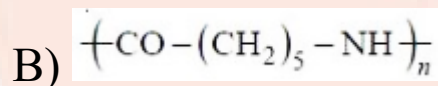
151. Match the following

List-I

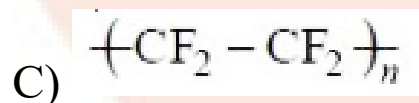
List-II



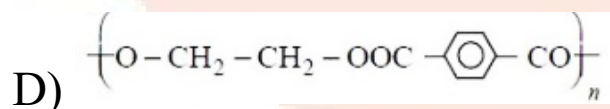
I) Ethylene glycol and Terephthalic acid



II) Phenol and formaldehyde



III) Caprolactam



IV) Hexamethylene diamine and adipic acid

V) Tetrafluoroethene

<u>A</u>	<u>B</u>	<u>C</u>	<u>D</u>
(1) IV	III	V	I

	<u>A</u>	<u>B</u>	<u>C</u>	<u>D</u>
(2)	III	II	IV	I

	<u>A</u>	<u>B</u>	<u>C</u>	<u>D</u>
(3)	IV	III	II	I

	<u>A</u>	<u>B</u>	<u>C</u>	<u>D</u>
(4)	II	IV	III	V

152. Which of the following statements about DNA is **not** correct?

- (1) It has a double helix structure
- (2) Adenine forms hydrogen bonds with thymine and cytosine forms hydrogen bonds with guanine
- (3) The two strands in a DNA molecule are not complementary to each other
- (4) It contains the pentose sugar, 2-deoxyribose

153. Which of the following is **not** an analgesic?

- (1) Ofloxacin
- (2) Paracetamol
- (3) Morphine
- (4) Codeine

154. Which one of the following statements are correct?

- a) The C-Cl bond in chlorobenzene is shorter than in chloromethane
- b) It is difficult to replace chlorine from chlorobenzene than from benzyl chloride
- c) The C-Cl bond in chlorobenzene has some double bond character.
- d) Chlorobenzene on chlorination gives m-dichlorobenzene

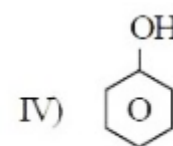
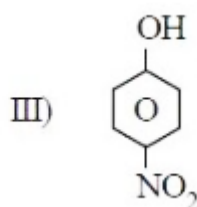
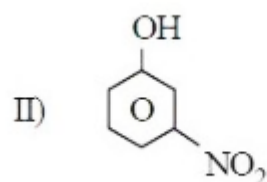
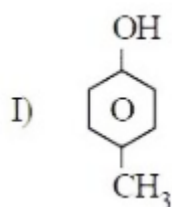
(1) a, b, c

(2) a, d only

(3) b, c, d

(4) c, d only

155. Arrange the following in increasing order of acidic character



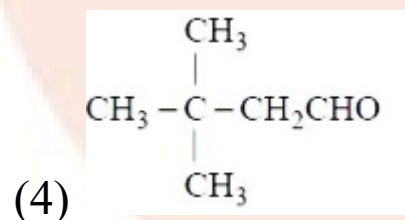
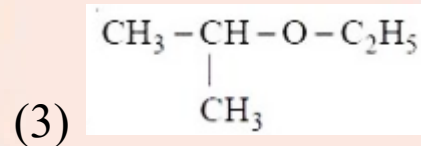
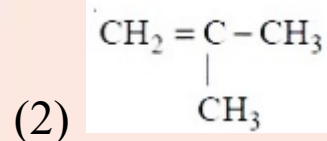
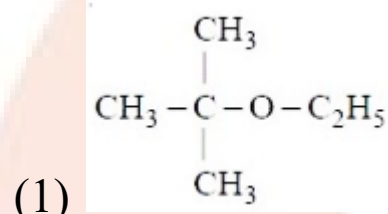
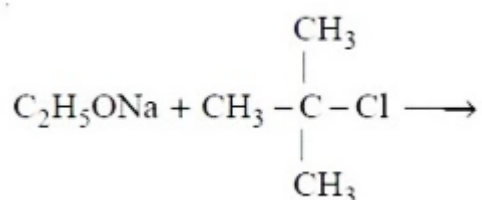
(1) I < II < III < IV

(2) I < IV < II < III

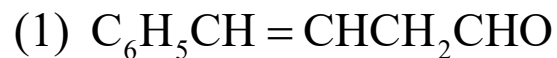
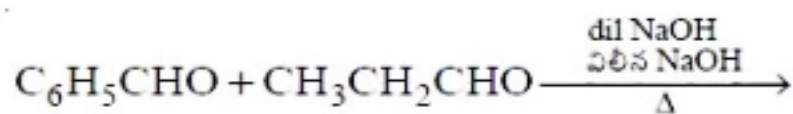
(3) IV < III < II < I

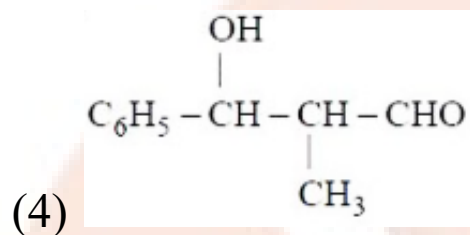
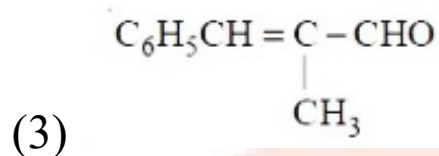
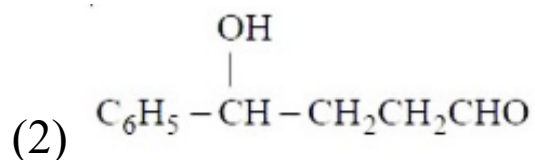
(4) $\text{II} < \text{III} < \text{IV} < \text{I}$

156. The major product obtained in the following reaction is



157. The product of the following reaction is,

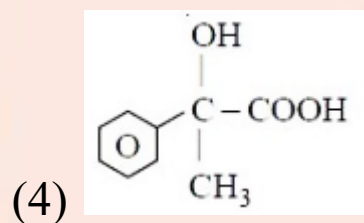
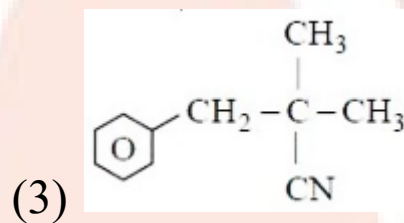
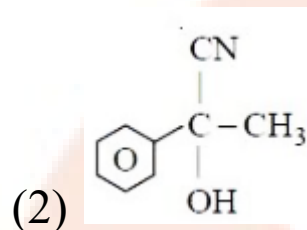
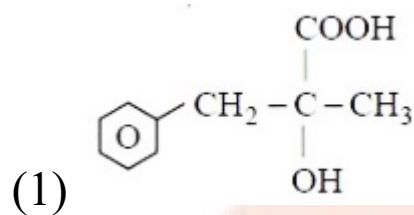
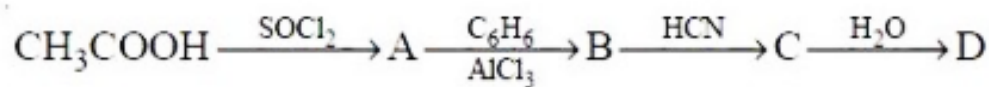




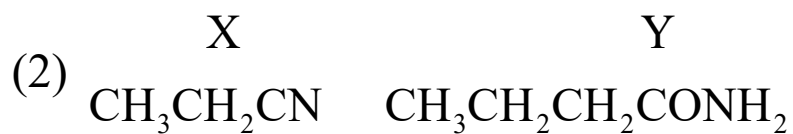
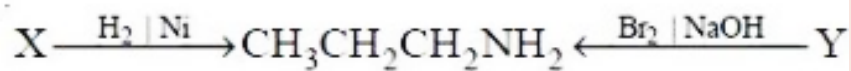
158. If general formula of oxime and semicarbazone is $\text{>C}=\text{N}-\text{Z}$, what is 'Z' in oxime (I) and semicarbazone (II)?

- | | |
|-----------------------|-------------------|
| I | II |
| (1) NHCONH_2 | OH |
| I | II |
| (2) OH | NH_2 |
| I | II |
| (3) OH | NHCONH_2 |
| I | II |
| (4) NH_2 | OH |

159. In the following reaction sequence, the product D is



160. Identify X and Y in the following reactions





AP EAMCET-2019-Engineering-20 Apr-Shift-2
Answer Key

1	3	33	3	65	3	97	3	129	2
2	2	34	4	66	4	98	2	130	4
3	1	35	1	67	3	99	1	131	3
4	3	36	2	68	1	100	2	132	1
5	3	37	2	69	4	101	1	133	2
6	1	38	1	70	2	102	3	134	1
7	2	39	2	71	1	103	4	135	4
8	2	40	1	72	1	104	2	136	1
9	1	41	3	73	3	105	1	137	3
10	4	42	2	74	1	106	2	138	3
11	4	43	2	75	3	107	4	139	1
12	3	44	1	76	3	108	3	140	2
13	1	45	3	77	4	109	4	141	3
14	4	46	4	78	2	110	3	142	2
15	4	47	1	79	4	111	2	143	4
16	2	48	1	80	4	112	2	144	2
17	3	49	1	81	4	113	1	145	4

18	2	50	4	82	2	114	2	146	1
19	3	51	3	83	4	115	4	147	2
20	4	52	1	84	1	116	3	148	4
21	1	53	3	85	4	117	1	149	4
22	4	54	1	86	3	118	4	150	2
23	2	55	1	87	4	119	3	151	1
24	1	56	4	88	3	120	2	152	3
25	3	57	2	89	4	121	3	153	1
26	2	58	3	90	2	122	4	154	1
27	2	59	2	91	1	123	2	155	2
28	2	60	4	92	3	124	1	156	2
29	1	61	3	93	4	125	4	157	3
30	1	62	3	94	2	126	3	158	3
31	4	63	3	95	3	127	2	159	4
32	2	64	1	96	4	128	4	160	2

AP EAMCET-2019-Engineering-20 Apr-Shift-2
Solutions – Mathematics

1 Consider the function.

$$f(x) = \frac{3x+5}{7x-3}$$

Let $y = f(x)$

So,

$$y = \frac{3x+5}{7x-3}$$

Hence,

$$x = \frac{3y+5}{7y-3}$$

As $f(x)$ is a bijective function and

$$f^{-1}(x) = f(x).$$

Thus,

$$(f \circ f)(x) = x$$

And,

$$(f \circ f \circ f \circ f)(x) = x$$

But,

$$(f \circ f \circ f) \neq x.$$

2. It is given that $f : A \rightarrow B$, $g : B \rightarrow C$ are the functions such that $g \circ f : A \rightarrow C$ is onto.

Then it is necessary that g is onto.

Thus, the correct option is (2).

3. Consider the given series,

$$\frac{3}{5} + \frac{21}{25} + \frac{117}{125} + \dots n$$

The above expression can be simplified as,

$$\begin{aligned} \frac{3}{5} + \frac{21}{25} + \frac{117}{125} + \dots n &= \left(1 - \frac{2}{5}\right) + \left(1 - \frac{4}{25}\right) + \left(1 - \frac{8}{125}\right) + \dots n \text{ terms} \\ &= n - \left[\frac{2}{5} + \left(\frac{2}{5}\right)^2 + \left(\frac{2}{5}\right)^3 + \dots n \text{ terms} \right] \\ &= n - \frac{\frac{2}{5} \left(1 - \left(\frac{2}{5}\right)^n\right)}{1 - \frac{2}{5}} \\ &= n + \frac{2^{n+1}}{3 \times 5^n} - \frac{2}{3} \end{aligned}$$

4. Consider the matrix,

$$\begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix} - x \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

It is solved as,

$$\begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix} - x \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{bmatrix} 2-x & 2 & 1 \\ 1 & 3-x & 1 \\ 1 & 2 & 2-x \end{bmatrix}$$

The matrix is singular. Therefore,

$$\begin{vmatrix} 2-x & 2 & 1 \\ 1 & 3-x & 1 \\ 1 & 2 & 2-x \end{vmatrix} = 0$$

Apply operation $C_1 \rightarrow C_1 + C_2 + C_3$

$$\begin{vmatrix} 5-x & 2 & 1 \\ 5-x & 3-x & 1 \\ 5-x & 2 & 2-x \end{vmatrix} = 0$$

$$(5-x) \begin{vmatrix} 1 & 2 & 1 \\ 1 & 3-x & 1 \\ 1 & 2 & 2-x \end{vmatrix} = 0$$

Apply operation $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$.

$$(5-x) \begin{vmatrix} 1 & 2 & 1 \\ 0 & 1-x & 0 \\ 0 & 0 & 1-x \end{vmatrix} = 0$$

$$(5-x)(1-x)^2 = 0$$

$$x = 1, 5, 5$$

Thus, the sum of the required value of x is,

$$\text{Sum} = 1 + 1 + 5$$

$$= 7$$

5. Consider the matrix.

$$A = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Therefore,

$$|A| = \cos^2 \alpha + \sin^2 \alpha$$

$$= 1$$

Now,

$$|\text{adj} A| = |A|^{3-1}$$

$$= 1^2$$

$$= 1$$

And,

$$\begin{aligned}\operatorname{adj}(\operatorname{adj} A) &= |A|^{3-2} \cdot A \\ &= 1 \cdot A \\ &= A\end{aligned}$$

Therefore,

$$\begin{aligned}(\operatorname{adj} A)^{-1} &= \frac{\operatorname{adj}(\operatorname{adj} A)}{|\operatorname{adj} A|} \\ &= \frac{A}{1} \\ &= A\end{aligned}$$

6. The given system of equation are,

$$x + 7ay + 2az = 0,$$

$$a + 6by + 2bz = 0$$

And,

$$x + 5cy + 2cz = 0$$

The system has non trivial solution. Therefore,

$$\begin{vmatrix} 1 & 7a & 2a \\ 1 & 6b & 2b \\ 1 & 5c & 2c \end{vmatrix} = 0$$

$$2bc - 4ac + 2ab =$$

$$2ac = bc + ab$$

$$b = \frac{2ac}{a+c}$$

Or,

$$\frac{2}{b} = \frac{1}{a} + \frac{1}{c}$$

Hence, a, b, c are in Harmonic progression.

7. It is given that,

$$|z - z_1| + |z - z_2| = k$$

And,

$$|z_1 - z_2| < k$$

Sum of the distance of the point z from z_1 and z_2 is constant (k) and more than the distance between points z_1 and z_2 . Thus, the locus of the point z is an ellipse.

8. It is given that,

$$\left\{ z \in \mathbb{C} : \arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4} \right\}$$

Let the complex number be $z = x + iy$

Thus,

$$\begin{aligned} \frac{z-1}{z+1} &= \frac{(x-1)+iy}{(x+1)+iy} \times \frac{(x+iy)-iy}{(x+iy)-iy} \\ &= \frac{x^2+y^2-1}{(x+1)^2+y^2} + i \left(\frac{2y}{(x+1)^2+y^2} \right) \end{aligned}$$

Therefore,

$$\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$$

$$\left(\frac{2y}{x^2+y^2-1} \right) = 1$$

$$x^2 + y^2 - 2y - 1 = 0$$

The complex plane represents a circle.

9. It is given that,

$$|z_1| = |z_2| = 1$$

And,

$$\operatorname{Re}(z_1 \bar{z}_2) = 0$$

Let,

$$\begin{aligned} z_1 &= a + ib \\ &= \cos \alpha + i \cos \beta \end{aligned}$$

And,

$$\begin{aligned} z_2 &= c + id \\ &= \cos \beta + i \sin \beta \end{aligned}$$

Thus,

$$\operatorname{Re}(z_1 \bar{z}_2) = 0$$

$$ac + bd = 0$$

$$\cos \alpha \cos \beta + \sin \alpha \sin \beta = 0$$

$$\cos(\alpha - \beta) = 0$$

Therefore,

$$\alpha - \beta = \frac{\pi}{2}, -\frac{\pi}{2}$$

The pair of complex number,

$$\begin{aligned} w_1 &= a + ic \\ &= \cos \alpha + i \cos \beta \end{aligned}$$

And,

$$\begin{aligned} w_2 &= b + id \\ &= \sin \alpha + i \sin \beta \end{aligned}$$

Hence,

$$\begin{aligned}\operatorname{Re}(w_1 \bar{w}_2) &= ab + cd \\ &= \cos \alpha \sin \alpha + \cos \beta \sin \beta \\ &= \frac{1}{2} 2 \sin(\alpha + \beta) \cos(\alpha - \beta) \\ &= 0\end{aligned}$$

This implies $\alpha - \beta = \frac{\pi}{2}, -\frac{\pi}{2}$.

10. Consider the expression.

$$(1+x)^n = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$

Substitute x for i in the above expression.

$$\begin{aligned}(1+i)^n &= \left[(a_0 - a_2 + a_4 - a_6 + \cdots) + \right. \\ &\quad \left. i(a_1 - a_3 + a_5 - a_7 + \cdots) \right] \\ \left[\sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \right]^n &= \left[(a_0 - a_2 + a_4 - a_6 + \cdots) + \right. \\ &\quad \left. i(a_1 - a_3 + a_5 - a_7 + \cdots) \right]\end{aligned}$$

Compare the real and imaginary parts.

$$2^{\frac{n}{2}} \cos \frac{n\pi}{4} = (a_0 - a_2 + a_4 - a_6 + \dots)$$

$$2^{\frac{n}{2}} \cos \frac{n\pi}{4} = k \cos \frac{n\pi}{4}$$

$$k = 2^{\frac{n}{2}}$$

11. Consider the equation.

$$\sqrt{x+2} - \sqrt{x-2} = \sqrt{4x-2}$$

Square both side,

$$x+2+x-2-2\sqrt{x^2-4} = 4x-2$$

$$1-x = \sqrt{x^2-4}$$

Now $x > 2$,

So,

$$1-x < 0$$

And,

$$\sqrt{x^2-4} > 0$$

Thus, there is no solution.

12. Consider the function.

$$y = \frac{x}{x^2 - 5x + 9}$$

The range of the function is determined as,

$$yx^2 - (5y + 1)x + 9y = 0.$$

The roots are real. Therefore,

$$(5y + 1)^2 - 36y^2 \geq 0$$

$$25y^2 + 10y + 1 - 36y^2 \geq 0$$

$$(11y + 1)(y - 1) \leq 0$$

$$y \in \left[-\frac{1}{11}, 1 \right]$$

13. Consider the cubic equations.

$$x^3 - ax^2 + bx - c = 0$$

The sum of the roots of the equation is,

$$\alpha + \beta + \gamma = a$$

And,

$$\alpha\beta + \beta\gamma + \gamma\alpha = b$$

The product of the roots is,

$$\alpha\beta\gamma = c$$

Therefore,

$$\begin{aligned}
\sum \alpha^2(\beta + \gamma) &= \sum \alpha^2(a - \alpha) \\
&= a \sum \alpha^2 - \sum \alpha^3 \\
&= \left[a \left[(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) \right] - \right. \\
&\quad \left. \left[(\alpha + \beta + \gamma)(\alpha^2 + \beta^2 + \gamma^2 - \alpha\beta - \beta\gamma - \gamma\alpha) \right] + \right. \\
&\quad \left. 3\alpha\beta\gamma \right] \\
&= a[a^2 - 2b] - [a(a^2 - 3b) + 3c]
\end{aligned}$$

Solve further,

$$\sum \alpha^2(\beta + \gamma) = ab - 3c$$

14. Consider the equation.

$$6x^6 - 25x^5 + 31x^4 - 31x^2 + 25x - 6 = 0$$

Simplify the equation as follows,

$$\begin{aligned}
6(x^6 - 1) - 25x(x^4 - 1) + 31x^2(x^2 - 1) &= 0 \\
(x^2 - 1)[6(x^4 + x^2 + 1) - 25(x^2 + 1) + 31x^2] &= 0 \\
(x^2 - 1)[6x^4 - 25x^3 + 37x^2 - 25x + 6] &= 0
\end{aligned}$$

So,

$$x = \pm 1$$

Or,

$$6x^4 - 25x^3 + 37x^2 - 25x + 6 = 0$$

$$6(x^4 + 1) - 25x(x^2 + 1) + 37x^2 = 0$$

$$6\left(x + \frac{1}{x}\right)^2 - 25\left(x + \frac{1}{x}\right) + 37 = 0$$

$$3\left(x + \frac{1}{x}\right)\left[2\left(x + \frac{1}{x}\right) - 5\right] - 5\left[2\left(x + \frac{1}{x}\right) - 5\right] = 0$$

Solve further,

$$\left[2\left(x + \frac{1}{x}\right) - 5\right]\left[3\left(x + \frac{1}{x}\right) - 5\right] = 0$$

Thus,

$$2\left(x + \frac{1}{x}\right) - 5 = 0$$

Or,

$$3\left(x + \frac{1}{x}\right) - 5$$

Therefore,

$$2x^2 - 5x + 2 = 0$$

$$2x(x - 2) - 1(x - 2) = 0$$

$$(x - 2)(2x - 1) = 0$$

$$x = 2, \frac{1}{2}$$

And,

$$3x^2 - 5x + 3 = 0$$

It has no solution because $D < 0$.

The sum of all rational roots is,

$$\begin{aligned}\text{Sum} &= 1 - 1 + 2 + \frac{1}{2} \\ &= 2.5\end{aligned}$$

15. A number is divisible by 6 if it is divisible by both 2 and 3.

Thus, the last digit can be filled by 0, 2, 4 only.

The first place can be filled by 5 choices except 0 and second and third by 6 choices.

The fourth place is filled such that it must be of the form $3k$ or $3k + 1$ or $3k + 2$. Thus it can be filled by only two choices 0 or 3.

Hence the total numbers are,

$$\begin{aligned}\text{Number} &= 5 \times 6 \times 6 \times 2 \times 3 \\ &= 1080\end{aligned}$$

16. The number of ways to select 5 person out of 8 is 8C_5 .

Thus, the required number of ways is calculated as,

Number of ways $= {}^8C_5 \times 4! \times 2!$

$$\begin{aligned}
 &= \frac{8 \times 7 \times 6}{3 \times 2} \times 4 \times 3 \times 2 \times 2 \\
 &= 42 \times 64 \\
 &= 2688
 \end{aligned}$$

17. It is given that 15^k divides $47!$.

Find the exponent of 5 in $47!$.

$$\left[\frac{47}{5} \right] + \left[\frac{47}{5^2} \right] + \dots, \{ \text{where } x \text{ is greatest integer of } x \}$$

$$\begin{aligned}
 \text{exponent 5 of in } 47! &= 9 + 1 \\
 &= 10
 \end{aligned}$$

Thus, the value of k is 10.

18. Consider the equation.

$$(1 + x + x^2)^n = c_0 + c_1x + c_2x^2 + \dots$$

Substitute $\frac{1}{x}$ for x in the above equation.

$$\frac{(1 - x + x^2)^n}{x^{2n}} = \frac{c_0x^{2n} - c_1x^{2n-1} + c_2x^{2n-2} + c_3x^{2n-3} + \dots}{x^{2n}}$$

$$(1 - x + x^2)^n = c_0x^{2n} - c_1x^{2n-1} + c_2x^{2n-3} + \dots$$

Now,

$$\begin{aligned}
 c_0 x^{2n} + c_1 x^{2n-1} + c_2 x^{2n-3} + \dots &= \text{Coeff of } x^{2n+1} \text{ in } (1+x+x^2)^n (1-x+x^2)^n \\
 &= \text{Coeff of } x^{2n+1} \text{ in } \left((1+x^2)^2 - x^2 \right)^n \\
 &= \text{Coeff of } x^{2n+1} \text{ in } (1+x^4+x^2)^n \\
 &= 0
 \end{aligned}$$

Since the coefficient of x^k in the expansion of $(1+x^4+x^2)^n$ exist if k is even other it will be zero and $2n+1$ is odd

19. If x is so small that x^2 and higher power of x can be neglected, then,

$$(1+x)^n = 1+nx$$

Thus,

$$\begin{aligned}
 \left(1 + \frac{3}{4}x\right)^{1/2} \left(1 - \frac{2}{3}x\right)^{-2} &= \left(1 + \frac{3}{4}x\right) \left(1 - \frac{2}{3}x\right) \\
 &= 1 + \frac{3}{8}x + \frac{4x}{3} \\
 &= \frac{24 + 41x}{24}
 \end{aligned}$$

20. Consider the equation.

$$\frac{x^4}{(x-a)(x-b)(x-c)} = P(x) + \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$$

The above equation is solved as,

$$x^4 = \left[P(x)(x-a)(x-b)(x-c) + A(x-b)(x-c) + B(x-a)(x-c) + C(x-a)(x-b) \right]$$

Substitute 0 for x in the above expression.

$$P(0)abc = [Abc + Bac + Cab]$$

$$P(0) = \frac{A}{a} + \frac{B}{b} + \frac{C}{c}$$

Now at $x = a$.

$$A = \frac{a^4}{(a-b)(a-c)}$$

At $x = b$

$$B = \frac{b^4}{(b-c)(b-a)}$$

At $x = c$

$$C = \frac{c^4}{(c-a)(c-b)}$$

Thus,

$$\begin{aligned}
 P(0) &= \frac{a^3}{(a-b)(a-c)} + \frac{b^3}{(b-c)(b-a)} + \frac{c^3}{(c-a)(c-a)} \\
 &= \frac{a^3(c-b) + bc(c^2 - b^2) + a(b^3 - c^3)}{(a-b)(b-c)(c-a)} \\
 &= \frac{a^3 + bc(c+b) - a(b^2 + c^2 + bc)}{(a-b)(a-c)} \\
 &= \frac{a(a^2 - b^2) - c^2(a-b) - bc(a-b)}{(a-b)(a-c)}
 \end{aligned}$$

Solve further,

$$\begin{aligned}
 P(0) &= \frac{(a^2 + ba) - c^2 - bc}{(a-c)} \\
 &= a + b + c
 \end{aligned}$$

Therefore,

$$P(0) + (a-b)(a-c)A = a + b + c + a^4$$

21. It is known that

$$\sin \alpha = p$$

$$\frac{2 \tan \frac{\alpha}{2}}{1 + \tan^2 \frac{\alpha}{2}} = p$$

The sum of the roots is,

$$\begin{aligned}\text{Sum} &= \tan \frac{\alpha}{2} + \cot \frac{\alpha}{2} \\ &= \frac{\tan^2 \frac{\alpha}{2} + 1}{\tan \frac{\alpha}{2}} \\ &= \frac{2}{p}\end{aligned}$$

The product of the roots is,

$$\begin{aligned}\text{Product} &= \tan \frac{\alpha}{2} \times \cot \frac{\alpha}{2} \\ &= 1\end{aligned}$$

Hence, the required quadratic equation is,

$$\begin{aligned}x^2 - \frac{2}{p}x + 1 &= 0 \\ px^2 - 2x + p &= 0\end{aligned}$$

22. Consider the expression.

$$\cos 66^\circ + \sin 84^\circ$$

It is solved as,

$$\begin{aligned}\cos 66^\circ + \sin 84^\circ &= 2\cos 36^\circ \cos 30^\circ \\ &= \sqrt{3}\cos 36^\circ \\ &= \sqrt{3}\left(\frac{\sqrt{5}+1}{4}\right)\end{aligned}$$

23. Consider the expression.

$$\tan \theta + \tan 2\theta + \sqrt{3} \tan \theta \tan 2\theta = \sqrt{3}$$

It can be simplified as,

$$\frac{\tan \theta + \tan 2\theta}{1 - \tan \theta \tan 2\theta} = \tan \frac{\pi}{3}$$

$$\tan(3\theta) = \tan \frac{\pi}{3}$$

$$3\theta = n\pi + \frac{\pi}{3}, n \in \mathbb{Z}$$

$$\theta = (3n+1)\frac{\pi}{9}, n \in \mathbb{Z}$$

24. Consider the equation.

$$\sin^{-1}\left(\frac{3x}{5}\right) + \sin^{-1}\left(\frac{4x}{5}\right) = \sin^{-1}(x)$$

The above equation can be rewritten as,

$$\sin^{-1}\left(\frac{3x}{5}\sqrt{1-\frac{16x^2}{25}} + \frac{4x}{5}\sqrt{1-\frac{9x^2}{25}}\right) = \sin^{-1}x$$

$$\frac{3x}{5}\sqrt{1-\frac{16x^2}{25}} + \frac{4x}{5}\sqrt{1-\frac{9x^2}{25}} = x$$

Thus,

$$x = 0$$

Or,

$$\frac{3\sqrt{25-16x^2}}{25} + \frac{4\sqrt{25-9x^2}}{25} = 1$$

$$9(25-16x^2) + 16(25-9x^2) + 24\sqrt{25-16x^2}\sqrt{25-9x^2} = 625$$

$$24\sqrt{25-16x^2}\sqrt{25-9x^2} = 625 - 625x^2$$

$$(25-16x^2)(25-9x^2) = 144x^4$$

Solve further,

$$625 - 625x^2 = 0$$

$$x = \pm 1$$

The sum of the values of x is,

$$\text{Sum} = 0 + 1 + -1$$

$$= 0$$

25. Consider the expression.

$$x = \log \left[\cot \left(\frac{\pi}{4} + \theta \right) \right]$$

$$e^x = \frac{\cot \pi/4 \cot \pi/6 - 1}{\cot \pi/4 + \cot \pi/6}$$

$$e^x = \frac{\sqrt{3} - 1}{\sqrt{3} + 1}$$

$$e^{-x} = \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$$

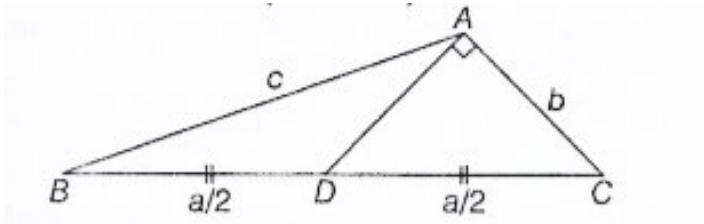
Now,

$$\begin{aligned} \sinh(x) &= \frac{e^x - e^{-x}}{2} \\ &= \frac{\frac{\sqrt{3} - 1}{\sqrt{3} + 1} - \frac{\sqrt{3} + 1}{\sqrt{3} - 1}}{2} \\ &= \frac{(3 + 1 - 2\sqrt{3}) - (3 + 1 + 2\sqrt{3})}{4} \\ &= -\frac{4\sqrt{3}}{4} \end{aligned}$$

Solve further,

$$\sinh(x) = -\sqrt{3}$$

26. Consider the figure shown below.



In $\triangle CAD$.

$$AD^2 = \frac{a^2}{4} - b^2$$

And from Apollonius theorem,

$$c^2 + b^2 = 2 \left(AD^2 + \frac{a^2}{4} \right)$$

Therefore,

$$c^2 + b^2 = 2 \left(\frac{a^2}{4} - b^2 + \frac{a^2}{4} \right)$$

$$c^2 + b^2 = a^2 - 2b^2$$

$$a^2 = 3b^2 + c^2$$

Thus,

$$\begin{aligned}
 3c \cos A \cos C + 2a^2 &= 3ca \left(\frac{b^2 + c^2 - a^2}{2bc} \right) \left(\frac{a^2 + b^2 - c^2}{2ab} \right) + 2a^2 \\
 &= \frac{3}{4b^2} \left[(b^2 + c^2 - 3b^2 - c^2)(3b^2 + c^2 + b^2 - c^2) \right] + 2a^2 \\
 &= \frac{3}{4b^2} (-2b^2)(4b^2) + 2a^2 \\
 &= -6b^2 + 2a^2
 \end{aligned}$$

Solve further,

$$\begin{aligned}
 3c \cos A \cos C + 2a^2 &= 2(a^2 - 3b^2) \\
 &= 2c^2
 \end{aligned}$$

27. It is known that,

$$\begin{aligned}
 \alpha^{-2} + \beta^{-2} + \gamma^{-2} &= \frac{a^2}{4\Delta^2} + \frac{b^2}{4\Delta^2} + \frac{c^2}{4\Delta^2} \\
 &= \frac{a^2 + b^2 + c^2}{4\Delta^2} \\
 &= \frac{(a^2 + b^2 - c^2) + (b^2 + c^2 - a^2) + (a^2 + c^2 - b^2)}{4 \left(\frac{abc}{4R} \right)^2} \\
 &= \frac{2ab \cos C + 2bc \cos A + 2ac \cos B}{\frac{a^2 b^2 c^2}{4R^2}}
 \end{aligned}$$

Solve further,

$$\alpha^{-2} + \beta^{-2} + \gamma^{-2} = \frac{\cos C}{\left(\frac{abc}{4R}\right) \frac{c}{2R}} + \frac{\cos A}{\left(\frac{abc}{4R}\right) \frac{a}{2R}} + \frac{\cos B}{\left(\frac{abc}{4R}\right) \frac{b}{2R}}$$

$$= \frac{1}{\Delta} (\cot A + \cot B + \cot C)$$

28. Consider the expression.

$$(r_2 + r_3) \cot\left(\frac{B+C}{2}\right)$$

The above expression is solved as,

$$(r_2 + r_3) \cot\left(\frac{B+C}{2}\right) = \left(\frac{\Delta}{s-b} + \frac{\Delta}{s-c}\right) \tan \frac{A}{2}$$

$$= \frac{\Delta(s-c+s-b)}{(s-c)(s-b)} \frac{\Delta}{s(s-a)}$$

$$= 2s - b - c$$

$$= a$$

29. The given vectors are,

$$\overline{AB} = \bar{i} + 3\bar{j} + 4\bar{k}$$

$$\overline{AC} = 5\bar{i} + \bar{j} + 2\bar{k}$$

Therefore,

$$\begin{aligned}\overline{AG} &= \frac{(\bar{i} + 3\bar{j} + 4\bar{k}) + (5\bar{i} + \bar{j} + 2\bar{k})}{3} \\ &= 2\bar{i} + \frac{4}{3}\bar{j} + 2\bar{k}\end{aligned}$$

Thus,

$$\begin{aligned}|\overline{AG}| &= \sqrt{4 + \frac{16}{9} + 4} \\ &= \sqrt{\frac{88}{9}} \\ &= \frac{2}{3}\sqrt{22}\end{aligned}$$

30. Let point P be the intersection of lines

$$\bar{r} = (\bar{i} + 2\bar{j} - \bar{k}) + \lambda(2\bar{i} + 3\bar{j} + 4\bar{k}) \quad \text{and}$$

$$\bar{r} = (-\bar{i} - 3\bar{j} + 7\bar{k}) + \mu(\bar{i} + 2\bar{j} - \bar{k})$$

Thus,

$$2\lambda + 1 = \mu - 1 \quad \dots\dots (I)$$

$$3\lambda + 2 = -3 + 2\mu \quad \dots\dots (II)$$

And,

$$4\lambda - 1 = 7 - \mu \quad \dots\dots (III)$$

On solving the above equations,

$$\lambda = 1$$

$$\mu = 4$$

Thus the required point of intersection is,

$$\bar{P} = (3\bar{i} + 5\bar{j} + 3\bar{k})$$

31. The position vector of A is $\bar{i} + \bar{j} + \bar{k}$, B is $4\bar{i} - \bar{j} + 2\bar{k}$, C is $5\bar{i} + \bar{j}$ and D is $7\bar{i} + 2\bar{j} + 3\bar{k}$.

Thus,

$$\overline{AB} = 3\bar{i} - 2\bar{j} + \bar{k}$$

And,

$$\overline{CD} = 2\bar{i} + \bar{j} + 3\bar{k}$$

Therefore, the projection of $\overline{AB}$ on $\overline{CD}$ is,

$$\begin{aligned}\text{Projection} &= \frac{|6 - 2 + 3|}{\sqrt{4 + 1 + 9}} \\ &= \frac{7}{\sqrt{14}} \\ &= \sqrt{\frac{7}{2}}\end{aligned}$$

32. It is given that,

$$\frac{1}{2}|\vec{a} - \vec{b}| = \sin \lambda \theta$$

Square both sides.

$$|\vec{a}|^2 + |\vec{b}|^2 - 2|\vec{a}||\vec{b}|\cos \theta = 4\sin^2(\lambda \theta)$$

$$1 + 1 - 2\cos \theta = 4\sin^2(\lambda \theta)$$

$$2(1 - \cos \theta) = 4\sin^2(\lambda \theta)$$

$$4\sin^2\left(\frac{\theta}{2}\right) = 4\sin^2(\lambda \theta)$$

Solve further,

$$\lambda = \frac{1}{2}$$

Therefore,

$$\begin{aligned} 4\lambda^2 &= 4\left(\frac{1}{2}\right)^2 \\ &= 1 \end{aligned}$$

33. Consider the give vectors.

$$\vec{a} = 2\vec{i} - 3\vec{j} + 4\vec{k}$$

$$\vec{b} = \vec{i} + 2\vec{j} - 2\vec{k}$$

And,

$$\vec{c} = 3\vec{i} - \vec{j} + \vec{k}$$

Thus,

$$\vec{a} + \vec{b} + \vec{c} = 6\vec{i} - 2\vec{j} + 3\vec{k}$$

$$\vec{a} - \vec{b} + \vec{c} = 4\vec{i} - 6\vec{j} + 7\vec{k}$$

And,

$$\vec{a} + \vec{b} - \vec{c} = \vec{k}$$

Hence, the required volume of the parallelepiped is calculated

as,

$$\begin{aligned} V &= \left| \begin{vmatrix} 6 & -2 & 3 \\ 4 & -6 & 7 \\ 0 & 0 & 1 \end{vmatrix} \right| \\ &= |1(-36 + 8)| \\ &= |-28| \\ &= 28 \end{aligned}$$

34. Consider the expression.

$$\vec{a} \times (\vec{a} \times \vec{c}) + \vec{b} = \vec{0}$$

It is solved as,

$$(\vec{a} \cdot \vec{c})\vec{a} - (\vec{a} \cdot \vec{a})\vec{c} = -\vec{b}$$

Let $\vec{a} \cdot \vec{c} = x$

Thus,

$$x\bar{a} - \bar{c} = -\bar{b}$$

Square both sides

$$x^2|\bar{a}|^2 + |\bar{c}|^2 - 1x|\bar{a} \cdot \bar{c}| = |\bar{b}|^2$$

$$x^2 + 4 - 2x^2 = 1$$

$$x^2 = 3$$

$$(\bar{a} \cdot \bar{c})^2 = 3$$

35. According to the given data,

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
2	-8	64
3	-7	49
5	-5	25
7	-3	9
11	1	1
13	3	9
17	7	49
22	12	144
$\sum x_i = 80$		$\sum (x_i - \bar{x})^2 = 350$

Thus,

$$\begin{aligned}\bar{x} &= \frac{80}{8} \\ &= 10\end{aligned}$$

Variance is calculated as,

$$\begin{aligned}\text{Variance} &= \frac{\sum (x_i - \bar{x})^2}{8} \\ &= \frac{350}{8} \\ &= 43.75\end{aligned}$$

36. The expression for the coefficient of variance is given by,

$$CV = \frac{\sigma}{\mu} \times 100$$

Therefore,

$$\begin{aligned}60 &= \frac{21}{\mu} \times 100 \\ \mu &= \frac{21 \times 10}{6} \\ &= 35\end{aligned}$$

The new mean on addition of 15 to every observation is,

$$\begin{aligned}\mu' &= 35 + 15 \\ &= 50\end{aligned}$$

But,

$$\sum |\mu' - x_i'| = \sum |\mu - x_i|$$

Now,

$$21 = \sqrt{\frac{\sum (\mu - x_i)^2}{n}}$$
$$21^2 = \frac{\sum (\mu - x_i)^2}{n}$$

$$\sum (\mu - x_i)^2 = 21^2 \times n$$

Thus,

$$\sigma' = \sqrt{\frac{\sum (\mu' - x_i')^2}{n}}$$
$$= \sqrt{\frac{21^2 \times n}{n}}$$
$$= 21$$

Therefore, the new coefficient of variance is,

$$CV_{\text{New}} = \frac{21}{50} \times 100$$
$$= 42$$

37. The required probability is calculated as,

$$\begin{aligned}
 P &= \frac{9+8+7+6+5+4+3+2+1}{10 \times 10} \\
 &= \frac{45}{100} \\
 &= \frac{9}{20}
 \end{aligned}$$

38. Let E_1 be the event of selecting word LONDON and E_2 be the event of selecting the word CLIFTON.

Thus, the required probability is calculated as,

$$\begin{aligned}
 P\left(\frac{E_1}{A}\right) &= \frac{P(E_1) \cdot P\left(\frac{A}{E_1}\right)}{P(E_1) \cdot P\left(\frac{A}{E_1}\right) + P(E_2) \cdot P\left(\frac{A}{E_2}\right)} \\
 &= \frac{\frac{1}{2} \cdot \frac{2}{5}}{\left(\frac{1}{2} \cdot \frac{2}{5}\right) + \left(\frac{1}{2} \cdot \frac{1}{6}\right)} \\
 &= \frac{2 \times 6}{12 + 5} \\
 &= \frac{12}{17}
 \end{aligned}$$

39. The given probability of the function is,

$$P(X = k) = Ck^2$$

Therefore,

$$\sigma^2 = E(X^2) - \mu^2$$

$$\sigma^2 + \mu^2 = E(X^2)$$

$$\begin{aligned}\sigma^2 + \mu^2 &= \sum (x_i^2) P(x_i) \\ &= 0(c(0)^2) + 1(c(1)^2) + 4(c(2)^2) + 9(c(3)^2) + 16(c(4)^2) \\ &= c + 16c + 81c + 256c\end{aligned}$$

Now,

$$\begin{aligned}\sum P(x_i) &= 1 \\ c[0^2 + 1^2 + 2^2 + 3^2 + 4^2] &= 1 \\ c &= \frac{1}{30}\end{aligned}$$

Solve further,

$$\begin{aligned}\sigma^2 + \mu^2 &= 354c \\ &= \frac{354}{30} \\ &= 11.8\end{aligned}$$

40. According to the given data,

$$\lambda = \frac{200}{250}$$
$$= \frac{4}{5}$$

Now,

$$P(X = x) = \frac{e^{-\lambda} (\lambda^x)}{x!}$$

$$P(X = 0) = e^{-\frac{4}{5}}$$

Therefore, the required probability is calculated as,

$$\begin{aligned} (P(X = 0))^5 &= \left(e^{-\frac{4}{5}} \right)^5 \\ &= e^{-4} \end{aligned}$$

41. Consider the given equation.

$$\sqrt{(x-2)^2 + y^2} + \sqrt{(x+2)^2 + y^2} = 4$$

Sum of the distance of point $P(x, y)$ with respect to points $A(2, 0)$ and $B(-2, 0)$ is equal to the distance between the points $A(2, 0)$ and $B(-2, 0)$.

$$AP + BP = AB$$

Therefore, the locus of point $P(x, y)$ is a line segment.

42. Consider the given equation.

$$2x^2 + 2y^2 - 8x - 12y + 18 = 0$$

It can be simplified as,

$$2(x-2)^2 + 2(y-3)^2 = 8$$

$$(x-2)^2 + (y-3)^2 = 4$$

After shifting the origin to $(2,3)$, the transformed equation becomes $x^2 + y^2 = 4$ and after rotation of axes through an angle of 45° about the point, the transformed equation is $x^2 + y^2 = 4$.

43. Let a line cuts the coordinate axis at A and B. So,

$$\frac{x}{a} + \frac{y}{b} = 1$$

Now, $A(a,0)$, $B(0,b)$ and the line passes through the point $P(4,2)$.

Thus,

$$\frac{4}{a} + \frac{2}{b} = 1$$

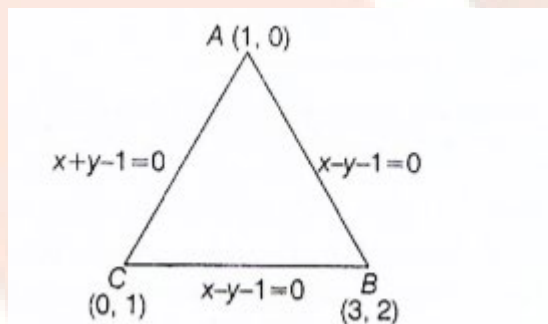
The coordinate of centre of the circumcentre of ΔOAB is midpoint of hypotenuse of the right angled ΔOAB and its midpoint is AB. So, the circumcentre is $\left(\frac{a}{2}, \frac{b}{2}\right)$.

Therefore,

$$\frac{2}{x} + \frac{1}{y} = 1$$

$$2x^{-1} + y^{-1} = 1$$

44. Consider the figure shown below.



The points of intersection of the lines are $A(1, 0)$, $B(3, 2)$ and $C(0, 1)$.

The centroid of ΔABC is calculated as,

$$\text{Centroid} = \left(\frac{1+3+0}{3}, \frac{0+2+1}{3} \right)$$

$$= \left(\frac{4}{3}, 1 \right)$$

45. The equation of perpendicular bisector of the line joining the points $A(-2, -1)$ and $B(6, -1)$ is,

$$x = 2$$

The equation of perpendicular bisector of the line joining the points $B(6, -1)$ and $C(2, 5)$ is,

$$y - 2 = \frac{4}{6}(x - 4)$$

$$2x - 3y = 2$$

The point of intersection of perpendicular bisector of the circumcentre of $\triangle ABC$.

$$C\left(2, \frac{2}{3}\right)$$

Thus, the required quadratic equation is,

$$x^2 - \left(2 + \frac{2}{3}\right)x + 2\left(\frac{2}{3}\right) = 0$$

$$3x^2 - 8x + 4 = 0$$

46. The point of intersection of the lines $x + y + 1 = 0$ and

$$2x - 3y + 4 = 0 \text{ is } A\left(-\frac{7}{5}, \frac{2}{5}\right).$$

If the coordinate of the image of point $P(-2,0)$ with respect to line $x + y + 1 = 0$ is (h,k) , then

$$\frac{h+2}{1} = \frac{k-0}{1} = 2 \times \frac{-2+1}{2} = 1$$

So,

$$h = -1$$

$$k = 1$$

Thus,

$$(h,k) = (-1,1)$$

The equation of required bisector of angle is line joining the points $A\left(\frac{-7}{5}, \frac{2}{5}\right)$ and $B(-1,1)$ is

$$y - 1 = \frac{\frac{3}{5}}{\frac{5}{2}}(x + 1)$$

$$2y - 2 = 3x + 3$$

$$3x - 2y + 5 = 0$$

47. The given equation of lines are,

$$3x^2 + 2hxy - 3y^2 = 0$$

And,

$$3x^2 + 2hxy - 3y^2 + 2x - 4y + c = 0$$

The point of intersection of the lines given by

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \text{ is } \left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2} \right)$$

Thus, the point of intersection of the second line is,

$$A(x, y) = A\left(\frac{2h - 3}{9 + h^2}, -\frac{h + 6}{9 + h^2}\right)$$

The equation of diagonal of square passes through origin is,

$$y = \frac{h + 6}{3 - 2h}x.$$

The equation of diagonal of square that does passes through the origin is

$$2x - 4y + c = 0$$

The diagonal are perpendicular if,

$$\frac{1}{2} \left(\frac{h + 6}{3 - 2h} \right) = -1$$

$$h + 6 = 4h - 6$$

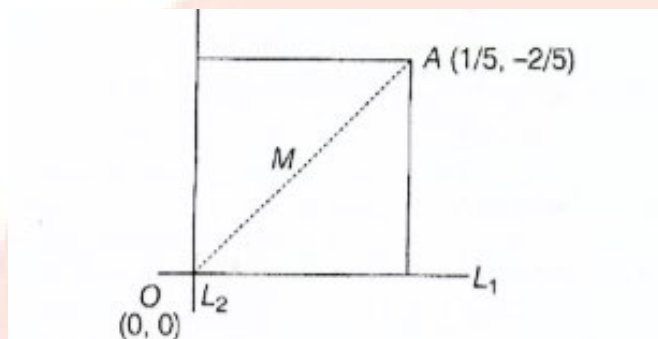
$$3h = 12$$

$$h = 4$$

Thus,

$$\begin{aligned}\text{Point } A &= \left(\frac{8-3}{9+16}, -\frac{10}{9+16} \right) \\ &= \left(\frac{1}{5}, -\frac{2}{5} \right)\end{aligned}$$

Consider the figure shown below.



The midpoint of line OA is,

$$M = \left(\frac{1}{10}, -\frac{1}{5} \right)$$

Therefore,

$$\frac{2}{10} + \frac{4}{5} + c = 0$$

$$c = -1$$

Therefore, $(h, c) = (4, -1)$.

48. Consider the equation of curve,

$$x^2 + xy + y^2 + x + 3y + 1 = 0$$

The equation of line joining the origin to the point of intersection of curve and the line is calculated as,

$$x^2 + xy + y^2 + (x + 3y)\left(\frac{x + y}{-2}\right) + \left(\frac{x + y}{-2}\right)^2 = 0$$

$$x^2 + y^2 + xy - \frac{1}{2}(x^2 + 4xy + 3y^2) + \frac{1}{4}(x^2 + 2xy + y^2) = 0$$

$$3x^2 - 2xy - y^2 = 0$$

The equation of the bisector of the angle between the pair of straight line is,

$$\frac{x^2 - y^2}{3 + 1} = \frac{xy}{-1}$$

$$x^2 + 4xy - y^2 = 0$$

49. Consider the equation of circle.

$$x^2 + y^2 - 2x - 4y - 20 = 0$$

The equation of tangent at the point B(1,7) on the circle is,

$$x + 7y - (x + 1) - 2(y + 7) - 20 = 0$$

$$5y = 35$$

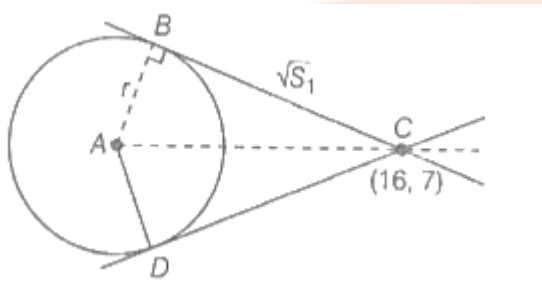
$$y = 7$$

The equation of tangent at the point D(4,-2) on the circle is,

$$4x - 2y - (x + 4) - 2(y - 2) - 20 = 0$$

$$3x - 4y = 20$$

Consider the figure shown below.



The point of intersection of tangents is $C(16, 7)$.

Thus, the area of quadrilateral ABCD is,

$$\begin{aligned} \text{Area} &= 2 \times \frac{1}{2} r \sqrt{S_1} \\ &= 2 \times \frac{1}{2} \times (5) \sqrt{256 + 49 - 32 - 28 - 20} \\ &= 5 \times \sqrt{225} \\ &= 75 \end{aligned}$$

50. The all conjugate of lines $5x + 7y - 78 = 0$ with respect to the circle $x^2 + y^2 + 6x + 8y - 96 = 0$ passes through the pole of the given line with respect to given circle.

Let required pole be $P(x_1, y_1)$ with respect to the circle $T = 0$.

$$xx_1 + yy_1 + 3(x + x_1) + 4(y + y_1) - 96 = 0$$
$$(x_1 + 3)x + (y_1 + 4)y + (3x_1 + 4y_1 - 96) = 0$$

The above line represent the line $5x + 7y - 78 = 0$ only.

Hence,

$$\frac{x_1 + 3}{5} = \frac{y_1 + 4}{7} = \frac{3x_1 + 4y_1 - 96}{-78} = k$$

Therefore,

$$x_1 = 5k - 3$$

$$y_1 = 7k - 4$$

And,

$$3x_1 + 4y_1 = 96 - 78k$$

Thus,

$$3(5k - 3) + 4(7k - 4) = 96 - 78k$$

$$121k = 121$$

$$k = 1$$

So,

$$x_1 = 2$$

$$y_1 = 3$$

Therefore, the required point of concurrence of all conjugate lines of the given line with respect to the given circle is $(2, 3)$.

51. Consider the given equation of circles.

$$S_1 : x^2 + y^2 + 4x - 6y - 12 = 0$$

Its centre is $(-2, 3)$ and radius is $r_1 = 5$

And,

$$S_2 : x^2 + y^2 - 8x + 10y + 5 = 0$$

Its centre is $(4, -5)$ and radius is $r_2 = 6$.

So,

$$\begin{aligned} r_1 + r_2 &= 5 + 6 \\ &= 11 \end{aligned}$$

And,

$$\begin{aligned} C_1C_2 &= \sqrt{(4+2)^2 + (-5-3)^2} \\ &= \sqrt{36 + 64} \\ &= \sqrt{100} \\ &= 10 \end{aligned}$$

Since,

$$r_1 + r_2 > C_1C_2$$

Therefore, there are only two common tangents to circle S_1 and S_2 .

52. Consider the given circles.

$$S_1 : (x-1)^2 + (y-3)^2 = r^2$$

And,

$$S_2 : (x-4)^2 + (y+1)^2 = 3^2$$

If the circle intersect at two distinct points, then

$$|r_1 - r_2| < C_1 C_2 < (r_1 + r_2)$$

$$|r - 3| < \sqrt{(4-1)^2 + (-1-3)^2} < (r + 3)$$

$$|r - 3| < 5 < (r + 3)$$

So,

$$r + 3 > 5$$

$$r > 2$$

And,

$$|r - 3| < 5$$

$$r - 3 \in (-5, 5)$$

$$r \in (-2, 8)$$

Thus, $2 < r < 8$.

53. Consider the circles.

$$S_1 = x^2 + y^2 + 6x - 2y + k = 0$$

And,

$$S_2 = x^2 + y^2 + 2x - 6y - 15 = 0$$

The circle S_1 bisects the circumference of the circle S_2 , then the common circles S_1 and S_2 will pass through the circle $S_2(-1,3)$.

The equation of common chord of the circle S_1 and S_2 is

$$4x + 4y + (k + 15) = 0$$

Since the chord passes through point $(-1,3)$. Thus,

$$-4 + 12 + k + 15 = 0$$

$$k = -23$$

54. Consider the given circles.

$$S_1 : x^2 + y^2 + 4y - 7 = 0$$

$$S_2 : 2x^2 + 2y^2 + 3x + 5y - 9 = 0$$

And,

$$S_3 : x^2 + y^2 + y = 0$$

The radical axis of circle S_1 and S_2 is $x - y - 1 = 0$.

The radical axis of circle S_2 and S_3 is $x + y - 3 = 0$.

Thus, the coordinate of radical centre is $(2,1)$ and the radius is,

$$r = \sqrt{4 + 1 + 8 - 7}$$

$$= \sqrt{6}$$

Hence, the equation of the required circle is,

$$(x - 2)^2 + (y - 1)^2 = 6$$

$$x^2 + y^2 - 4x - 2y - 1 = 0$$

55. Consider the given equation of parabola.

$$2y^2 + 5x - 6y + 1 = 0$$

It can be simplified as,

$$2(y^2 - 3y) + 5x + 1 = 0$$

$$2\left(y^2 - 3y + \left(\frac{3}{2}\right)^2 - \left(\frac{3}{2}\right)^2\right) + 5x + 1 = 0$$

$$2\left(y - \left(\frac{3}{2}\right)\right)^2 - \frac{9}{2} + 5x + 1 = 0$$

$$\left(y - \frac{3}{2}\right)^2 = 4\left(-\frac{5}{8}\right)\left(x - \frac{7}{10}\right)$$

The coordinate of the vertex of parabola is $\left(\frac{7}{10}, \frac{3}{2}\right)$ and the

coordinate of the focus of the parabola is $\left(\frac{3}{40}, \frac{3}{2}\right)$

56. Consider the equation of parabola

$$y^2 = 9x$$

The equation of normal chord at a point t on the parabola is

$$tx + y = 2\left(\frac{9}{4}\right)t + \frac{9}{4}t^3$$

The chord subtends right angle at vertex of parabola $V(0,0)$.

Thus,

$$y^2 - 9x \left(\frac{tx + y}{\frac{9}{2}t + \frac{9}{4}t^3} \right) = 0$$

The sum of the coefficient of x^2 and y^2 terms should be zero.

Hence,

$$1 - \frac{9t}{\frac{9}{2}t + \frac{9}{4}t^3} = 0$$

$$1 - \frac{1}{\frac{1}{2} + \frac{t^2}{4}} = 0$$

$$\frac{1}{2} + \frac{t^2}{4} = 1$$

$$t^2 = 2$$

Therefore,

$$t = \pm\sqrt{2}$$

57. Let the equation of ellipse be,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

The length of the latus rectum is,

$$\frac{2b^2}{a} = 4$$

The distance between the foci is,

$$2ae = 4\sqrt{2}$$

$$a\sqrt{1 - \frac{b^2}{a^2}} = 2\sqrt{2}$$

$$a^2 - b^2 = 8$$

$$a^2 - 2a - 8 = 0$$

Solve further,

$$(a - 4)(a + 2) = 0$$

$$a = 4, -2$$

Since, $a > 0$, $a = 4$.

Hence,

$$b^2 = 8$$

Therefore, the required ellipse is,

$$\frac{x^2}{16} + \frac{y^2}{8} = 1$$
$$x^2 + 2y^2 = 16$$

58. Consider the equation of ellipse

$$\frac{x^2}{4} + \frac{y^2}{1} = 1$$

So,

$$c^2 = 4(4) + 1$$

$$c^2 = 16 + 1$$

$$c^2 = 17$$

$$c = \pm\sqrt{17}$$

Now consider the equation.

$$x^3 - x^2 - 17x + 17 = 0$$

$$x^2(x-1) - 17(x-1) = 0$$

$$(x-1)(x^2 - 17) = 0$$

$$x = \pm\sqrt{17}, 1$$

Thus, option (3) is correct.

59. Consider the hyperbola.

$$9x^2 - 16y^2 = 144$$

Its eccentricity is e_1

And,

$$9x^2 - 16y^2 = -144$$

Its eccentricity is e_2 and is a conjugate hyperbola.

Therefore,

$$\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1$$

$$\frac{e_1^2 + e_2^2}{e_1^2 e_2^2} = 1$$

$$\frac{e_1^2 e_2^2}{e_1^2 + e_2^2} = 1$$

60. Consider the points.

$$A = (2, 4, -1)$$

$$B = (3, 6, -1)$$

And,

$$C = (4, 5, 1)$$

Let $D = (x, y, z)$

The midpoint of AC is equal to the midpoint of BD .

$$\left(\frac{2+4}{2}, \frac{4+5}{2}, \frac{-1+1}{2} \right) = \left(\frac{3+x}{2}, \frac{6+y}{2}, \frac{-1+z}{2} \right)$$
$$\left(3, \frac{9}{2}, 0 \right) = \left(\frac{3+x}{2}, \frac{6+y}{2}, \frac{-1+z}{2} \right)$$

Therefore,

$$x = 3$$

$$y = 3$$

$$z = 1$$

Hence, the fourth vertex is $D(3, 3, 1)$.

61. Consider the points.

$$A = (2, 3, -1)$$

$$B = (3, 5, -3)$$

The DR's of AB are,

$$\begin{aligned} \text{DR's } AB &= (3-2, 5-3, -3+1) \\ &= (1, 2, -2) \end{aligned}$$

And,

$$C = (1, 2, 3)$$

$$D = (3, y, 7)$$

$$\begin{aligned}\text{DR's } CD &= (3-1, y-2, 7-3) \\ &= (2, y-2, 4)\end{aligned}$$

It is given that $AB \perp CD$

Hence,

$$\begin{aligned}a_1a_2 + b_1b_2 + c_1c_2 &= 0 \\ (1)(2) + 2(y-2) + (-2)(4) &= 0 \\ 2y - 10 &= 0 \\ y &= 5\end{aligned}$$

62. It is given that

$$\text{X-intercept}(a) = 2$$

$$\text{Y-intercept}(b) = 3$$

$$\text{Z-intercept}(c) = 4$$

Therefore, the equation of the plane is,

$$\frac{x}{2} + \frac{y}{3} + \frac{z}{4} = 1$$

$$6x + 4y + 3z = 12$$

The given points are $A(-1, 6, 2)$, $B(1, 2, 3)$ and $C(-2, 3, 4)$.

Thus,

$$\begin{aligned}\text{DR's of } BC &= (-2-1, 3-2, 4-3) \\ &= (-3, 1, 1)\end{aligned}$$

Therefore, the equation of plane passing through $A(-1, 6, 2)$ and

DR's $(-3, 1, 1)$ is,

$$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$$

$$-3(x + 1) + 1(y - 6) + 1(z - 2) = 0$$

$$-3x + y + z - 11 = 0$$

$$-3x + y + z = -11$$

Hence, the angle between the planes is,

$$\begin{aligned}\cos \theta &= \frac{|(6)(-3) + (4)(1) + (3)(1)|}{\sqrt{36 + 16 + 9} \sqrt{9 + 1 + 1}} \\ &= \frac{|-18 + 4 + 3|}{\sqrt{61} \sqrt{11}} \\ &= \frac{\sqrt{11}^2}{\sqrt{61} \sqrt{11}} \\ &= \sqrt{\frac{11}{61}}\end{aligned}$$

Solve further,

$$\theta = \cos^{-1} \left(\sqrt{\frac{11}{61}} \right)$$

63. Consider the equation.

$$\alpha = \lim_{x \rightarrow 0} \frac{x \cdot 2^x - x}{1 - \cos x}$$

And,

$$\beta = \lim_{x \rightarrow 0} \frac{x \cdot 2^x - x}{\sqrt{1+x^2} - \sqrt{1-x^2}}$$

Simplify the first equation as,

$$\begin{aligned} \alpha &= \lim_{x \rightarrow 0} \frac{x \cdot 2^x - x}{1 - \cos x} \\ &= \lim_{x \rightarrow 0} \frac{x(2^x - 1)}{2 \sin^2 \frac{x}{2}} \\ &= \lim_{x \rightarrow 0} \frac{(2^x - 1)}{\frac{2 \sin^2 \frac{x}{2}}{x}} \\ &= \frac{1}{2} \lim_{x \rightarrow 0} \frac{\left(\frac{2^x - 1}{x} \right)}{\frac{\sin^2 \frac{x}{2}}{x^2}} \end{aligned}$$

Solve further,

$$\alpha = \frac{1}{2} \lim_{x \rightarrow 0} \frac{\left(\frac{2^x - 1}{x} \right)}{\left(\frac{\sin \frac{1}{2} x}{\frac{x}{2}} \right)^2}$$

$$= \frac{1}{2} \left[\lim_{x \rightarrow 0} \frac{2^x - 1}{x} \times \frac{1}{\lim_{x \rightarrow 0} \left(\frac{\sin \frac{1}{2} x}{\frac{x}{2}} \right)^2 \times \frac{1}{4}} \right]$$

$$= \frac{1}{2} \times \log 2 \times \frac{1}{1 \times \frac{1}{4}}$$

$$= 2 \log 2$$

Also,

$$\begin{aligned}\beta &= \lim_{x \rightarrow 0} \frac{x \cdot 2^x - x}{\sqrt{1+x^2} - \sqrt{1-x^2}} \times \frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} + \sqrt{1-x^2}} \\&= \lim_{x \rightarrow 0} \frac{x(2^x - 1)(\sqrt{1+x^2} + \sqrt{1-x^2})}{1+x^2 - (1-x^2)} \\&= \lim_{x \rightarrow 0} \frac{x(2^x - 1)(\sqrt{1+x^2} + \sqrt{1-x^2})}{2x^2} \\&= \frac{1}{2} \lim_{x \rightarrow 0} \frac{(2^x - 1)(\sqrt{1+x^2} + \sqrt{1-x^2})}{x}\end{aligned}$$

Solve further,

$$\beta = \frac{1}{2} \times \log 2 \times 2$$

$$\beta = \log 2$$

$$2\beta = 2 \log 2$$

$$2\beta = \alpha$$

64. Consider the piecewise function,

$$f(x) = \begin{cases} \frac{x - |x|}{x} & , \text{ when } x < 0 \\ 5x^2 + a & , \text{ when } 0 \leq x \leq 1 \\ b \left(\frac{x^2 - 1}{x^2 - 3x + 2} \right) & , \text{ when } 1 < x < 3 \\ -14 & , \text{ when } x \geq 3 \end{cases}$$

The function is continuous at $x = 0$ and $x = 3$.

LHL at $x = 0$ is equal to RHL at $x = 0$.

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x)$$

$$\lim_{x \rightarrow 0^-} \frac{x - |x|}{x} = \lim_{x \rightarrow 0^+} 5x^2 + a$$

$$\lim_{x \rightarrow 0^-} \frac{x - (-x)}{x} = \lim_{x \rightarrow 0^+} 5x^2 + a$$

$$\lim_{x \rightarrow 0^-} \frac{2x}{x} = \lim_{x \rightarrow 0^+} 5x^2 + a$$

Solve further,

$$2 = a$$

$$a = 2$$

The function is continuous at $x = 3$.

LHL at $x = 3$ is equal to RHL at $x = 3$.

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x)$$

$$\lim_{x \rightarrow 3^-} b \left(\frac{x^2 - 1}{x^2 - 3x + 2} \right) = \lim_{x \rightarrow 3^+} -14$$

$$\lim_{x \rightarrow 3^-} b \left(\frac{(x+1)(x-1)}{(x-1)(x-2)} \right) = \lim_{x \rightarrow 3^+} -14$$

$$\lim_{x \rightarrow 3^-} b \left(\frac{(x+1)}{(x-2)} \right) = \lim_{x \rightarrow 3^+} -14$$

Solve further,

$$b \frac{(3+1)}{(3-2)} = -14$$

$$4b = -14$$

$$b = \frac{-14}{4}$$

$$b = -\frac{7}{2}$$

Therefore,

$$(a, b) = \left(2, -\frac{7}{2} \right)$$

65. Consider the given equation.

$$f(x) = |x - 0.5| + |x - 1| + \tan x$$

Clearly $f(x)$ is not differentiable at $x = 0.5$, 1 and $\frac{\pi}{2}$ for $x \in (0, 2)$.

Therefore, the number of non differentiable points of $f(x)$ are 3.

66. It is given that,

$$x = \sec \theta - \cos \theta$$

$$y = \sec^{10} \theta - \cos^{10} \theta$$

And,

$$(x^2 + 4) \left(\frac{dy}{dx} \right)^2 = k(y^2 + 4)$$

Therefore,

$$\frac{dy}{d\theta} = 10(\sec^{10} \theta + \cos^{10} \theta) \tan \theta$$

And,

$$\frac{dx}{d\theta} = (\sec \theta + \cos \theta) \tan \theta$$

Hence,

$$\begin{aligned}\frac{dy}{dx} &= 10 \frac{\sec^{10} \theta + \cos^{10} \theta}{\sec \theta + \cos \theta} \\ \left(\frac{dy}{dx}\right)^2 &= 100 \left(\frac{\sec^{10} \theta + \cos^{10} \theta}{\sec \theta + \cos \theta}\right)^2 \\ &= 100 \left(\frac{\sec^{20} \theta + \cos^{20} \theta + 2}{\sec^2 \theta + \cos^2 \theta + 2}\right) \\ &= 100 \left(\frac{(\sec^{10} \theta - \cos^{10} \theta)^2 + 4}{(\sec \theta - \cos \theta)^2 + 4}\right)\end{aligned}$$

Solve further,

$$\begin{aligned}\left(\frac{dy}{dx}\right)^2 &= 100 \left(\frac{y^2 + 4}{x^2 + 4}\right) \\ (x^2 + 4) \left(\frac{dy}{dx}\right)^2 &= 100(y^2 + 4) \\ &= k(y^2 + 4)\end{aligned}$$

Therefore,

$$k = 100$$

67. Consider the equation.

$$y = e^{\sin^{-1} x}$$

Differentiate with respect to x .

$$y_1 = e^{\sin^{-1}x} \cdot \frac{1}{\sqrt{1-x^2}}$$

$$y_1 \sqrt{1-x^2} = e^{\sin^{-1}x}$$

$$y_1 \sqrt{1-x^2} = y$$

Differentiate with respect to x

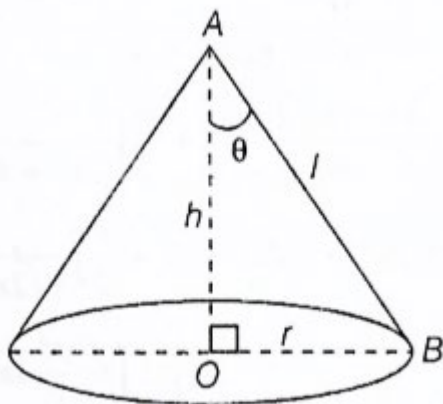
$$y_2 \cdot \sqrt{1-x^2} + y_1 \frac{1}{2\sqrt{1-x^2}}(-2x) = y_1$$

$$\frac{y_2(1-x^2) - xy_1}{\sqrt{1-x^2}} = y_1$$

$$y_2(1-x^2) - xy_1 = y_1 \sqrt{1-x^2}$$

$$y_2(1-x^2) - xy_1 = y$$

68. Consider the given figure.



In $\triangle AOB$

$$\tan 45^\circ = \frac{r}{h}$$

$$1 = \frac{r}{h}$$

$$r = h$$

Also,

$$l^2 = r^2 + h^2$$

$$l^2 = 2h^2$$

$$l = \sqrt{2}h$$

Therefore, the lateral surface area,

$$S = \pi r l$$

$$= \pi h \sqrt{2}h$$

$$= \sqrt{2}\pi (20.025)^2$$

$$= 401\sqrt{2}\pi$$

69. The rate of change of radius is,

$$\frac{dr}{dt} = 0.04 \text{ cm/sec}$$

The volume of the sphere is given by,

$$V = \frac{4}{3}\pi r^3$$

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$$

The surface area of the sphere is given as,

$$S = 4\pi r^2$$

$$\frac{dS}{dt} = 8\pi r \frac{dr}{dt}$$

Therefore,

$$\frac{\frac{dV}{dt}}{\frac{dS}{dt}} = \frac{4\pi r^2 \frac{dr}{dt}}{8\pi r \frac{dr}{dt}}$$

$$\begin{aligned} \frac{dV}{dS} &= \frac{r}{2} \\ &= \frac{10}{2} \\ &= 5 \text{ cm} \end{aligned}$$

70. Consider the function

$$f(x) = a \cdot \log|x| + bx^2 + x$$

Differentiate with respect to x

$$\begin{aligned} f'(x) &= a \frac{1}{|x|} \cdot \frac{x}{|x|} + 2bx + 1 \\ &= \frac{ax}{|x|^2} + 2bx + 1 \end{aligned}$$

It is given that $x = -1$ is one of the extremity of $f(x)$

$$f'(-1) = 0$$

$$\frac{a(-1)}{1} + 2b(-1) + 1 = 0$$

$$-a - 2b + 1 = 0$$

Also,

$$f'(-2) = 0$$

$$a\frac{(2)}{4} + 2b(2) + 1 = 0$$

$$a + 8b + 2 = 0$$

On solving both the equations,

$$a = 2$$

$$b = -\frac{1}{2}$$

Therefore,

$$(a, b) = \left(2, -\frac{1}{2}\right)$$

71. Consider the function.

$$f(x) = x^3 + px^2 + qx$$

And,

$$f(0) = f(2)$$

$$0 = 2^3 + 4p + 2q$$

$$2p + q + 4 = 0$$

Now,

$$f'(x) = 3x^2 + 2px + q$$

And,

$$f'\left(1 + \frac{1}{\sqrt{3}}\right) = 0$$

$$3\left(1 + \frac{1}{\sqrt{3}}\right)^2 + 2p\left(1 + \frac{1}{\sqrt{3}}\right) + q = 0$$

$$2p\left(1 + \frac{1}{\sqrt{3}}\right) + q + 4 + \frac{6}{\sqrt{3}} = 0$$

On solving both the equations,

$$p = -3$$

$$q = 2$$

Therefore,

$$\begin{aligned} p^2 + q^2 &= (-3)^2 + 2^2 \\ &= 9 + 4 \\ &= 13 \end{aligned}$$

72. Consider the integral

$$I = \int \frac{x^2}{2x^2 + 6\sqrt{2}x + 9} dx$$

It is solved as,

$$\begin{aligned} I &= \int \frac{\frac{1}{2}(2x^2 + 6\sqrt{2}x + 9) - \left(3\sqrt{2}x + \frac{9}{2}\right)}{2x^2 + 6\sqrt{2}x + 9} dx \\ &= \int \frac{1}{2} dx - \int \frac{\left(3\sqrt{2}x + \frac{9}{2}\right)}{2x^2 + 6\sqrt{2}x + 9} dx \\ &= \frac{x}{2} - \int \frac{\left(3\sqrt{2}x + \frac{9}{2}\right)}{2x^2 + 6\sqrt{2}x + 9} dx \end{aligned}$$

Now,

$$3\sqrt{2}x + \frac{9}{2} = A(4x + 6\sqrt{2}) + B$$

Compare the coefficients of like terms.

$$4A = 3\sqrt{2}$$

$$A = \frac{3\sqrt{2}}{4}$$

And,

$$6\sqrt{2}A + B = \frac{9}{2}$$

$$B = -\frac{9}{2}$$

Therefore,

$$3\sqrt{2}x + \frac{9}{2} = 3\frac{\sqrt{2}}{4}(4x + 6\sqrt{2}) + \frac{9}{4}$$

Hence,

$$\begin{aligned} I &= \frac{x}{2} - \int \frac{3\frac{\sqrt{2}}{4}(4x + 6\sqrt{2}) + \frac{9}{4}}{2x^2 + 6\sqrt{2}x + 9} dx \\ &= \frac{x}{2} - 3\frac{\sqrt{2}}{4} \int \frac{(4x + 6\sqrt{2})}{2x^2 + 6\sqrt{2}x + 9} dx + \frac{9}{4} \int \frac{1}{2x^2 + 6\sqrt{2}x + 9} dx \\ &= \frac{x}{2} - \frac{3\sqrt{2}}{4} \log|2x^2 + 6\sqrt{2}x + 9| + \frac{9}{2} \int \frac{1}{(\sqrt{2}x + 3)^2} dx \\ &= \frac{x}{2} - \frac{3\sqrt{2}}{4} \log|(\sqrt{2}x + 3)^2| + \frac{9}{2} \frac{(\sqrt{2}x + 3)^{-2+1}}{-2+1} \cdot \frac{1}{\sqrt{2}} + c' \end{aligned}$$

Solve further,

$$\begin{aligned}
 I &= \frac{x}{2} - \frac{3\sqrt{2}}{2} \log \left| \left(\sqrt{2x+3} \right) \right| + \frac{9}{2\sqrt{2}} \frac{\left(\sqrt{2x+3} \right)^{-1}}{-1} + c' \\
 &= \frac{1}{2\sqrt{2}} \left[\left(\sqrt{2x+3} \right) - 6 \log \left| \sqrt{2x+3} \right| - \frac{9}{\sqrt{2x+3}} \right] + c
 \end{aligned}$$

73. Consider the given integral.

$$I = \int \frac{\sqrt{1-x^2} \sin^{-1} x + x}{\sqrt{1-x^2}} dx$$

It is solved as,

$$\begin{aligned}
 I &= \int \sin^{-1} x dx + \int \frac{x}{\sqrt{1-x^2}} dx \\
 &= \sin^{-1} x \cdot x - \int \frac{1}{\sqrt{1-x^2}} \cdot x dx + \int \frac{x}{\sqrt{1-x^2}} dx + c \\
 &= x \sin^{-1} x + c
 \end{aligned}$$

74. Consider the equation.

$$\int x^3 e^{2x} dx = \frac{e^{2x}}{8} f(x) + c$$

Simplify the above equation as follows

$$x^3 \cdot \frac{e^{2x}}{2} - \int 3x^2 \cdot \frac{e^{2x}}{2} dx = \frac{e^{2x}}{8} f(x) + c$$

$$\frac{1}{2} x^3 e^{2x} - \frac{3}{2} \left[x^2 \frac{e^{2x}}{2} - \int 2x \cdot \frac{e^{2x}}{2} dx \right] = \frac{e^{2x}}{8} f(x) + c$$

$$\frac{1}{2} x^3 e^{2x} - \frac{3}{4} x^2 e^{2x} + \frac{3}{2} \int x \cdot e^{2x} dx = \frac{e^{2x}}{8} f(x) + c$$

$$\frac{1}{2} x^3 e^{2x} - \frac{3}{4} x^2 e^{2x} + \frac{3}{2} \left[x \frac{e^{2x}}{2} - \int 1 \cdot \frac{e^{2x}}{2} dx \right] = \frac{e^{2x}}{8} f(x) + c$$

Simplify further,

$$\frac{e^{2x}}{8} [4x^3 - 6x^2 + 6x - 3] + c_1 = \frac{e^{2x}}{8} f(x) + c$$

Therefore, on comparison

$$f(x) = 4x^3 - 6x^2 + 6x - 3$$

$$1 = 4x^3 - 6x^2 + 6x - 3$$

$$4x^3 - 6x^2 + 6x - 4 = 0$$

$$2(x^3 - 1) - 3x(x - 1) = 0$$

Solve further,

$$2(x - 1)(x^2 + x + 1) - 3x(x - 1) = 0$$

$$(x - 1)(2x^2 + 2x + 2 - 3x) = 0$$

$$(x - 1)(2x^2 - x + 2) = 0$$

$x = 1$ is a real root.

The sum of non-real complex roots from the above equation is $\frac{1}{2}$

75. Consider the integral.

$$\begin{aligned} I &= \int \sin^5 x \cos^5 x dx \\ &= \int \sin^4 x \sin x \cos^5 x dx \\ &= \int (1 - \cos^2 x)^2 \cos^5 x dx \end{aligned}$$

Let,

$$\cos x = t$$

$$-\sin x dx = dt$$

Therefore,

$$\begin{aligned} I &= \int (1 - t)^2 t^5 (-dt) \\ &= - \left[\frac{t^{10}}{10} - 2 \frac{t^8}{8} + \frac{t^6}{6} \right] + c \\ &= - \frac{t^6}{60} [6t^4 - 15t^2 + 10] + c \\ &= - \frac{\cos^6 x}{60} [6\cos^4 x - 15\cos^2 x + 10] + c \end{aligned}$$

Solve further,

$$\begin{aligned}
 I &= -\frac{\cos^6 x}{60} \left[6\sin^4 x - 12\sin^2 x + 6 - 15 + 15\sin^2 x + 10 \right] + c \\
 &= -\frac{\cos^6 x}{60} \left[6\sin^4 x - 3\sin^2 x + 1 \right] + c
 \end{aligned}$$

76. Consider the limit.

$$y = \lim_{x \rightarrow \infty} \left[\frac{1}{n} + \frac{n^2}{(n+1)^3} + \frac{n^2}{(n+2)^3} + \frac{n^2}{(n+3)^3} + \dots + \frac{1}{125n} \right]$$

It is solved as,

$$\begin{aligned}
 y &= \lim_{x \rightarrow \infty} \left[\frac{n^2}{(n+0)^3} + \frac{n^2}{(n+1)^3} + \frac{n^2}{(n+2)^3} + \dots + \frac{n^2}{(n+4n)^3} \right] \\
 &= \lim_{x \rightarrow \infty} \sum_{r=0}^{4n} \frac{n^2}{(n+r)^3}
 \end{aligned}$$

Let,

$$\frac{r}{n} = x$$

$$\frac{1}{n} = dx$$

Therefore,

$$\begin{aligned}y &= \int_0^4 \frac{dx}{(1+x)^3} \\&= \left[\frac{(1+x)^{-2}}{-2} \right]_0^4 \\&= -\frac{1}{2} \left[\frac{1}{5^2} - 1 \right] \\&= \frac{12}{25}\end{aligned}$$

77. Consider the integral.

$$I = \int_{e^{-1}}^{e^2} \left| \frac{\log x}{x} \right| dx$$

It can be simplified as,

$$I = \int_{e^{-1}}^1 \frac{\log x}{x} dx + \int_1^{e^2} \frac{\log x}{x} dx$$

Let,

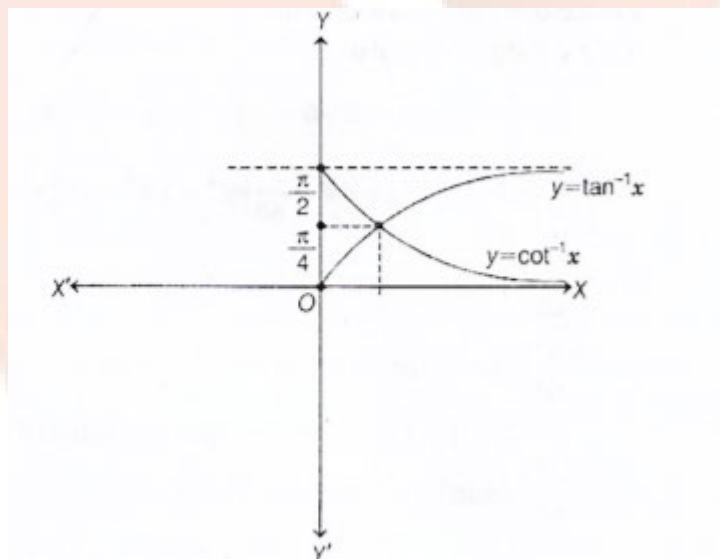
$$\log x = t$$

$$\frac{dx}{x} = dt$$

Thus,

$$\begin{aligned} I &= \int_{-1}^0 t dt + \int_0^1 t dt \\ &= \left[\frac{t^2}{2} \right]_{-1}^0 + \left[\frac{t^2}{2} \right]_0^1 \\ &= -\frac{1}{2}(0-1) + \frac{1}{2}(1-0) \\ &= \frac{1}{2} \end{aligned}$$

78. Consider the graph of $y = \tan^{-1}x$ and $y = \cot^{-1}x$.



The required area is calculated as,

$$\begin{aligned}
 A &= 2 \left[\frac{\pi}{4} - \int_0^1 \tan^{-1} x dx \right] \\
 &= \frac{\pi}{2} - 2 \left[\left(x \tan^{-1} x \right)_0^1 - \int_0^1 \frac{x}{1+x^2} dx \right] \\
 &= \frac{\pi}{2} - 2 \left[\frac{\pi}{4} - \frac{1}{2} \left[\log_e (1+x^2) \right]_0^1 \right] \\
 &= \frac{\pi}{2} - \frac{\pi}{2} + \log_e 2
 \end{aligned}$$

Solve further,

$$A = \log_e 2$$

79. The equation of family of parabolas with focus and the x axis is,

$$y^2 = 4a(x - a)$$

Differentiate with respect to x .

$$2y \frac{dy}{dx} = 4a$$

$$a = \frac{y}{2} \frac{dy}{dx}$$

Therefore,

$$y^2 = 2y \frac{dy}{dx} \left(x - \frac{y}{2} \right) \frac{dy}{dx}$$

$$y^2 = y \frac{dy}{dx} (2x - y) \frac{dy}{dx}$$

$$y \left(\frac{dy}{dx} \right)^2 - 2x \left(\frac{dy}{dx} \right) + y = 0$$

Hence, the order of the differential equation is 1.

80. Consider the differential equation.

$$yy' = x \left[\frac{y^2}{x^2} + \frac{\phi \left(\frac{x^2}{y^2} \right)}{\phi' \left(\frac{x^2}{y^2} \right)} \right]$$

The above equation can be written as,

$$\frac{y}{x} \frac{dy}{dx} = \frac{y^2}{x^2} + \frac{\phi \left(\frac{x^2}{y^2} \right)}{\phi' \left(\frac{x^2}{y^2} \right)}$$

Let,

$$y = vx$$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

Therefore,

$$v \left(v + x \frac{dv}{dx} \right) = v^2 + \frac{\phi(v^2)}{\phi'(v^2)}$$

$$vx \frac{dv}{dx} = \frac{\phi(v^2)}{\phi'(v^2)}$$

$$\int v \frac{\phi'(v^2)}{\phi(v^2)} dv = \int \frac{dx}{x}$$

$$\frac{1}{2} \log_e \phi(v^2) = \log_e x + \log_e (\sqrt{c})$$

Solve further,

$$\phi(v^2) = cx^2$$

$$\phi\left(\frac{y^2}{x^2}\right) = cx^2$$

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Solutions – Physics

81. As the physical quantity is obtained from the ratio of thermal conductivity and universal gravitational constant therefore, the dimensional formula of it is,

$$\frac{[k]}{[G]} = \frac{[M^1 L^1 T^{-3} K^{-1}]}{[M^{-1} L^3 T^{-2}]} \\ = [M^2 L^{-2} T^{-1} K^{-1}]$$

Compare the above dimensional formula with the given dimensional formula,

$$[M^{2a} L^{4b} T^{2c} K^d] = [M^2 L^{-2} T^{-1} K^{-1}]$$

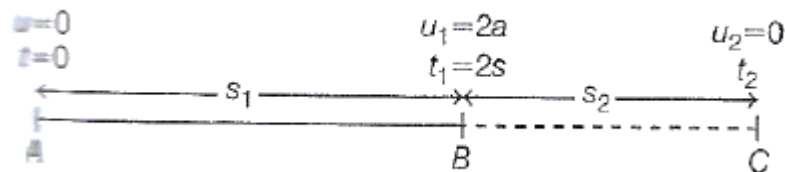
$$\text{Thus, } a = 1, b = -\frac{1}{2}, c = -\frac{1}{2} \text{ and } d = \frac{1}{2}$$

The values of required expression is calculated as,

$$\frac{a+b}{c+b} - d = \frac{1 - \frac{1}{2}}{-\frac{1}{2} - \frac{1}{2}} - (-1)$$

$$\frac{a+b}{c+b} - d = \frac{1}{2}$$

82. Consider the diagram shown below.



The velocity at point B after 2 s is,

$$v_1 = u + at_1$$

$$v_1 = 0 + a(2)$$

$$v_1 = 2a$$

The time t_2 is,

$$v_2 = v_1 - at_2$$

$$0 = 2a - at_2$$

$$t_2 = 2s$$

Total time of travel from A to C is,

$$t = t_1 + t_2$$

$$t = (2 + 2)s$$

$$t = 4s$$

The distance between A and B is,

$$AB = ut_1 + \frac{1}{2}at_1^2$$

$$AB = (0)(2) + \frac{1}{2}a(2)^2$$

$$AB = 2a$$

Thus, $AB = BC = 2a$, and $AC = 4a$

When body return from C to A; $u_1 = 0$ and $Ac = 4a$

The time taken to return from C to A is,

$$AC = u_1t' + \frac{1}{2}at'^2$$

$$4a = (0)t' + \frac{1}{2}at'^2$$

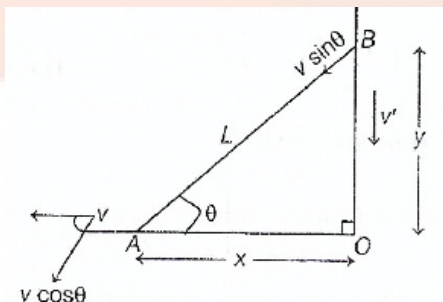
$$t' = 2\sqrt{2} \text{ s}$$

Thus total time taken by body is,

$$t_0 = t + t'$$

$$t_0 = (4 + 2\sqrt{2}) \text{ s}$$

83. Consider the diagram as shown below,



From the definition of velocity,

$$v = \frac{dx}{dt}, \text{ and } v' = -\frac{dy}{dt}$$

From the figure,

$$x^2 + y^2 = L^2$$

Differentiating w.r.t. time, we get

$$2x\left(\frac{dx}{dt}\right) + 2y\left(\frac{dy}{dt}\right) = 0$$

$$2xv + 2yv' = 0$$

$$v' = -\frac{x}{y}v$$

$$v' = -v \cot \theta$$

84. Total time to reach on the horizontal surface is $t = 2 + 4 = 6$ s

The equation of horizontal projectile motion ($\theta_1 = 0^\circ$) after 6 s is,

$$\tan 0^\circ = \frac{v \sin \theta - g(6)}{v \cos \theta}$$

$$v \sin \theta = 60$$

The equation of vertical projectile motion ($\theta_2 = 30^\circ$) after 4 s is,

$$\tan 30^\circ = \frac{v \sin \theta - g(4)}{v \cos \theta}$$

$$\tan 30^\circ = \frac{60 - g(4)}{v \cos \theta}$$

$$\frac{1}{\sqrt{3}} = \frac{60 - 40}{v \cos \theta}$$

$$v \cos \theta = 20\sqrt{3}$$

The magnitude of initial velocity is,

$$v = \sqrt{(v \sin \theta)^2 + (v \cos \theta)^2}$$

$$v = \sqrt{(60)^2 + (20\sqrt{3})^2}$$

$$v = 40\sqrt{3} \text{ m/s}$$

85. The acceleration of the body is calculated as,

$$a = \frac{F}{m}$$

$$a = \frac{(2.6\hat{i} + 1.6\hat{j})}{2}$$

$$a = (1.3\hat{i} + 0.8\hat{j}) \text{ m/s}^2$$

The velocity vector is,

$$v = \int a dt$$

$$v = \int (1.3\hat{i} + 0.8\hat{j}) dt$$

$$v = (1.3\hat{i} + 0.8\hat{j})t + c$$

Apply the given condition in above expression, i.e. at $t = 0$,

$$v = (3.6\hat{i} - 4.8\hat{j}) \text{ m/s},$$

$$(3.6\hat{i} - 4.8\hat{j}) = (1.3\hat{i} + 0.8\hat{j})(0) + c$$

$$c = (3.6\hat{i} - 4.8\hat{j})$$

Thus, the velocity is,

$$v = (1.3\hat{i} + 0.8\hat{j})t + (3.6\hat{i} - 4.8\hat{j})$$

$$v = (1.3t + 3.6)\hat{i} + (0.8t - 4.8)\hat{j}$$

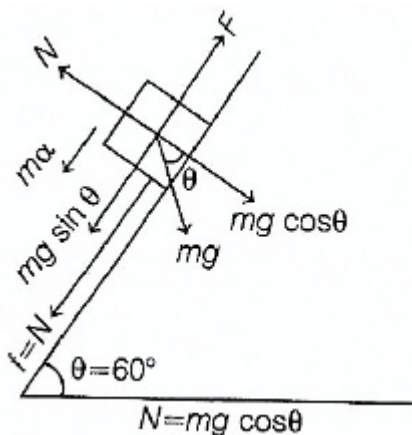
According to question, $v_y = 0$

Thus,

$$0.8t - 4.8 = 0$$

$$t = 6 \text{ s}$$

86. Consider the free body diagram of the system,



The expression of the net acceleration force down the inclined plane is,

$$F = mg \sin \theta - \mu mg \cos \theta$$

The expression of net retarding force is,

$$F_1 = mg \sin \theta + \mu mg \cos \theta$$

Given condition, $F_1 = 2F$

Substituting the values, we get

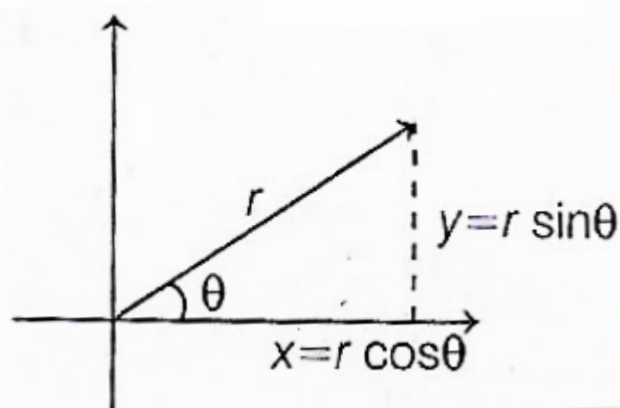
$$mg \sin \theta + \mu mg \cos \theta = 2(mg \sin \theta - \mu mg \cos \theta)$$

$$3\mu g \cos \theta = mg \sin \theta$$

$$\tan 60^\circ = 3\mu \quad (\text{As } \theta = 60^\circ)$$

$$\mu = \frac{1}{\sqrt{3}}$$

87. Consider the diagram,



The expression of force is,

$$\vec{F} = K \left[\frac{x}{(x^2 + y^2)^{3/2}} \hat{i} + \frac{y}{(x^2 + y^2)^{3/2}} \hat{j} \right]$$

Substitute $x = r \cos \theta$ and $y = r \sin \theta$ in above equation.

$$\vec{F} = K \left[\frac{r \cos \theta}{(r^2 \cos^2 \theta + r^2 \sin^2 \theta)^{3/2}} \hat{i} + \frac{r \sin \theta}{(r^2 \cos^2 \theta + r^2 \sin^2 \theta)^{3/2}} \hat{j} \right]$$

$$\vec{F} = K \left[\frac{r \cos \theta}{r^3 (1)^{3/2}} \hat{i} + \frac{r \sin \theta}{r^3 (1)^{3/2}} \hat{j} \right]$$

$$\vec{F} = \frac{K}{r^2} [\cos \theta \hat{i} + \sin \theta \hat{j}]$$

The displacement of particle having circular motion will be

$$s = 0$$

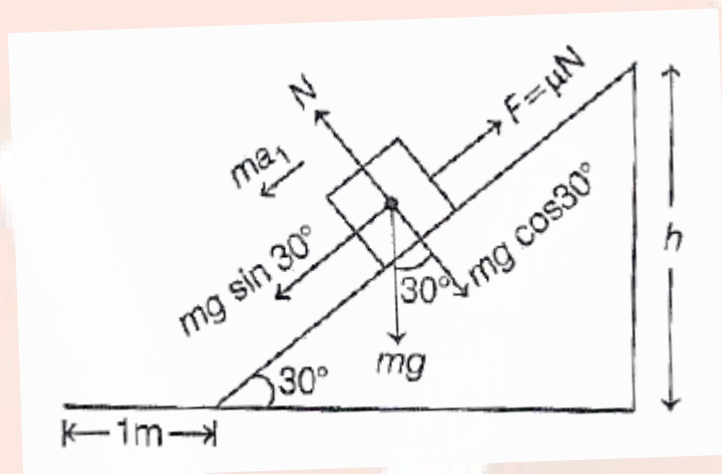
The expression of the work done is,

$$W = F \times s$$

$$W = F \times (0)$$

$$W = 0$$

88. The free body diagram of the system is shown below,



The acceleration of the block is,

$$a = \mu g$$

$$a = (0.2)(10)$$

$$a = 2 \text{ m/s}^2$$

The initial velocity of the block is calculated as,

$$v^2 = u^2 - 2as$$

$$(0)^2 = u^2 - 2(2)(1)$$

$$u = 2 \text{ m/s}$$

The net accelerating force down the inclined plane is,

$$F = ma_1$$

$$mg \sin 30^\circ - \mu mg \cos 30^\circ = ma_1$$

$$a_1 = g(\sin 30^\circ - \mu \cos 30^\circ)$$

$$a_1 = 5(1 - \sqrt{3}\mu)$$

The displacement is calculated as,

$$v_1^2 = u_1^2 + 2a_1S$$

$$(2)^2 = (0)^2 + 2[5(1 - \sqrt{3}\mu)]S \quad (\text{As, } v_1 = u = 2 \text{ m/s, } u_1 = 0)$$

$$S = \frac{2}{5(1 - \sqrt{3}\mu)}$$

$$S = 0.612 \text{ m}$$

The work done by friction force is,

$$W = (\mu mg \cos 30^\circ)S + \frac{1}{2}mu^2$$

$$W = (0.2)\left(\frac{100}{1000}\right)(10)(\cos 30^\circ)(0.612) + \frac{1}{2}\left(\frac{100}{1000}\right)(2)^2$$

$$W = 0.306 \text{ J}$$

89. The expression of kinetic energy for rotating disc is given as,

$$KE_1 = \frac{1}{2}mv_1^2 + \frac{1}{2}I\omega^2$$

$$KE_1 = \frac{1}{2}mv_1^2 + \frac{1}{2}\left(\frac{mR^2}{2}\right)\left(\frac{v_1}{R}\right)^2$$

$$KE_1 = \frac{3}{4}mv_1^2$$

The expression of the kinetic energy for the sliding disc is,

$$KE_2 = \frac{1}{2}mv_2^2$$

Given condition, $KE_1 = KE_2$

Substitute values.

$$\frac{3}{4}mv_1^2 = \frac{1}{2}mv_2^2$$

$$\frac{v_1^2}{v_2^2} = \frac{2}{3}$$

$$v_1 : v_2 = \sqrt{2} : \sqrt{3}$$

90. The expression of the angular acceleration is,

$$\alpha = \frac{\tau}{I}$$

$$\alpha = \frac{2RF \sin 90^\circ}{\frac{2}{3}MR^2 + MR^2}$$

$$\alpha = \left(\frac{6}{5}\right) \frac{F}{RM}$$

The tangential acceleration is calculated as,

$$a_t = R\alpha$$

$$a_t = R \left(\frac{6}{5}\right) \frac{F}{RM}$$

$$a_t = \frac{6}{5} \frac{F}{M}$$

91. The time period of the pendulum while moving upward is,

$$T_1 = 2\pi \sqrt{\frac{l}{g+a}} \quad \text{..... (I)}$$

The time period of the pendulum while moving downward is,

$$T_2 = 2\pi \sqrt{\frac{l}{g-a}} \quad \text{.....(II)}$$

Dividing equation (I) by (II),

$$\frac{T_1}{T_2} = \sqrt{\frac{g-a}{g+a}}$$

$$\frac{1}{2} = \sqrt{\frac{10-a}{10+a}}$$

$$\frac{1}{4} = \frac{10-a}{10+a}$$

$$a = 6 \text{ m/s}^2$$

92. The expression of the mass of the earth is,

$$M_e = \rho V$$

$$M_e = \rho \left(\frac{4}{3} \pi R^3 \right)$$

The gravitational force on the body of mass m due to earth is,

$$F = \frac{GM_e m}{(R+h)^2}$$

$$F = \frac{G\rho \left(\frac{4}{3} \pi R^3 \right) m}{(R+h)^2}$$

$$F = \frac{4}{3} \pi G R \rho m \left(1 + \frac{h}{R} \right)^{-2}$$

$$\text{or, } F = \frac{4}{3} \pi G R \rho m \left(1 - \frac{2h}{R} \right)$$

Thus the difference in weight is,

$$F = \frac{4}{3}\pi GR\rho m - \frac{4}{3}\pi G\rho m R\left(\frac{2h}{R}\right)$$

$$F = \frac{4}{3}\pi GR\rho m - \frac{8}{3}\pi\rho Gmh$$

The error in weight is,

$$\text{Error} = \frac{8}{3}\pi\rho Gmh$$

93. The expression of the stress is,

$$\sigma = \frac{\text{Tension}}{\text{Area}}$$

$$\sigma = \frac{m\omega^2 l}{A}$$

The expression of strain is,

$$\varepsilon = \frac{\sigma}{Y}$$

$$\frac{\Delta l}{l} = \frac{1}{Y}\left(\frac{m\omega^2 l}{A}\right)$$

Substituting the values, we get

$$\frac{2 \times 10^{-3}}{1} = \frac{(1)\omega^2(1)}{(2 \times 10^{31})(0.01 \times 10^{-4})}$$

$$\omega = 20 \text{ rad/sec}$$

94. The expression of the volume flow rate is,

$$Q = Av$$

$$Q = A\sqrt{2gh}$$

Substitute values.

$$(100 \times 10^{-6} \text{ m}^3/\text{s}) = (2 \times 10^{-4} \text{ m}^2) \sqrt{2(10)h}$$

$$h = 0.125 \text{ m}$$

$$h = 12.5 \text{ cm}$$

95. The piece of wood begins to sink when its weight is equal to the weight of benzene displaced.

The equilibrium equation is,

$$V_w \rho_w g = V_b \rho_b g$$

$$\frac{\rho_w}{1 + \gamma_w \Delta T} = \frac{\rho_b}{1 + \gamma_b \Delta T}$$

Substitute values,

$$\frac{880}{1 + (1.2 \times 10^{-3})\Delta T} = \frac{900}{1 + (1.5 \times 10^{-3})\Delta T}$$

$$\Delta T \approx 83.3^\circ\text{C}$$

Let the temperature at which pieces of wood will just sink in the benzene is T_2 . Then $\Delta T = T_2 - T_1$. Thus,

$$(T_2 - T_1) = 83.3^\circ\text{C}$$

$$(T_2 - 0^\circ\text{C}) = 83.3^\circ\text{C}$$

$$T_2 = 83.3^\circ\text{C}$$

96. The expression of the equivalent resistance is,

$$R_{eq} = R_1 + R_2 + R_3$$

$$R_{eq} = \frac{l_1}{k_1 A_1} + \frac{l_2}{k_2 A_2} + \frac{l_3}{k_3 A_3}$$

Substitute values.

$$R_{eq} = \frac{1 \times 10^{-2}}{(0.8)(2.6)} + \frac{5 \times 10^{-2}}{(0.08)(2.6)} + \frac{1 \times 10^{-2}}{(0.8)(2.6)}$$

$$R_{eq} = \frac{1}{4}$$

The heat flow is calculated as,

$$H = \frac{(T_2 - T_1)}{R_{eq}}$$

$$H = \frac{[18 - (-2)]}{(1/4)}$$

$$H = 80 \text{ W}$$

97. The expression of change in internal energy due to adiabatic process is,

$$\Delta U = nC_v \Delta T$$

$$\Delta U = n \left(\frac{R}{\gamma - 1} \right) (T_2 - T_1)$$

Substitute values.

$$\Delta U = 5 \left(\frac{8.3}{1.4 - 1} \right) (673 - 273)$$

$$\Delta U = 41.5 \text{ kJ}$$

98. The temperature at point A is,

$$T_A = \frac{3p_0V_0}{R}$$

The temperature at point B is,

$$T_B = \frac{30p_0V_0}{R}$$

The expression of the change in internal energy due to adiabatic process is,

$$\Delta U = nC_v \Delta T$$

$$\Delta U = \frac{3}{2} R \Delta T$$

The expression of the work done in process A to B is,

$$W = 18 p_0 V_0$$

$$W = \frac{2}{3} R \Delta T$$

From 1st law of thermodynamics for process A to B,

$$Q = \Delta U + W$$

$$nC \Delta T = \frac{3}{2} R \Delta T + \frac{2}{3} R \Delta T$$

$$(1) C \Delta T = \frac{3}{2} R \Delta T + \frac{2}{3} R \Delta T$$

$$C = \frac{13}{6} R$$

99. The molecular mass of the mixture is calculated as,

$$M_{\text{mix}} = \frac{n_{\text{He}}M_{\text{He}} + n_{\text{H}}M_{\text{H}}}{n_{\text{He}} + n_{\text{H}}}$$

$$M_{\text{mix}} = \frac{(2)(4) + (2)(2)}{2 + 2}$$

$$M_{\text{mix}} = 3 \text{ gm/mole}$$

The degree of freedom is,

$$f = \frac{(2)(3) + (2)(2)}{2 + 2}$$

$$f = 4$$

The degree of freedom of the mixture is,

$$f_{\text{mix}} = 1 + \frac{2}{f}$$

$$f_{\text{mix}} = 1 + \frac{2}{4}$$

$$f_{\text{mix}} = 1.5$$

The expression of the speed of sound is,

$$v = \sqrt{\frac{f_{\text{mix}}RT}{M}}$$

Substitute values.

$$v = \sqrt{\frac{(1.5)\left(\frac{25}{3}\right)\left(\frac{972}{5}\right)}{3 \times 10^{-3}}}$$

$$v = 900$$

So,

$$n \times 100 = 900$$

$$n = 9$$

100. The expression for the frequency of siren recorded by train A for observer A is,

$$f_A = f_s \left(\frac{v_s + v_A}{v_s} \right)$$

Substitute values.

$$5.5 = 5 \left(1 + \frac{v_A}{v_s} \right)$$

$$v_A = 0.1v_s \quad \dots\dots (I)$$

The expression for the frequency of siren recorded by train B for observer B is,

$$f_B = f_s \left(\frac{v_s + v_B}{v_s} \right)$$

$$6 = 5 \left(1 + \frac{v_B}{v_s} \right)$$

$$v_B = 0.2v_s \quad \text{..... (II)}$$

From equation (I) and (II),

$$\frac{v_B}{v_A} = 2$$

101. The expression of the speed of transverse wave is,

$$v = \sqrt{\frac{T}{\mu}}$$

$$v = \sqrt{\frac{T}{m}} \times l$$

$$l = \frac{v^2 m}{T} \quad \text{..... (I)}$$

The expression of the change in length is,

$$\Delta l = \frac{Tl}{AY} \quad \text{..... (II)}$$

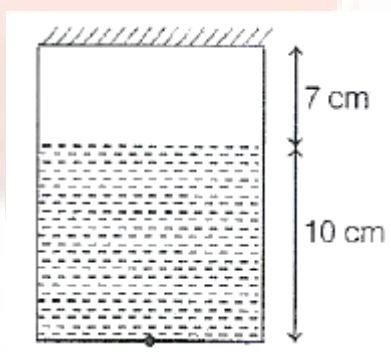
From equation (I) and (II),

$$\Delta l = \frac{v^2 m}{AY}$$

$$\Delta l = \frac{(80)^2 (5 \times 10^{-3})}{(1 \times 10^{-6})(4 \times 10^{11})}$$

$$\Delta l = 8 \times 10^{-5} \text{ m}$$

102. Consider the figure,



The apparent depth is calculated as,

$$d_A = \frac{\text{Real depth}}{\mu}$$

$$d_A = \frac{10}{1.33}$$

$$d_A = 7.5 \text{ cm}$$

The distance of image from the mirror is calculated as,

$$d = (7 \text{ cm}) + d_A$$

$$d = (7 \text{ cm}) + (7.5 \text{ cm})$$

$$d = 14.5 \text{ cm}$$

103. The expression of the angular deviation of n th fringe is,

$$\theta = \frac{n\lambda}{d}$$

Substitute values.

$$12^\circ \times \frac{\pi}{180^\circ} = \frac{5 \times \lambda}{9 \times 10^{-6}}$$
$$\lambda = 3798 \times 10^{-10} \text{ m}$$
$$\lambda = 3798 \text{ \AA}$$

104. The electric force acting on a charge $-q$ at a point P is calculated as,

$$F_p = -\frac{\sigma}{2\epsilon_0} \hat{k} - \frac{2\sigma}{2\epsilon_0} \hat{k} - \frac{\sigma}{2\epsilon_0} \hat{k}$$

$$F_p = -\frac{4\sigma}{2\epsilon_0} \hat{k}$$

$$F_p = -\frac{2\sigma}{\epsilon_0} \hat{k}$$

105. The charge on any particle is expressed in terms of integral multiple of basic unit of the charge. It is written as

$$q = \pm ne \quad (\text{where } n = 1, 2, 3, \dots)$$

Thus electric charge is quantised and it cannot be written as integral multiple of charge on half electron.

106. The expression of the electric potential difference between two points is,

$$\Delta V = -\int E \cdot dr$$

Substitute values and integrate above equation,

$$\Delta V = -\int (x\hat{i} - 2y\hat{j} + z\hat{k}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k})$$

$$\Delta V = -\int_2^0 xdx + 2\int_1^2 ydy - \int_0^4 zdz$$

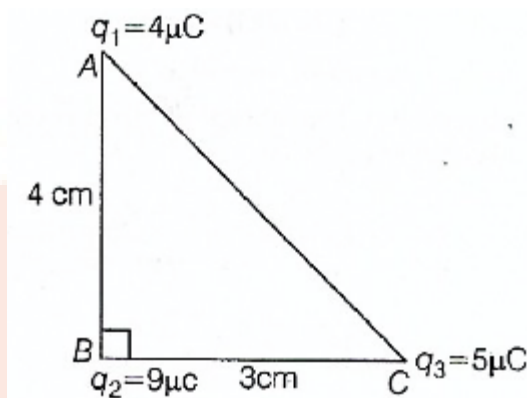
$$\Delta V = \left[\frac{x^2}{2} \right]_2^0 + \left[y^2 \right]_1^2 - \left[\frac{z^2}{2} \right]_0^4$$

$$\Delta V = -3 \text{ V}$$

Thus, the magnitude of electric potential difference is,

$$|\Delta V| = 3 \text{ V}$$

107. Consider the diagram,



By using triangle property, the distance AC is,

$$AC = \sqrt{(AB)^2 + (BC)^2}$$

$$AC = 5 \text{ cm}$$

$$AC = 5 \times 10^{-2} \text{ m}$$

The initial electric potential energy of the three charges is,

$$U = \frac{kq_1q_2}{AB} + \frac{kq_1q_3}{AC} + \frac{kq_2q_3}{BC}$$

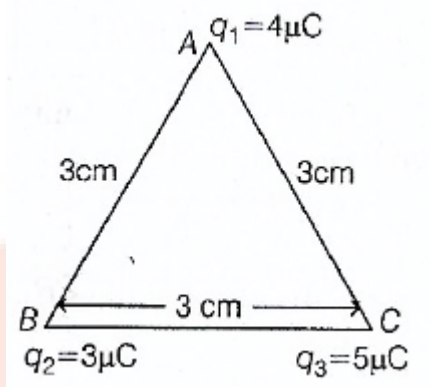
$$U = k \left[\frac{q_1q_2}{AB} + \frac{q_1q_3}{AC} + \frac{q_2q_3}{BC} \right]$$

Substitute values.

$$U = (9 \times 10^9) \left[\frac{4 \times 3 \times 10^{-12}}{4 \times 10^{-2}} + \frac{4 \times 5 \times 10^{-12}}{5 \times 10^{-2}} + \frac{3 \times 5 \times 10^{-12}}{3 \times 10^{-2}} \right]$$

$$U \approx 10.8 \text{ J}$$

When the three charges located on an equilateral triangle of side 3 cm as shown in figure below,



The final potential energy is calculated as,

$$U' = (9 \times 10^9) \left[\frac{4 \times 3 \times 10^{-12}}{3 \times 10^{-2}} + \frac{4 \times 5 \times 10^{-12}}{3 \times 10^{-2}} + \frac{3 \times 5 \times 10^{-12}}{3 \times 10^{-2}} \right]$$

$$U' \approx 14.1 \text{ J}$$

The work done in moving charges at points A and C is calculated as,

$$W = U' - U$$

$$W = 14.1 - 10.8$$

$$W = 3.3 \text{ J}$$

108. When $C = 100 \Omega$, the balanced equation for the wheat stone bridge is,

$$\frac{R_A}{R_B} = \frac{100}{R} \quad \dots\dots (I)$$

When $C = 121 \Omega$, the balanced equation for the wheat stone bridge is,

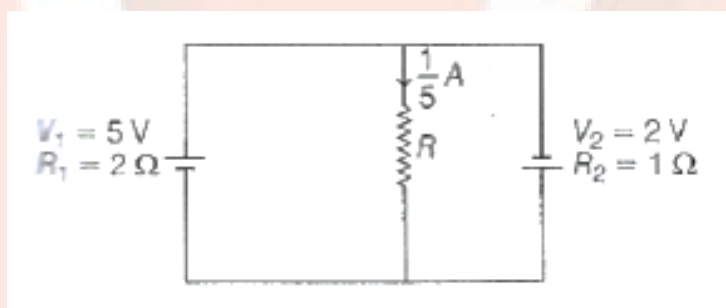
$$\frac{R_B}{R_A} = \frac{121}{R} \quad \dots\dots (II)$$

From equation (I) and (II), we get

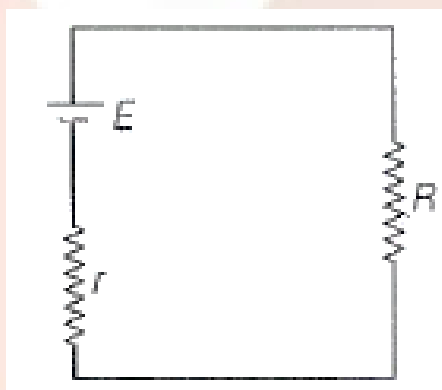
$$\frac{100}{R} = \frac{R}{121}$$

$$R = 110 \, \Omega$$

109. Consider the given figure,



The equivalent circuit of the given circuit is shown below.



The equivalent emf is calculated as,

$$e = \frac{\frac{V_1}{R_1} - \frac{V_2}{R_2}}{\frac{1}{R_1} + \frac{1}{R_2}}$$

$$e = \frac{\frac{5}{2} - \frac{2}{1}}{\frac{1}{2} + \frac{1}{1}}$$

$$e = \frac{1}{3} \text{ V}$$

The internal resistance is,

$$r = \frac{R_1 \cdot R_2}{R_1 + R_2}$$

$$r = \frac{2 \times 1}{2 + 1}$$

$$r = \frac{2}{3} \Omega$$

The current is given as,

$$I = \frac{e}{r + R}$$

Substitute values to find R ,

$$\frac{1}{5} = \frac{\frac{1}{3}}{\frac{2}{3} + R}$$

$$R = 1 \Omega$$

110. The magnetic force in uniform transverse magnetic field ($\theta = 90^\circ$) is,

$$F = qvB \sin \theta$$

$$F = e \sqrt{\frac{2eV}{m_e}} B \sin 90^\circ$$

$$F = e \sqrt{\frac{2eV}{m_e}} B \quad \text{..... (I)}$$

The expression of the magnetic force when electron accelerated through a potential difference $2V$ is,

$$F' = e \sqrt{\frac{2e(2V)}{m_e}} B \quad \text{..... (II)}$$

From equation (I) and (II), we get

$$F' = \sqrt{2}F$$

111. The current to be passed through the coil can be calculated as,

$$\tau = MB$$

$$\tau = IAB$$

$$2\sqrt{3} \times 10^{-5} = I(\sqrt{3} \times 10^{-4})(100 \times 10^{-3})$$

$$I = 2 \text{ A}$$

112. The rise in the temperature can be calculated as,

$$\Delta Q = ms\Delta\theta$$

$$10^{-2} \times 42 \times 60 = (6 \times 10^3 \times 0.1 \times 10^{-3})(4.2)\Delta\theta$$

$$\Delta\theta = 10^\circ\text{C}$$

113. Given that the radius of wire is halved, so its length will be $4l$ and the resistance becomes $16R$.

The number of turns is doubled in coil so its radius should also be double to accommodate the length of wire and the area of coil will become 4 times.

Thus,

$$V_2 = 8V_1$$

In first case, the power dissipated is,

$$P_1 = \frac{V_1^2}{R}$$

In second case, the power dissipated is,

$$P_2 = \frac{V_2^2}{16R}$$

$$P_2 = \frac{(8V_1)^2}{16R}$$

$$P_2 = \frac{4V_1^2}{R}$$

Thus,

$$\frac{P_1}{P_2} = \frac{1}{4}$$

$$P_1 : P_2 = 1 : 4$$

114. The expression of the average value of emf is,

$$E_{avg} = \frac{1}{T} \int_0^T [E^2 dt]$$

Substitute values.

$$E_{avg} = \frac{1}{T} \int_0^T \left[(6 \sin \omega t + 4 \sin 2\omega t)^2 dt \right]$$

$$E_{avg} = \frac{1}{T} \int_0^T \left[(36 \sin^2 \omega t + 16 \sin^2 2\omega t + 48 \sin \omega t \cdot \sin 2\omega t) dt \right]$$

$$E_{avg} = \frac{1}{T} \int_0^T \left[\left\{ 36 \sin^2 \omega t + 16 \sin^2 2\omega t + 48 \left(\cos \frac{\omega t}{2} - \cos \frac{3\omega t}{2} \right) \right\} dt \right]$$

$$E_{avg} = \left[36 \left(\frac{T}{2} \right) + 16 \left(\frac{T}{2} \right) + 48(0 - 0) \right]$$

$$E_{avg} = 26 \text{ V}$$

The rms value of emf is calculated as,

$$E_{rms} = \sqrt{E_{avg}}$$

$$E_{rms} = \sqrt{26} \text{ V}$$

115. The force due to radiation pressure can be calculated as,

$$F = \frac{I}{c}$$

$$F = \frac{100}{3 \times 10^8}$$

$$F = 3.3 \times 10^{-7} \text{ N}$$

116. The velocity of charged particle at time t is,

$$\vec{v} = v_0 \hat{i} + \frac{eE_0}{m} t \hat{j}$$

The magnitude of velocity is,

$$v = \sqrt{v_0^2 + \left(\frac{eE_0}{m} t \right)^2}$$

Using the formula of de-Broglie wavelength,

$$\lambda = \frac{h}{mv}$$

$$\lambda = \frac{h}{m \sqrt{v_0^2 + \left(\frac{eE_0}{m} t \right)^2}}$$

$$\lambda = \frac{h}{mv_0 \sqrt{1 + \frac{e^2 E_0^2 t^2}{m^2 v_0^2}}}$$

$$\lambda = \frac{\lambda_0}{\sqrt{1 + \frac{e^2 E_0^2 t^2}{m^2 v_0^2}}} \quad \left(\text{As, } \lambda_0 = \frac{h}{mv_0} \right)$$

117. The expression of speed of revolution of electron is,

$$v = \frac{1}{4\pi\epsilon_0} \frac{2\pi Ze^2}{nh}$$

The expression of kinetic energy of electron is,

$$KE = \left(\frac{1}{4\pi\epsilon_0} \right)^2 \frac{2\pi^2 m e^4 Z^2}{n^2 h^2}$$

The expression of kinetic energy of electron is,

$$KE = \left(\frac{1}{4\pi\epsilon_0} \right)^2 \frac{2\pi^2 m e^4 Z^2}{n^2 h^2}$$

The expression of total energy of electron is,

$$TE = - \left(\frac{1}{4\pi\epsilon_0} \right)^2 \frac{2\pi^2 m e^4 Z^2}{n^2 h^2}$$

The expression of frequency of electron is,

$$f = \left(\frac{1}{4\pi\epsilon_0} \right)^2 \frac{4\pi^2 Z^2 e^4 m}{n^3 h^3}$$

Thus option (1) is correct.

118. The expression of the number of atom left undecayed is,

$$N = N_0 \left(\frac{1}{2} \right)^{\frac{t}{T_{1/2}}} \quad \text{..... (I)}$$

The above expression for 20% decay is,

$$\left(1 - \frac{20}{100}\right) N_0 = N_0 \left(\frac{1}{2}\right)^{\frac{t_1}{18}}$$

$$0.8 = \left(\frac{1}{2}\right)^{\frac{t_1}{18}} \quad \text{..... (II)}$$

The equation (I) for 80% decay is,

$$\left(1 - \frac{80}{100}\right) N_0 = N_0 \left(\frac{1}{2}\right)^{\frac{t_2}{18}}$$

$$0.2 = \left(\frac{1}{2}\right)^{\frac{t_2}{18}} \quad \text{..... (III)}$$

From equation (II) and (III),

$$4 = \left(\frac{1}{2}\right)^{\frac{t_1 - t_2}{18}}$$

$$\log 4 = \log \left(\frac{1}{2}\right)^{\frac{t_1 - t_2}{18}}$$

$$\log 2^2 = \frac{t_1 - t_2}{18} \log 2^{\frac{t_2 - t_1}{18}}$$

$$2 = \frac{t_2 - t_1}{18}$$

$$t_2 - t_1 = 36 \text{ min}$$

119. The expression of the DC current gain is,

$$\beta = \frac{\alpha}{1 - \alpha}$$

The DC current gain for $\alpha = \frac{20}{21}$ is,

$$\beta = \frac{\frac{20}{21}}{1 - \frac{20}{21}}$$

$$\beta = 20$$

The DC current gain for $\alpha = \frac{100}{101}$ is,

$$\beta = \frac{\frac{100}{101}}{1 - \frac{100}{101}}$$

$$\beta = 100$$

Thus the range of dc current gain is, $\beta \in [20, 100]$

120. The expression of the maximum range of the transmission is,

$$d = \sqrt{2R_{\text{earth}} h_{\text{tower}}}$$

The expression of the population covered is,

$$P_c = nA$$

$$P_c = n\pi d^2$$

$$P_c = n\pi(2R_{\text{earth}} h_{\text{tower}})$$

Substitute values.

$$P_c = \left(\frac{1000}{\pi} \right) \pi (2 \times 6400 \times 5 \times 10^{-3})$$

$$P_c = 64000$$

AP EAMCET-2019-Engineering-20 Apr-Shift-2

Solutions – Chemistry

121. The formula of energy related with Bohr's orbit in the hydrogen atom is shown below.

$$E_n = -\frac{13.6}{n^2} \text{eV} \dots\dots (I)$$

Radius of Bohr atom is calculated as.

$$r_n = 0.529 \frac{n^2}{Z} \text{\AA} \dots\dots (II)$$

For hydrogen atom, $Z = 1$,

$$r_n = r_1 n^2$$

It is given that $r_n = 9r_1$

Therefore,

$$9r_1 = r_1 n^2$$

$$n^2 = 9$$

$$n = 3$$

Therefore, the energy related with the orbit of radius of $9r_1$ is as follows:

$$E_9 = -\frac{13.6}{9} \text{ eV}$$
$$= -1.51 \text{ eV}$$

122. The threshold frequency can be calculated as:

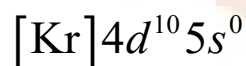
$$\frac{2hv_2}{3} = hv_0$$
$$\frac{2 \times h \times 2 \times 10^{16}}{3} = hv_0$$
$$v_0 = 1.33 \times 10^{16}$$

123. (I) Properties of elements are periodic function of their atomic numbers.

(II) In a periodic table, the count of non-metallic elements is less in number.

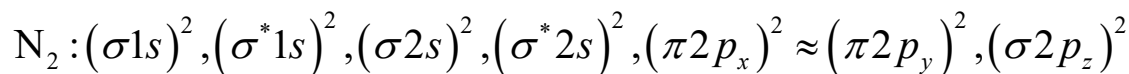
(III) The first ionization energy of elements do not change generally along a period.

(IV) The ground state electronic configuration of Pd is



Therefore, (I), (II) and (III) statements are correct.

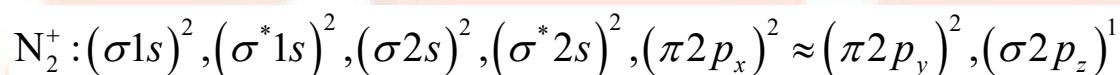
124. The bond order of $N_2(14)$ is shown below.



The Bond order can be calculated as:

$$B.O = \frac{10 - 4}{2} = 3$$

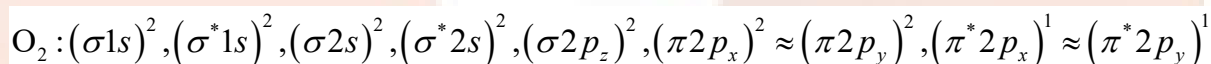
The bond order of $N_2^+(13)$ is shown below.



The Bond order can be calculated as:

$$B.O = \frac{9 - 4}{2} = 2.5$$

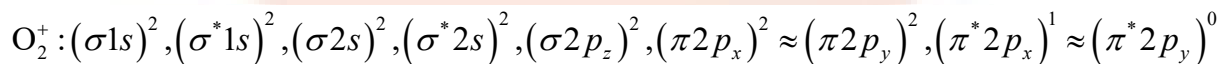
The bond order of $O_2(16)$ is shown below.



The Bond order can be calculated as:

$$B.O = \frac{10 - 6}{2} = 2$$

The bond order of $O_2^+(15)$ is shown below.



The Bond order can be calculated as:

$$B.O = \frac{10 - 5}{2} = 2.5$$

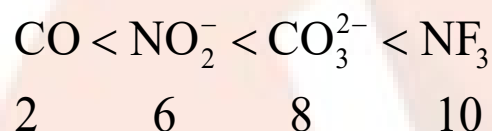
Bond length is reverse of bond order.

Therefore,

In N_2^+ , as bond order decreases, bond length increases.

In O_2^+ , as bond order increases, bond length decreases.

125. The order of lone pair of electrons is shown below.



126. Root mean square velocity for hydrogen is:

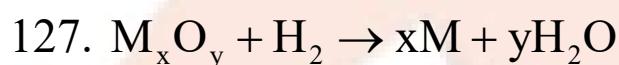
$$r_{\text{H}_2} = \sqrt{\frac{3RT_1}{M_{\text{H}_2}}}$$

Root mean square velocity for oxygen is:

$$r_{\text{O}_2} = \sqrt{\frac{3RT_2}{M_{\text{O}_2}}}$$

By combining both equations we get,

$$\begin{aligned}\frac{r_{\text{H}_2}}{r_{\text{O}_2}} &= \sqrt{\frac{M_{\text{O}_2} \times T_1}{M_{\text{H}_2} \times T_2}} \\ &= \sqrt{\frac{32 \times 50}{2 \times 800}} \\ &= 1\end{aligned}$$



The number of moles of metal is calculated as:

$$\begin{aligned}n_{\text{metal}} &= \frac{3.2}{64} \\ &= 0.05\end{aligned}$$

Mass of oxygen is equal to $3.6 - 3.2 = 0.4$

The number of moles of oxygen is calculated as:

$$\begin{aligned}n_{\text{oxygen}} &= \frac{0.4}{16} \\ &= 0.025\end{aligned}$$

The ratio can be calculated as:

$$\begin{aligned}\frac{n_{\text{metal}}}{n_{\text{oxygen}}} &= \frac{0.05}{0.025} \\ &= \frac{2}{1}\end{aligned}$$

Therefore, the formula of the oxide is M_2O .

128. The Gibbs free energy can be derived by using the expression shown below.

$$\Delta G = -2.303RT \log K$$

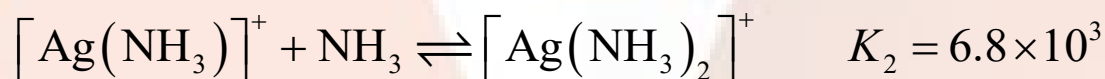
Substitute all the values in the above expression.

$$\begin{aligned}\Delta G &= -2.303 \times 0.008314 \text{ kJ/mol} \cdot \text{K} \times 300 \text{ K} \times \log(10) \\ &= -5.74 \text{ kJ/mol}\end{aligned}$$

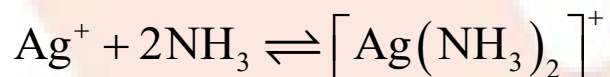
129. The first reaction is shown below.



The second reaction is shown below.



The third reaction with unknown value of K is shown below.



The value of K is obtained as follows:

$$\begin{aligned}K &= K_1 \times K_2 \\ &= 1.6 \times 10^3 \times 6.8 \times 10^3 \\ &= 1.088 \times 10^7\end{aligned}$$

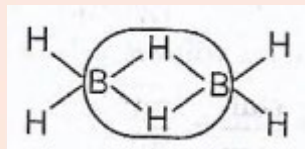
130. The pH of the buffer solution is calculated as:

$$\text{pH} = \text{pK}_a + \log \left[\frac{\text{salt}}{\text{acid}} \right]$$

Substitute the values as follows:

$$\begin{aligned} \text{pH} &= 4.76 + \log \left(\frac{0.5 \times 20}{10 \times 1} \right) \\ &= 4.76 \end{aligned}$$

131. (I) B_2H_6 is an electron deficient hydride. It has $3\text{C} - 2\text{e}^-$ electron deficient bridge bond as follows:



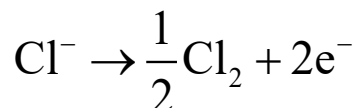
(II) NH_3 has lone pair of electrons hence; it is an electron rich hydride.

(III) NaH is not a covalent hydride. It is an ionic hydride as it belongs to alkali metal hydride.

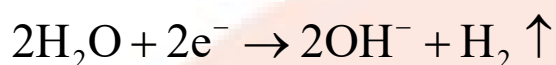
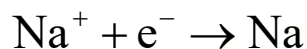
(IV) $\text{YbH}_{2.55}$ forms an interstitial hydride.

Therefore, statements (I), (II) and (IV) are correct.

132. In Castner-Kellner cell, the reaction occurs at anode is as follows.

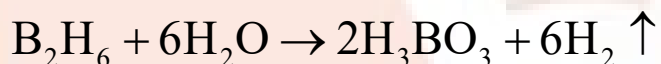
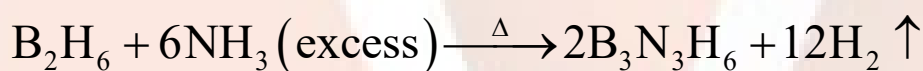


At cathode:



Therefore, by using aqueous solution of NaCl (B), the formation of NaOH (A) occurs.

133. The reactions are shown below.



Therefore,

X is $\text{B}_3\text{N}_3\text{H}_6$

Y is B_2H_6

Z is H_3BO_3

134. (I) Each carbon shows hybridization of sp^3 in diamond.

(II) The carbon atoms in graphite forms planar hexagonal layer.

(III) Silicones are having chemical formula of $(R_2SiO)_n$. They

are produced from non-alkyl groups which repel water.

(IV) In group 14 elements, the tendency of catenation goes on decreasing as elements bond strength decreases.

Therefore, the correct order is



Hence, statements (I), (II) and (III) are correct.

135. COD stands for Chemical Oxygen Demand which measures or checks the quality of water. It is not the measure of bacteria in water.

Therefore, statement (IV) is incorrect.

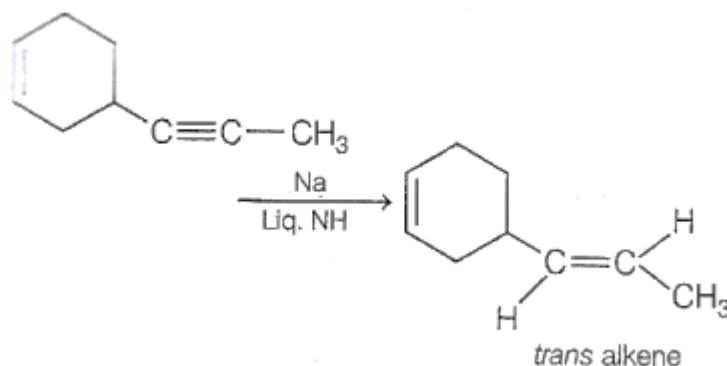
136. Structure (I) is highly stable because it has neutral charge.

Structure (II) is more stable than (III) because in structure (II), oxygen being an electronegative atom possesses a negative charge.

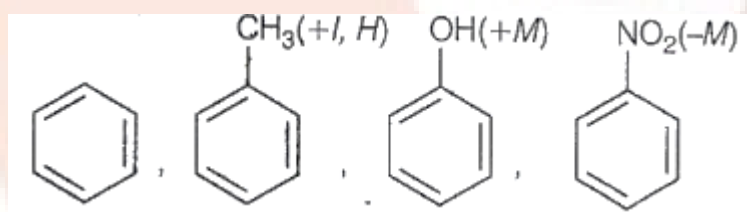
Therefore, the correct order is:



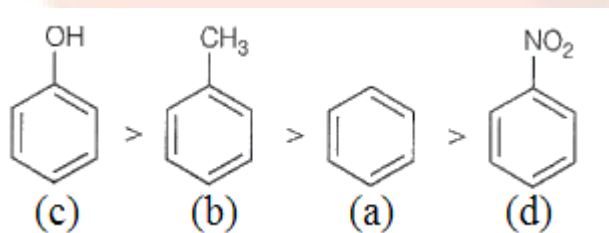
137. In the give reaction, anti-addition of H_2 occurs as shown below.



138. Electrophilic substitution reaction is directly proportional to
 $+M + I, +H$ effect and inversely proportional to
 $-M, -I, -H$ effect



Therefore, the correct order is:



139. In ionic solids, Schottky defect changes the density of the crystal as ions leave their lattice sites and creates a vacant place there. Hence, the density of the crystal decreases.

Therefore, statement (I) is not correct.

140. The depression in freezing point can be calculated as:

$$\Delta T_f = iK_f m$$

Where, m is the molality which is calculated by using the expression shown below.

$$m = \frac{\text{Moles of solute}}{\text{Mass of solvent (in g)}} \times 1000$$

$$\begin{aligned}\text{Mass of solvent} &= 1.2 \times 1.06 \\ &= 1.272 \text{ g}\end{aligned}$$

Therefore,

$$\begin{aligned}0.041^\circ\text{C} &= i \times 1.86 \times \frac{1.272 \times 1000}{60 \times 1000} \\ i &= \frac{0.041^\circ\text{C} \times 60 \times 1000}{1.86 \times 1000 \times 1.272 \text{ g}} \\ &= 1.04\end{aligned}$$

141. Osmotic pressure is:

$$\pi = \frac{WRT}{MV}$$

The osmotic pressure of urea can be calculated as:

$$\begin{aligned}\pi_1 &= \frac{1.5 \times 1000 \times 0.0821 \times 298}{60 \times 200} \\ &= 3.05 \text{ atm}\end{aligned}$$

The osmotic pressure of cane sugar can be calculated as:

$$\begin{aligned}\pi_2 &= \frac{3.42 \times 1000 \times 0.0821 \times 298}{342 \times 200} \\ &= 1.22 \text{ atm}\end{aligned}$$

For the resulting solution:

$$\begin{aligned}\pi &= \pi_1 + \pi_2 \\ &= 3.05 + 1.22 \\ &= 4.27 \text{ atm}\end{aligned}$$

142. The standard potential of the cell (E_{cell}°) for the given reaction is calculated as:

$$\begin{aligned}E_{\text{cell}}^{\circ} &= E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ} \\ &= E_{\text{Mg}^{+2}|\text{Mg}}^{\circ} - E_{\text{Sn}^{+2}|\text{Sn}}^{\circ}\end{aligned}$$

Apply Nernst equation,

$$E_{\text{cell}} = \left(E_{\text{Mg}^{+2}|\text{Mg}}^{\circ} - E_{\text{Sn}^{+2}|\text{Sn}}^{\circ} \right) - \frac{0.0591}{n} \log \frac{\text{Mg}^{+2}}{\text{Sn}^{+2}}$$

Now, substitute all the values.

$$\begin{aligned} E_{\text{cell}} &= \left(-0.14 - (-2.34) \right) - \frac{0.0591}{2} \log \frac{0.01}{0.1} \\ &= 2.2 - \frac{0.0592}{2} \times (-1) \\ &= 2.2 + 0.0295 \\ &= 2.23 \text{ V} \end{aligned}$$

143. For a first order reaction, the rate constant is derived by the formula as follows:

$$k = \frac{2.303}{t} \log \frac{a}{a-x}$$

Substitute all the values to calculate k .

$$\begin{aligned} k &= \frac{2.303}{40} \log \frac{1}{0.25} \\ &= 0.0575 \times 0.602 \\ &= 0.0346 \end{aligned}$$

$$\begin{aligned} \text{Rate} &= k[\text{A}] \\ &= 0.0346 \times 0.1 \\ &= 0.00346 \\ &= 3.46 \times 10^{-3} \text{ mol L}^{-1} \text{ min}^{-1} \end{aligned}$$

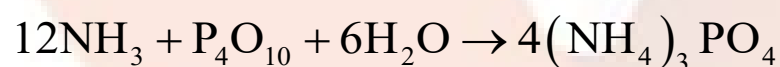
Therefore, the rate is approximate to $3.44 \times 10^{-3} \text{ mol L}^{-1} \text{ min}^{-1}$.

144. Both the adsorption processes such as chemical and physical adsorption are exothermic in nature. Therefore, statement (2) is incorrect.

145. High melting metals are not refined by liquation. Liquation method is a process which is used for refining of metals which are impure.

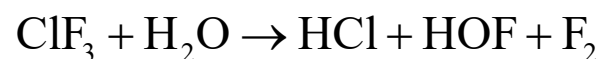
Therefore, statement (4) is incorrect.

146. From ammonia gas, moisture cannot be removed with the help of P_4O_{10} . It is due to the fact that P_4O_{10} undergoes a reaction with ammonia gas. The reaction is shown below.



Therefore, both Assertion (A) and Reason (R) are correct and (R) is the correct explanation of (A).

147. The following reaction does not occur:

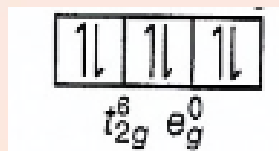


The correct reaction for the above equation is as follows:



148. Helium is used as a cryogenic agent. Therefore, statement (d) is incorrect.

149. In the coordination complex, $[\text{Co}(\text{CN})_6]^{3-}$, CN^- acts as a strong ligand which causes pairing of the electrons as shown below.



There are no unpaired electrons, therefore, $[\text{Co}(\text{CN})_6]^{3-}$ has minimum value of magnetic moment.

150. The overall reaction is shown below.



3 moles of MnO_4^{2-} produces 1 mol of MnO_2

Therefore,

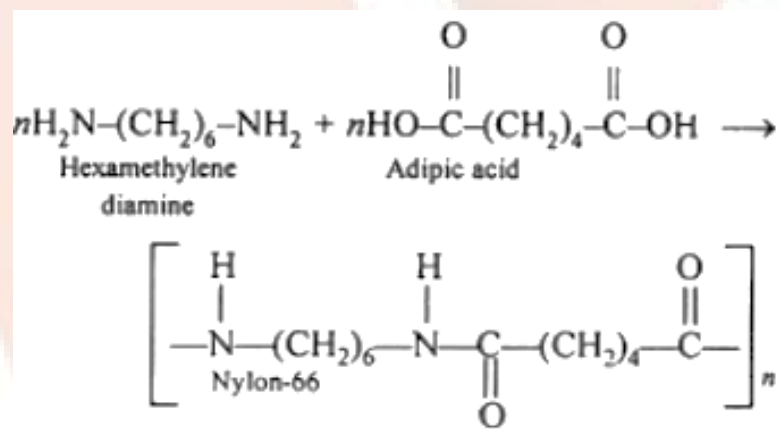
1 mole of MnO_4^{2-} produces $\frac{1}{3}$ mol of MnO_2

3 moles of MnO_4^{2-} produces 2 mol of MnO_4^-

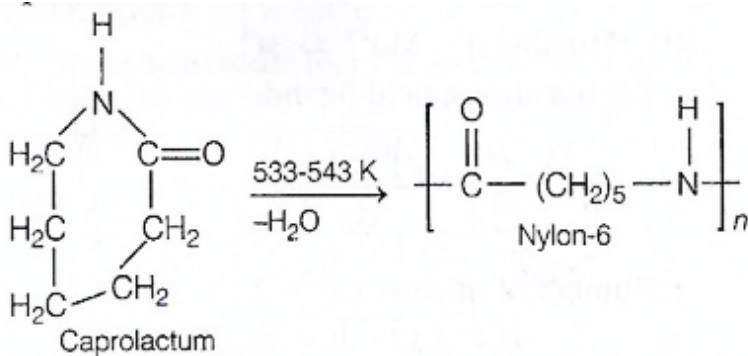
Therefore,

1 mole of MnO_4^{2-} produces $\frac{2}{3}$ mol of MnO_4^-

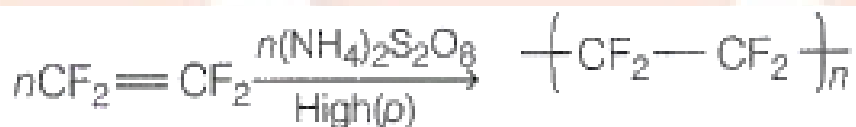
151. (A) Nylon 6,6 is formed by the reaction of hexamethylene diamine and adipic acid as follows:



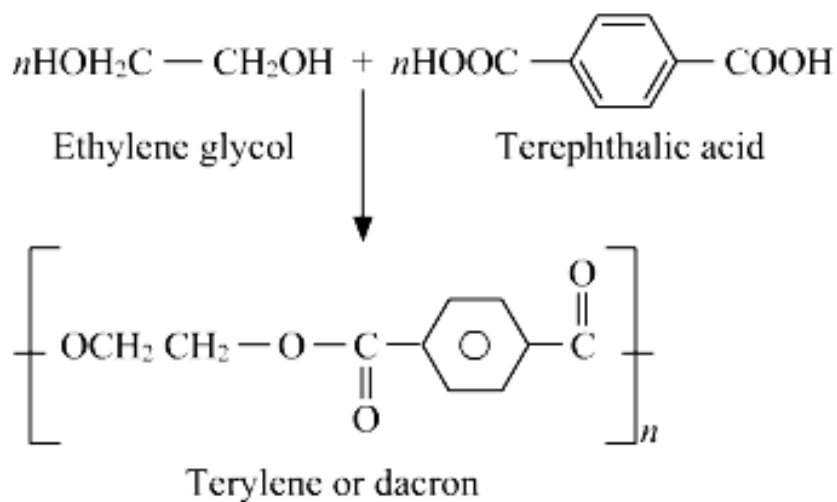
(B) Nylon 6 is formed by the reaction of caprolactam at 533-543 K.



(C) Polytetrafluoroethylene is formed from tetrafluoroethene monomer as follows:



(D) Dacron is formed by the reaction of terephthalic acid and ethylene glycol as shown below.



Therefore, A-IV, B-III, C-V and D-I

152. In a DNA molecule, the two strands are complementary to each other.

Therefore, statement (c) is not correct.

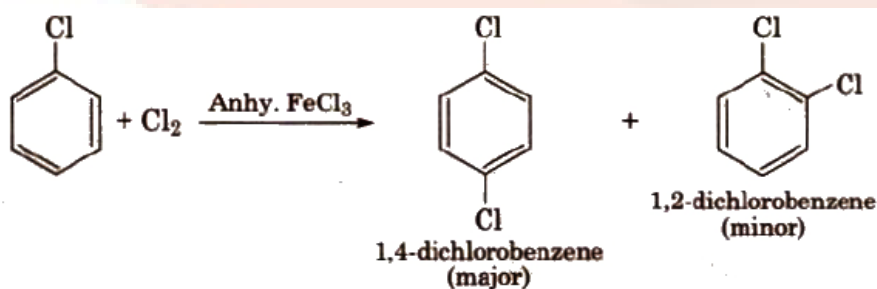
153. Ofloxacin is not an analgesic. It is an antibiotic.

154. (a) The carbon chlorine bond in chlorobenzene is shorter than in chloromethane because chlorobenzene has double bond character which makes bond length shorter.

(b) It becomes difficult to replace chlorine from chlorobenzene due to the presence of double bond character in chlorobenzene.

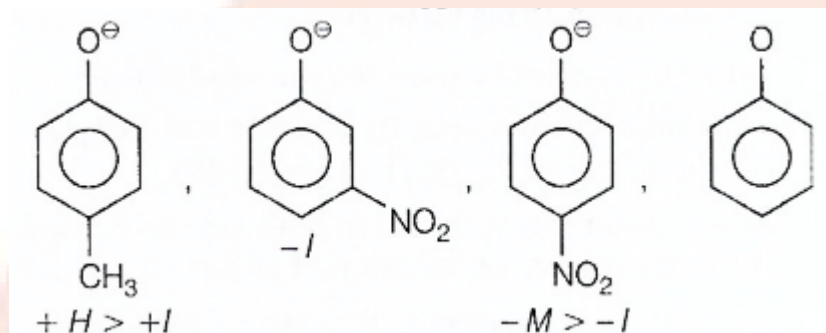
(c) The C-Cl bond in chlorobenzene has some double bond character.

(d) On chlorination of chlorobenzene, it forms ortho and para derivative of dichlorobenzene as follows:

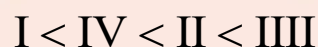


Therefore, statements (a),(b) and (c) are correct.

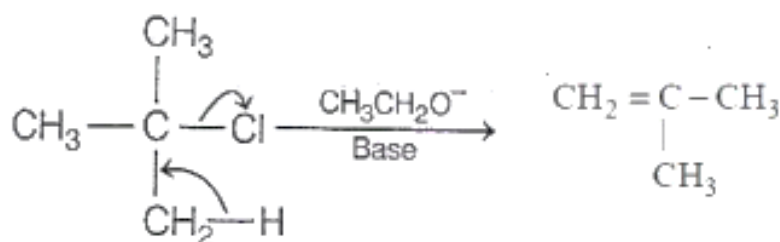
155. The strength of acidic character depends upon the stability of conjugate base. The stability of anion is directly proportional to $-M$, $-I$ effect and inversely proportional to $+I$, H effect.



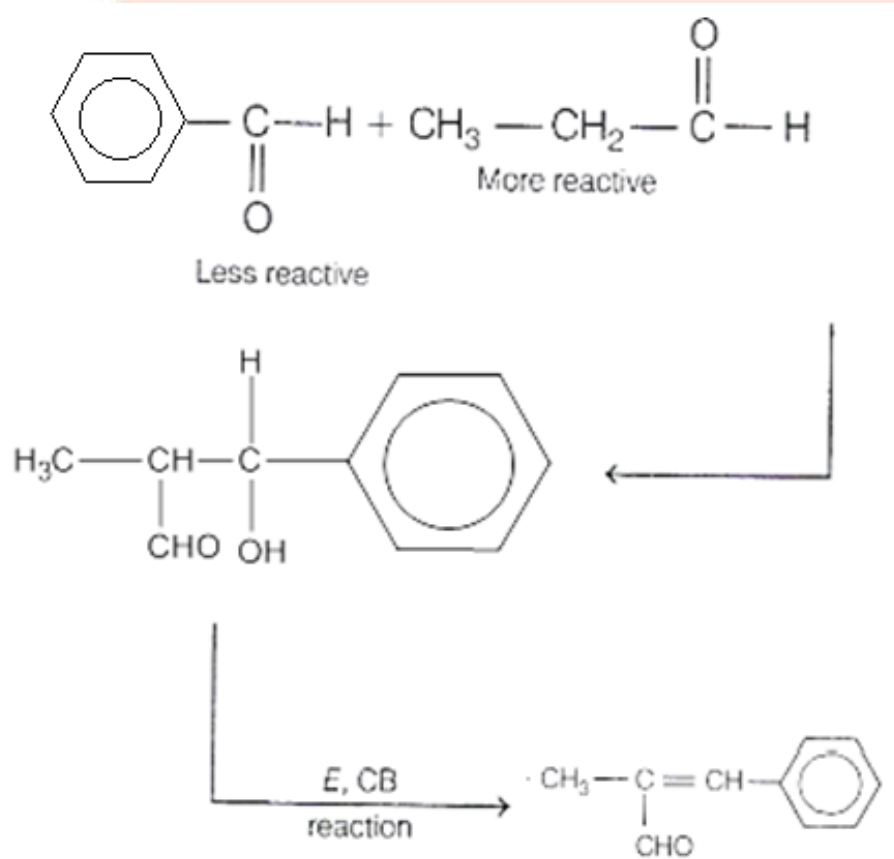
Therefore, the order of acidic character is shown below.



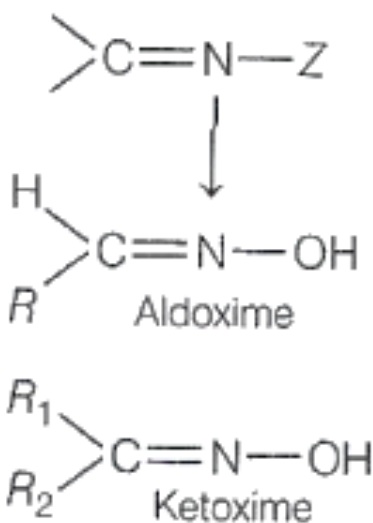
156. In the given reaction, alkyl halide is hindered and base is less bulkier. As a result, elimination reaction will occur. The major product obtained is shown below.



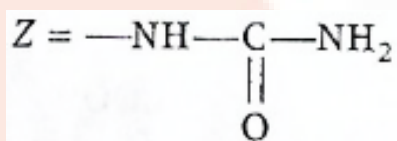
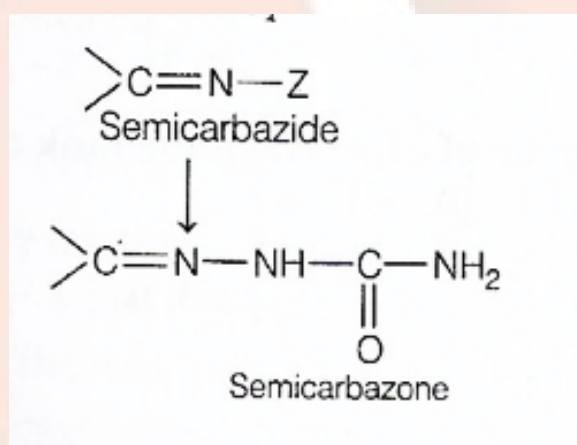
157. The major product formed is shown below.



158. The structures of oxime are shown below where Z is OH.

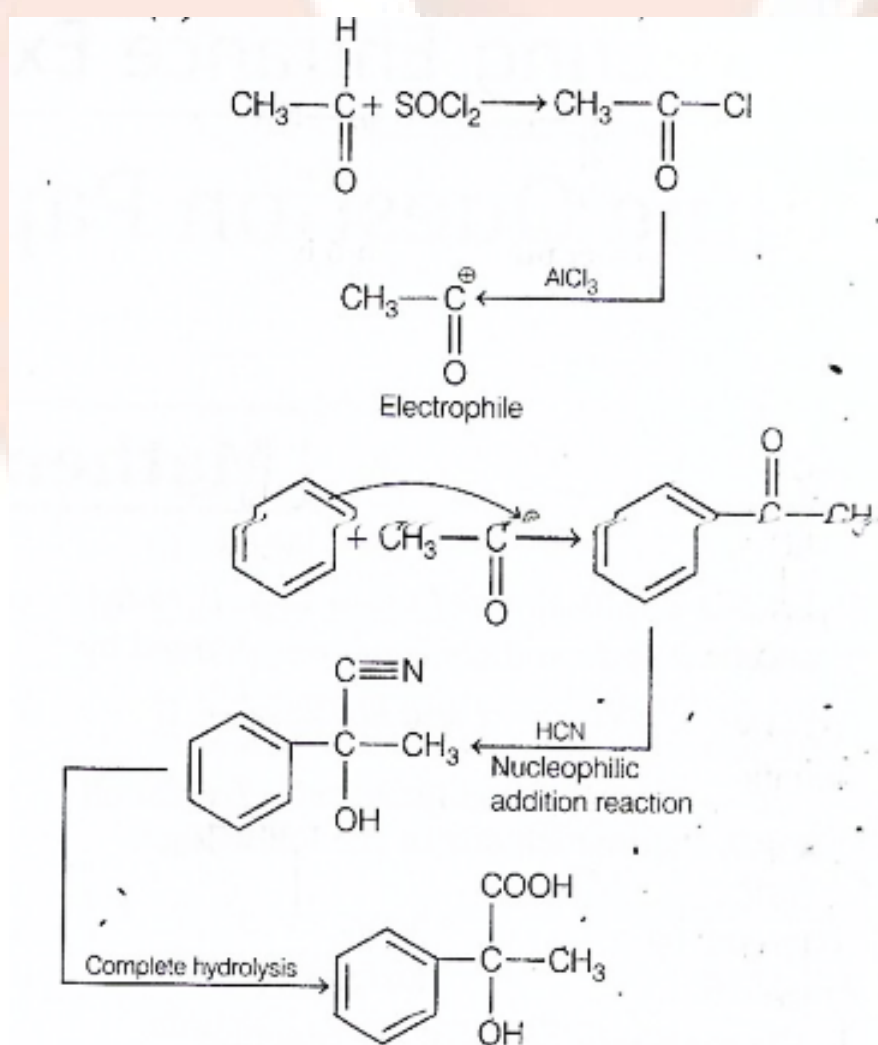


The structure of semicarbazone is shown below.



Where Z is

159. The product D is shown below.



160. The products X and Y are shown below.

